
Analytical geometry

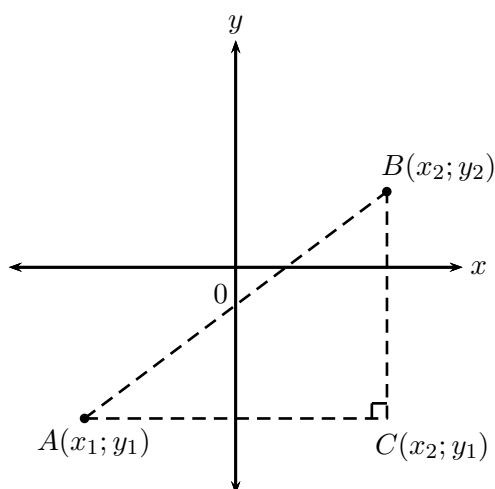
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Analytical geometry, also referred to as coordinate or Cartesian geometry, is the study of geometric properties and relationships between points, lines and angles in the Cartesian plane. Geometrical shapes are defined using a coordinate system and algebraic principles. In this chapter we deal with the equation of a straight line, parallel and perpendicular lines and inclination of a line.

4.1 Revision

EMBG7

Points $A(x_1; y_1)$, $B(x_2; y_2)$ and $C(x_2; y_1)$ are shown in the diagram below:



Theorem of Pythagoras

$$AB^2 = AC^2 + BC^2$$

Distance formula

Distance between two points:

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Notice that $(x_1 - x_2)^2 = (x_2 - x_1)^2$.

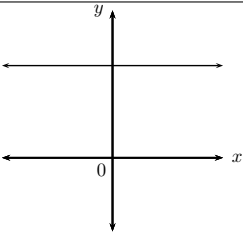
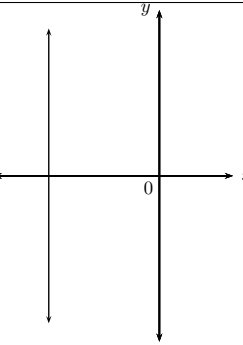
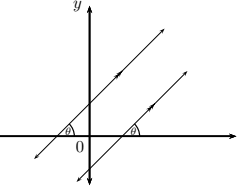
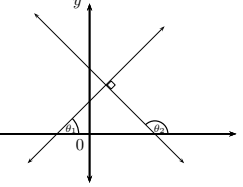
► See video: 22JD at www.everythingmaths.co.za

Gradient

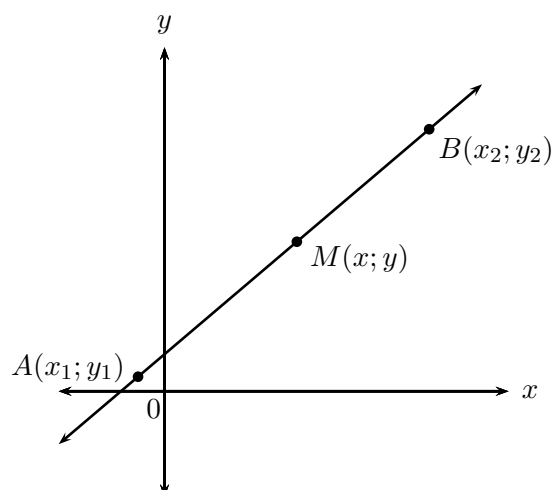
Gradient (m) describes the slope or steepness of the line joining two points. The gradient of a line is determined by the ratio of vertical change to horizontal change.

$$m_{AB} = \frac{y_2 - y_1}{x_2 - x_1} \quad \text{or} \quad m_{AB} = \frac{y_1 - y_2}{x_1 - x_2}$$

Remember to be consistent: $m \neq \frac{y_1 - y_2}{x_2 - x_1}$.

Horizontal lines		$m = 0$
Vertical lines		m is undefined
Parallel lines		$m_1 = m_2$
Perpendicular lines		$m_1 \times m_2 = -1$

Mid-point of a line segment



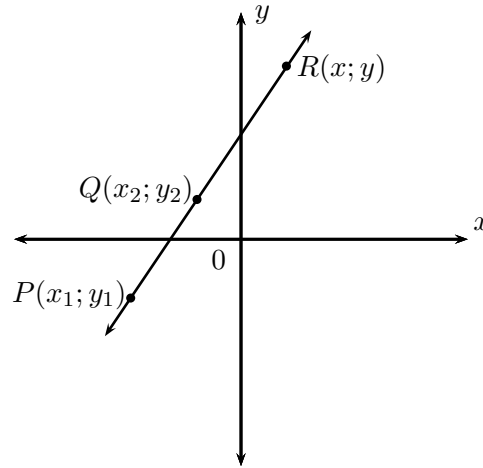
The coordinates of the mid-point $M(x; y)$ of a line between any two points $A(x_1; y_1)$ and $B(x_2; y_2)$:

$$M(x; y) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

► See video: 22JF at www.everythingmaths.co.za

Points on a straight line

The diagram shows points $P(x_1; y_1)$, $Q(x_2; y_2)$ and $R(x; y)$ on a straight line.



We know that $m_{PR} = m_{QR} = m_{PQ}$.

Using $m_{PR} = m_{PQ}$, we obtain the following for any point $(x; y)$ on a straight line

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

Worked example 1: Revision

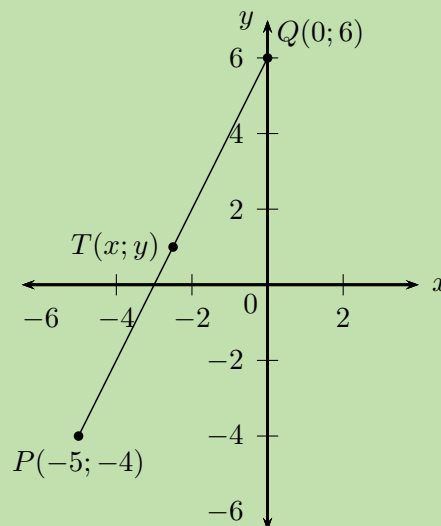
QUESTION

Given the points $P(-5; -4)$ and $Q(0; 6)$:

1. Determine the length of the line segment PQ .
2. Determine the mid-point $T(x; y)$ of the line segment PQ .
3. Show that the line passing through $R(1; -\frac{3}{4})$ and $T(x; y)$ is perpendicular to the line PQ .

SOLUTION

Step 1: Draw a sketch



Step 2: Assign variables to the coordinates of the given points

Let the coordinates of P be $(x_1; y_1)$ and $Q(x_2; y_2)$

$$x_1 = -5; \quad y_1 = -4; \quad x_2 = 0; \quad y_2 = 6$$

Write down the distance formula

$$\begin{aligned} PQ &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(0 - (-5))^2 + (6 - (-4))^2} \\ &= \sqrt{25 + 100} \\ &= \sqrt{125} \\ &= 5\sqrt{5} \end{aligned}$$

The length of the line segment PQ is $5\sqrt{5}$ units.

Step 3: Write down the mid-point formula and substitute the values

$$\begin{aligned} T(x; y) &= \left(\frac{x_1 + x_2}{2}; \frac{y_1 + y_2}{2} \right) \\ x &= \frac{x_1 + x_2}{2} \\ &= \frac{-5 + 0}{2} \\ &= -\frac{5}{2} \\ y &= \frac{y_1 + y_2}{2} \\ &= \frac{-4 + 6}{2} \\ &= \frac{2}{2} \\ &= 1 \end{aligned}$$

The mid-point of PQ is $T(-\frac{5}{2}; 1)$.

Step 4: Determine the gradients of PQ and RT

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} \\ m_{PQ} &= \frac{6 - (-4)}{0 - (-5)} \\ &= \frac{10}{5} \\ &= 2 \end{aligned}$$

$$\begin{aligned}
 m_{RT} &= \frac{-\frac{3}{4} - 1}{1 - (-\frac{5}{2})} \\
 &= \frac{-\frac{7}{4}}{\frac{7}{2}} \\
 &= -\frac{7}{4} \times \frac{2}{7} \\
 &= -\frac{1}{2}
 \end{aligned}$$

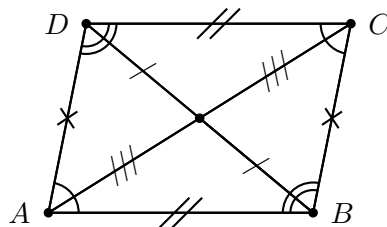
Calculate the product of the two gradients:

$$\begin{aligned}
 m_{RT} \times m_{PQ} &= -\frac{1}{2} \times 2 \\
 &= -1
 \end{aligned}$$

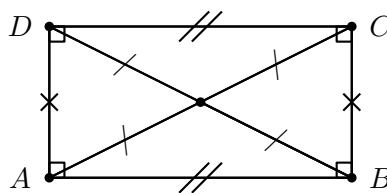
Therefore PQ is perpendicular to RT .

Quadrilaterals

- A quadrilateral is a closed shape consisting of four straight line segments.
- A parallelogram is a quadrilateral with both pairs of opposite sides parallel.

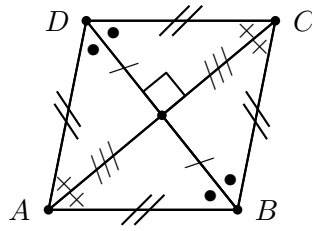


- Both pairs of opposite sides are equal in length.
 - Both pairs of opposite angles are equal.
 - The diagonals bisect each other.
- A rectangle is a parallelogram that has all four angles equal to 90° .

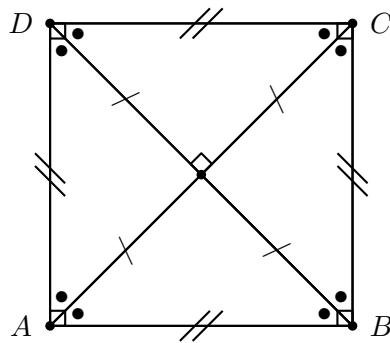


- Both pairs of opposite sides are equal and parallel.
- The diagonals bisect each other.
- The diagonals are equal in length.

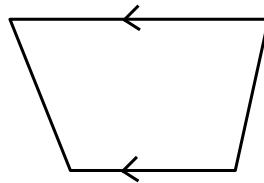
- A rhombus is a parallelogram that has all four sides equal in length.



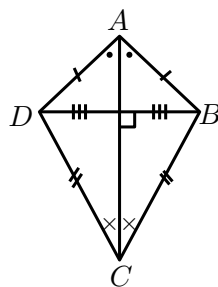
- Both pairs of opposite sides are equal and parallel.
 - The diagonals bisect each other at 90° .
 - The diagonals of a rhombus bisect both pairs of opposite angles.
- A square is a rhombus that has all four interior angles equal to 90° .



- Both pairs of opposite sides are equal and parallel.
 - The diagonals bisect each other at 90° .
 - The diagonals are equal in length.
 - The diagonals bisect both pairs of interior opposite angles (that is, all angles are 45°).
- A trapezium is a quadrilateral with one pair of opposite sides parallel.



- A kite is a quadrilateral with two pairs of adjacent sides equal.



- One pair of opposite angles are equal (the angles are between unequal sides).
- The diagonal between equal sides bisects the other diagonal.
- The diagonal between equal sides bisects the interior angles.
- The diagonals intersect at 90° .

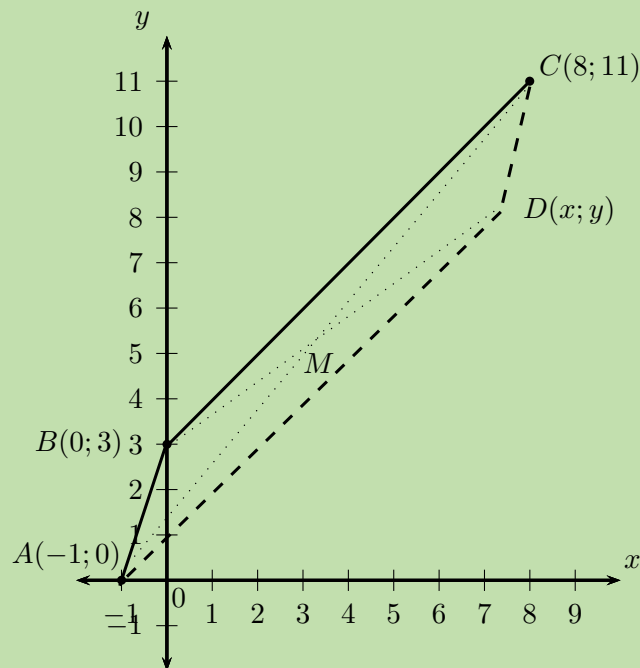
Worked example 2: Quadrilaterals

QUESTION

Points $A(-1;0)$, $B(0;3)$, $C(8;11)$ and $D(x;y)$ are points on the Cartesian plane. Determine $D(x;y)$ if $ABCD$ is a parallelogram.

SOLUTION

Step 1: Draw a sketch



The mid-point of AC will be the same as the mid-point of BD . We first find the mid-point of AC and then use it to determine the coordinates of point D .

Step 2: Assign values to $(x_1; y_1)$ and $(x_2; y_2)$

Let the mid-point of AC be $M(x; y)$

$$x_1 = -1; \quad y_1 = 0; \quad x_2 = 8; \quad y_2 = 11$$

Step 3: Write down the mid-point formula

$$M(x; y) = \left(\frac{x_1 + x_2}{2}; \frac{y_1 + y_2}{2} \right)$$

Step 4: Substitute the values and calculate the coordinates of M

$$\begin{aligned} M(x; y) &= \left(\frac{-1 + 8}{2}; \frac{0 + 11}{2} \right) \\ &= \left(\frac{7}{2}; \frac{11}{2} \right) \end{aligned}$$

Step 5: Use the coordinates of M to determine D

M is also the mid-point of BD so we use $M\left(\frac{7}{2}; \frac{11}{2}\right)$ and $B(0; 3)$ to find $D(x; y)$

Step 6: Substitute values and determine x and y

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$
$$\therefore \left(\frac{7}{2}, \frac{11}{2} \right) = \left(\frac{0 + x}{2}, \frac{3 + y}{2} \right)$$

$$\frac{7}{2} = \frac{0 + x}{2}$$
$$7 = 0 + x$$
$$\therefore x = 7$$

$$\frac{11}{2} = \frac{3 + y}{2}$$
$$11 = 3 + y$$
$$\therefore y = 8$$

Step 7: Alternative method: inspection

Since we are given that $ABCD$ is a parallelogram, we can use the properties of a parallelogram and the given points to determine the coordinates of D .

From the sketch we expect that point D will lie below C .

Consider the given points A , B and C :

- Opposite sides of a parallelogram are parallel, therefore BC must be parallel to AD and their gradients must be equal.
- The vertical change from B to C is 8 units up.
- Therefore the vertical change from A to D is also 8 units up ($y = 0 + 8 = 8$).
- The horizontal change from B to C is 8 units to the right.
- Therefore the horizontal change from A to D is also 8 units to the right ($x = -1 + 8 = 7$).

or

- Opposite sides of a parallelogram are parallel, therefore AB must be parallel to DC and their gradients must be equal.
- The vertical change from A to B is 3 units up.

- Therefore the vertical change from C to D is 3 units down ($y = 11 - 3 = 8$).
- The horizontal change from A to B is 1 unit to the right.
- Therefore the horizontal change from C to D is 1 unit to the left ($x = 8 - 1 = 7$).

Step 8: Write the final answer

The coordinates of D are $(7; 8)$.

Exercise 4 – 1: Revision

- Determine the length of the line segment between the following points:
 - $P(-3; 5)$ and $Q(-1; -5)$
 - $R(0,75; 3)$ and $S(0,75; -4)$
 - $T(2x; y - 2)$ and $U(3x + 1; y - 2)$
- Given $Q(4; 1)$, $T(p; 3)$ and length $QT = \sqrt{8}$ units, determine the value of p .
- Determine the gradient of the line AB if:
 - $A(-5; 3)$ and $B(-7; 4)$
 - $A(3; -2)$ and $B(1; -8)$
- Prove that the line PQ , with $P(0; 3)$ and $Q(5; 5)$, is parallel to the line $5y + 5 = 2x$.
- Given the points $A(-1; -1)$, $B(2; 5)$, $C(-1; -\frac{5}{2})$ and $D(x; -4)$ and $AB \perp CD$, determine the value of x .
- Calculate the coordinates of the mid-point $P(x; y)$ of the line segment between the points:
 - $M(3; 5)$ and $N(-1; -1)$
 - $A(-3; -4)$ and $B(2; 3)$
- The line joining $A(-2; 4)$ and $B(x; y)$ has the mid-point $C(1; 3)$. Determine the values of x and y .
- Given quadrilateral $ABCD$ with vertices $A(0; 3)$, $B(4; 3)$, $C(5; -1)$ and $D(1; -1)$.
 - Determine the equation of the line AD and the line BC .
 - Show that $AD \parallel BC$.
 - Calculate the lengths of AD and BC .
 - Determine the equation of the diagonal BD .
 - What type of quadrilateral is $ABCD$?

9. $MPQN$ is a parallelogram with points $M(-5; 3)$, $P(-1; 5)$ and $Q(4; 5)$. Draw a sketch and determine the coordinates of $N(x; y)$.
10. $PQRS$ is a quadrilateral with points $P(-3; 1)$, $Q(1; 3)$, $R(6; 1)$ and $S(2; -1)$ in the Cartesian plane.
- Determine the lengths of PQ and SR .
 - Determine the mid-point of PR .
 - Show that $PQ \parallel SR$.
 - Determine the equations of the line PS and the line SR .
 - Is $PS \perp SR$? Explain your answer.
 - What type of quadrilateral is $PQRS$?

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- 1a. [22JG](#) 1b. [22JH](#) 1c. [22JJ](#) 2. [22JK](#) 3a. [22JM](#) 3b. [22JN](#)
 4. [22JP](#) 5. [22JQ](#) 6a. [22JR](#) 6b. [22JS](#) 7. [22JT](#) 8. [22JV](#)
 9. [22JW](#) 10. [22JX](#)



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4.2 Equation of a line

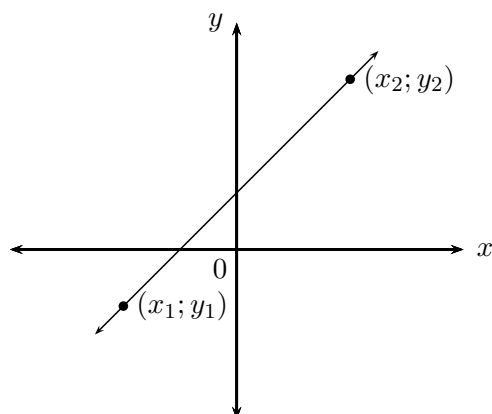
EMBG8

We can derive different forms of the straight line equation. The different forms are used depending on the information provided in the problem:

- The two-point form of the straight line equation: $\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$
- The gradient–point form of the straight line equation: $y - y_1 = m(x - x_1)$
- The gradient–intercept form of the straight line equation: $y = mx + c$

The two-point form of the straight line equation

EMBG9



Given any two points $(x_1; y_1)$ and $(x_2; y_2)$, we can determine the equation of the line passing through the two points using the equation:

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

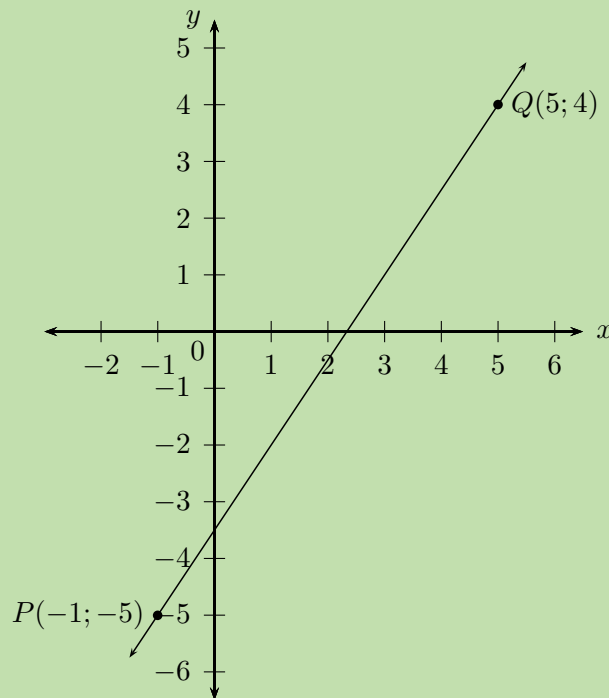
Worked example 3: The two-point form of the straight line equation

QUESTION

Find the equation of the straight line passing through $P(-1; -5)$ and $Q(5; 4)$.

SOLUTION

Step 1: Draw a sketch



Step 2: Assign variables to the coordinates of the given points

Let the coordinates of P be $(x_1; y_1)$ and $Q(x_2; y_2)$

$$x_1 = -1; \quad y_1 = -5; \quad x_2 = 5; \quad y_2 = 4$$

Step 3: Write down the two-point form of the straight line equation

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

Step 4: Substitute the values and make y the subject of the equation

$$\frac{y - (-5)}{x - (-1)} = \frac{4 - (-5)}{5 - (-1)}$$

$$\frac{y + 5}{x + 1} = \frac{9}{6}$$

$$y + 5 = \frac{3}{2}(x + 1)$$

$$y + 5 = \frac{3}{2}x + \frac{3}{2}$$

$$y = \frac{3}{2}x - \frac{7}{2}$$

Step 5: Write the final answer

$$y = \frac{3}{2}x - 3\frac{1}{2}$$

Exercise 4 – 2: The two-point form of the straight line equation

Determine the equation of the straight line passing through the points:

1. $(3; 7)$ and $(-6; 1)$
2. $(1; -\frac{11}{4})$ and $(\frac{2}{3}; -\frac{7}{4})$
3. $(-2; 1)$ and $(3; 6)$
4. $(2; 3)$ and $(3; 5)$
5. $(1; -5)$ and $(-7; -5)$
6. $(-4; 0)$ and $(1; \frac{15}{4})$
7. $(s; t)$ and $(t; s)$
8. $(-2; -8)$ and $(1; 7)$
9. $(2p; q)$ and $(0; -q)$

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1. [22JY](#) 2. [22JZ](#) 3. [22K2](#) 4. [22K3](#) 5. [22K4](#) 6. [22K5](#)
7. [22K6](#) 8. [22K7](#) 9. [22K8](#)



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We derive the gradient–point form of the straight line equation using the definition of gradient and the two-point form of a straight line equation

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

Substitute $m = \frac{y_2 - y_1}{x_2 - x_1}$ on the right-hand side of the equation

$$\frac{y - y_1}{x - x_1} = m$$

Multiply both sides of the equation by $(x - x_1)$

$$y - y_1 = m(x - x_1)$$

To use this equation, we need to know the gradient of the line and the coordinates of one point on the line.

► See video: 22K9 at www.everythingmaths.co.za

Worked example 4: The gradient–point form of the straight line equation

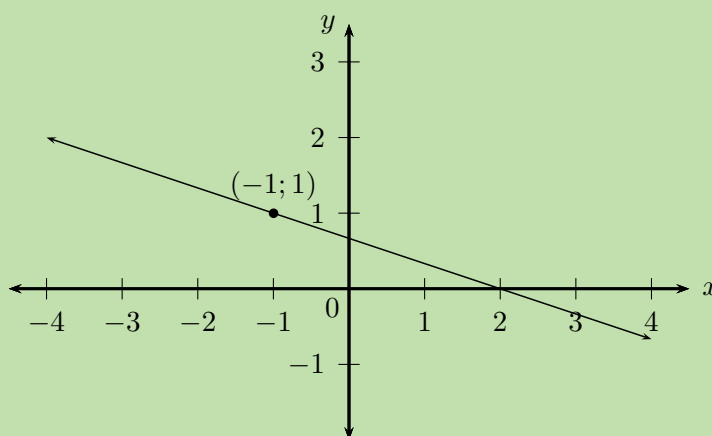
QUESTION

Determine the equation of the straight line with gradient $m = -\frac{1}{3}$ and passing through the point $(-1; 1)$.

SOLUTION

Step 1: Draw a sketch

We notice that $m < 0$, therefore the graph decreases as x increases.



Step 2: Write down the gradient–point form of the straight line equation

$$y - y_1 = m(x - x_1)$$

Substitute the value of the gradient

$$y - y_1 = -\frac{1}{3}(x - x_1)$$

Substitute the coordinates of the given point

$$\begin{aligned}y - 1 &= -\frac{1}{3}(x - (-1)) \\y - 1 &= -\frac{1}{3}(x + 1) \\y &= -\frac{1}{3}x - \frac{1}{3} + 1 \\&= -\frac{1}{3}x + \frac{2}{3}\end{aligned}$$

Step 3: Write the final answer

The equation of the straight line is $y = -\frac{1}{3}x + \frac{2}{3}$.

If we are given two points on a straight line, we can also use the gradient–point form to determine the equation of a straight line. We first calculate the gradient using the two given points and then substitute either of the two points into the gradient–point form of the equation.

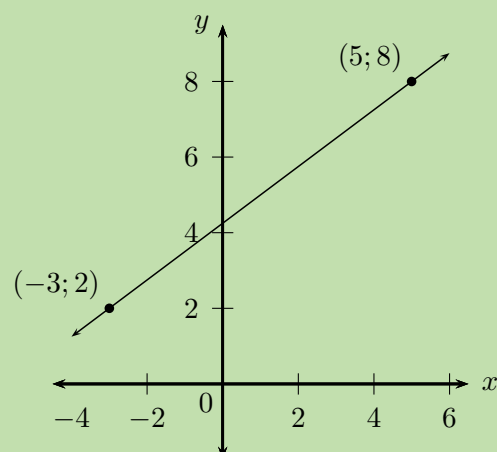
Worked example 5: The gradient–point form of the straight line equation

QUESTION

Determine the equation of the straight line passing through $(-3; 2)$ and $(5; 8)$.

SOLUTION

Step 1: Draw a sketch



Step 2: Assign variables to the coordinates of the given points

$$x_1 = -3; \quad y_1 = 2; \quad x_2 = 5; \quad y_2 = 8$$

Step 3: Calculate the gradient using the two given points

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{8 - 2}{5 - (-3)} \\ &= \frac{6}{8} \\ &= \frac{3}{4} \end{aligned}$$

Step 4: Write down the gradient–point form of the straight line equation

$$y - y_1 = m(x - x_1)$$

Substitute the value of the gradient

$$y - y_1 = \frac{3}{4}(x - x_1)$$

Substitute the coordinates of a given point

$$\begin{aligned} y - y_1 &= \frac{3}{4}(x - x_1) \\ y - 2 &= \frac{3}{4}(x - (-3)) \\ y - 2 &= \frac{3}{4}(x + 3) \\ y &= \frac{3}{4}x + \frac{9}{4} + 2 \\ &= \frac{3}{4}x + \frac{17}{4} \end{aligned}$$

Step 5: Write the final answer

The equation of the straight line is $y = \frac{3}{4}x + 4\frac{1}{4}$.

▶ See video: [22KB](#) at www.everythingmaths.co.za

Exercise 4 – 3: Gradient–point form of a straight line equation

Determine the equation of the straight line:

1. passing through the point $(-1; \frac{10}{3})$ and with $m = \frac{2}{3}$.
2. with $m = -1$ and passing through the point $(-2; 0)$.
3. passing through the point $(3; -1)$ and with $m = -\frac{1}{3}$.
4. parallel to the x -axis and passing through the point $(0; 11)$.
5. passing through the point $(1; 5)$ and with $m = -2$.
6. perpendicular to the x -axis and passing through the point $(-\frac{3}{2}; 0)$.
7. with $m = -0,8$ and passing through the point $(10; -7)$.
8. with undefined gradient and passing through the point $(4; 0)$.
9. with $m = 3a$ and passing through the point $(-2; -6a + b)$.

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1. [22KC](#) 2. [22KD](#) 3. [22KF](#) 4. [22KG](#) 5. [22KH](#) 6. [22KJ](#)
7. [22KK](#) 8. [22KM](#) 9. [22KN](#)



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The gradient–intercept form of a straight line equation EMBGC

Using the gradient–point form, we can also derive the gradient–intercept form of the straight line equation.

Starting with the equation

$$y - y_1 = m(x - x_1)$$

Expand the brackets and make y the subject of the formula

$$\begin{aligned}y - y_1 &= mx - mx_1 \\y &= mx - mx_1 + y_1 \\y &= mx + (y_1 - mx_1)\end{aligned}$$

We define constant c such that $c = y_1 - mx_1$ so that we get the equation

$$y = mx + c$$

This is also called the **standard form** of the straight line equation.

Notice that when $x = 0$, we have

$$\begin{aligned}y &= m(0) + c \\ &= c\end{aligned}$$

Therefore c is the y -intercept of the straight line.

Worked example 6: The gradient–intercept form of straight line equation

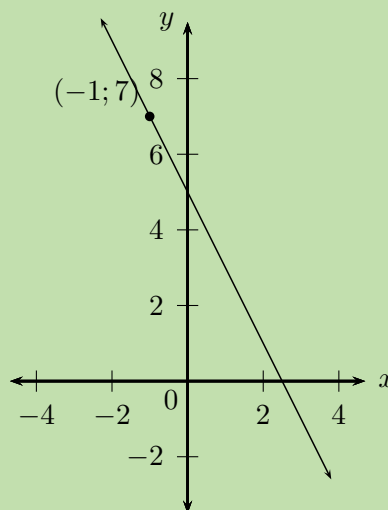
QUESTION

Determine the equation of the straight line with gradient $m = -2$ and passing through the point $(-1; 7)$.

SOLUTION

Step 1: Slope of the line

We notice that $m < 0$, therefore the graph decreases as x increases.



Step 2: Write down the gradient–intercept form of straight line equation

$$y = mx + c$$

Substitute the value of the gradient

$$y = -2x + c$$

Substitute the coordinates of the given point and find c

$$y = -2x + c$$

$$7 = -2(-1) + c$$

$$7 - 2 = c$$

$$\therefore c = 5$$

This gives the y -intercept $(0; 5)$.

Step 3: Write the final answer

The equation of the straight line is $y = -2x + 5$.

If we are given two points on a straight line, we can also use the gradient–intercept form to determine the equation of a straight line. We solve for the two unknowns m and c using simultaneous equations — using the methods of substitution or elimination.

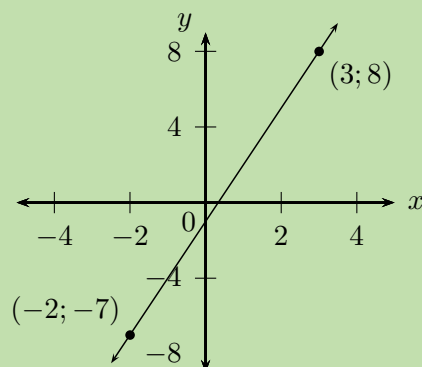
Worked example 7: The gradient–intercept form of straight line equation

QUESTION

Determine the equation of the straight line passing through the points $(-2; -7)$ and $(3; 8)$.

SOLUTION

Step 1: Draw a sketch



Step 2: Write down the gradient–intercept form of straight line equation

$$y = mx + c$$

Step 3: Substitute the coordinates of the given points

$$\begin{aligned}-7 &= m(-2) + c \\ -7 &= -2m + c \quad \dots (1)\end{aligned}$$

$$\begin{aligned}8 &= m(3) + c \\ 8 &= 3m + c \quad \dots (2)\end{aligned}$$

We have two equations with two unknowns; we can therefore solve using simultaneous equations.

Step 4: Make the coefficient of one of the variables the same in both equations

We notice that the coefficient of c in both equations is 1, therefore we can subtract one equation from the other to eliminate c :

$$\begin{aligned}-7 &= -2m + c \\ -(8 &= 3m + c) \\ -15 &= -5m \\ \therefore 3 &= m\end{aligned}$$

Substitute $m = 3$ into either of the two equations and determine c :

$$\begin{aligned}-7 &= -2m + c \\ -7 &= -2(3) + c \\ \therefore c &= -1\end{aligned}$$

or

$$\begin{aligned}8 &= 3m + c \\ 8 &= 3(3) + c \\ \therefore c &= -1\end{aligned}$$

Step 5: Write the final answer

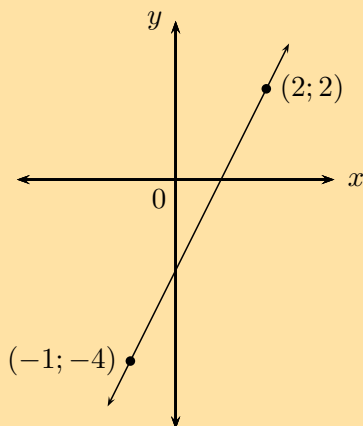
The equation of the straight line is $y = 3x - 1$.

Exercise 4 – 4: The gradient–intercept form of a straight line equation

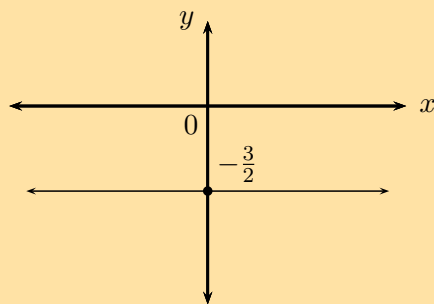
Determine the equation of the straight line:

1. passing through the point $(\frac{1}{2}; 4)$ and with $m = 2$.
2. passing through the points $(\frac{1}{2}; -2)$ and $(2; 4)$.
3. passing through the points $(2; -3)$ and $(-1; 0)$.
4. passing through the point $(2; -\frac{6}{7})$ and with $m = -\frac{3}{7}$.
5. which cuts the y -axis at $y = -\frac{1}{5}$ and with $m = \frac{1}{2}$.

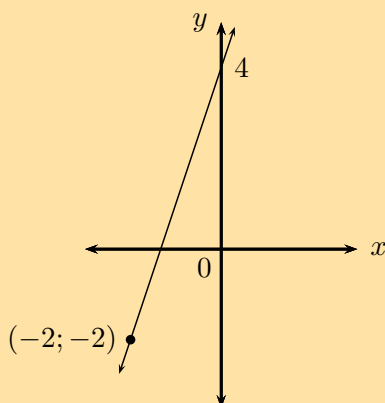
6.



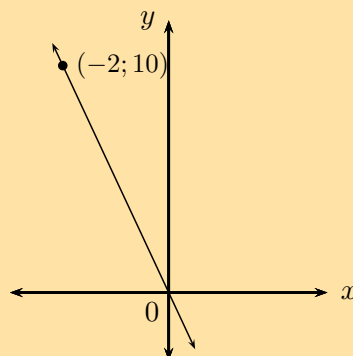
7.



8.



9.



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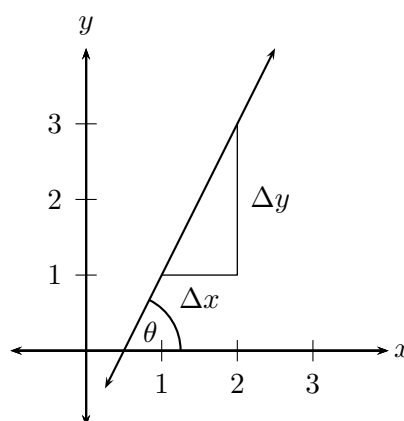
1. 22KP 2. 22KQ 3. 22KR 4. 22KS 5. 22KT 6. 22KV
7. 22KW 8. 22KX 9. 22KY



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The diagram shows that a straight line makes an angle θ with the positive x -axis. This is called the **angle of inclination** of a straight line.

We notice that if the gradient changes, then the value of θ also changes, therefore the angle of inclination of a line is related to its gradient. We know that gradient is the ratio of a change in the y -direction to a change in the x -direction:

$$m = \frac{\Delta y}{\Delta x}$$

From trigonometry we know that the tangent function is defined as the ratio:

$$\tan \theta = \frac{\text{opposite side}}{\text{adjacent side}}$$

And from the diagram we see that

$$\begin{aligned} \tan \theta &= \frac{\Delta y}{\Delta x} \\ \therefore m &= \tan \theta \quad \text{for } 0^\circ \leq \theta < 180^\circ \end{aligned}$$

Therefore the gradient of a straight line is equal to the tangent of the angle formed between the line and the positive direction of the x -axis.

Vertical lines

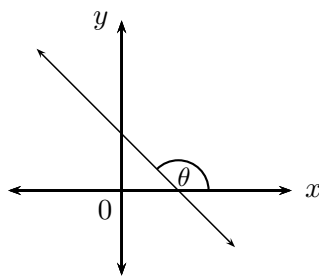
- $\theta = 90^\circ$
- Gradient is undefined since there is no change in the x -values ($\Delta x = 0$).
- Therefore $\tan \theta$ is also undefined (the graph of $\tan \theta$ has an asymptote at $\theta = 90^\circ$).

Horizontal lines

- $\theta = 0^\circ$
- Gradient is equal to 0 since there is no change in the y -values ($\Delta y = 0$).
- Therefore $\tan \theta$ is also equal to 0 (the graph of $\tan \theta$ passes through the origin $(0^\circ; 0)$).

Lines with negative gradients

If a straight line has a negative gradient ($m < 0$, $\tan \theta < 0$), then the angle formed between the line and the positive direction of the x -axis is obtuse.



From the CAST diagram in trigonometry, we know that the tangent function is negative in the second and fourth quadrant. If we are calculating the angle of inclination for a line with a negative gradient, we must add 180° to change the negative angle in the fourth quadrant to an obtuse angle in the second quadrant:

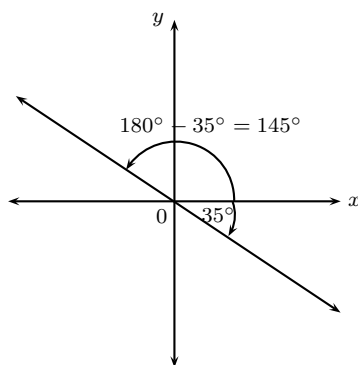
If we are given a straight line with gradient $m = -0,7$, then we can determine the angle of inclination using a calculator:

$$\begin{aligned}\tan \theta &= m \\ &= -0,7 \\ \therefore \theta &= \tan^{-1}(-0,7) \\ &= -35,0^\circ\end{aligned}$$

This negative angle lies in the fourth quadrant. We must add 180° to get an obtuse angle in the second quadrant:

$$\begin{aligned}\theta &= -35,0^\circ + 180^\circ \\ &= 145^\circ\end{aligned}$$

And we can always use our calculator to check that the obtuse angle $\theta = 145^\circ$ gives a gradient of $m = -0,7$.



Exercise 4 – 5: Angle of inclination

- Determine the gradient (correct to 1 decimal place) of each of the following straight lines, given that the angle of inclination is equal to:
 - 60°
 - 135°
 - 0°
 - 54°
 - 90°
 - 45°
 - 140°
 - 180°
 - 75°
- Determine the angle of inclination (correct to 1 decimal place) for each of the following:
 - a line with $m = \frac{3}{4}$
 - $2y - x = 6$
 - the line passes through the points $(-4; -1)$ and $(2; 5)$
 - $y = 4$
 - $x = 3y + \frac{1}{2}$
 - $x = -0,25$
 - the line passes through the points $(2; 5)$ and $(\frac{2}{3}; 1)$
 - a line with gradient equal to $0,577$

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- | | | | | | |
|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| 1a. 22KZ | 1b. 22M2 | 1c. 22M3 | 1d. 22M4 | 1e. 22M5 | 1f. 22M6 |
| 1g. 22M7 | 1h. 22M8 | 1i. 22M9 | 2a. 22MB | 2b. 22MC | 2c. 22MD |
| 2d. 22MF | 2e. 22MG | 2f. 22MH | 2g. 22MJ | 2h. 22MK | |



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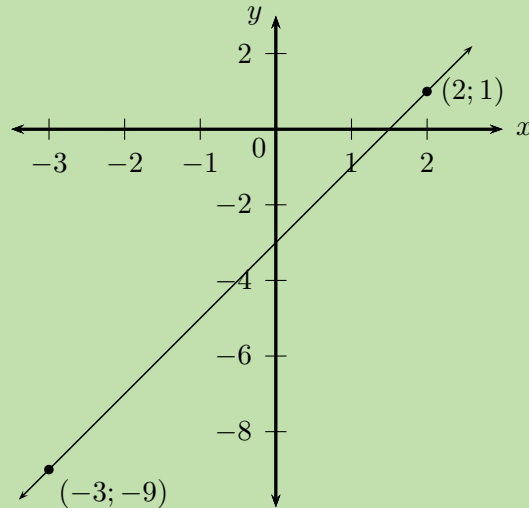
Worked example 8: Inclination of a straight line

QUESTION

Determine the angle of inclination (correct to 1 decimal place) of the straight line passing through the points $(2; 1)$ and $(-3; -9)$.

SOLUTION

Step 1: Draw a sketch



Step 2: Assign variables to the coordinates of the given points

$$x_1 = 2; \quad y_1 = 1; \quad x_2 = -3; \quad y_2 = -9$$

Step 3: Determine the gradient of the line

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{-9 - 1}{-3 - 2} \\ &= \frac{-10}{-5} \\ \therefore m &= 2 \end{aligned}$$

Step 4: Use the gradient to determine the angle of inclination of the line

$$\begin{aligned} \tan \theta &= m \\ &= 2 \\ \therefore \theta &= \tan^{-1} 2 \\ &= 63,4^\circ \end{aligned}$$

Important: make sure your calculator is in DEG (degrees) mode.

Step 5: Write the final answer

The angle of inclination of the straight line is $63,4^\circ$.

Worked example 9: Inclination of a straight line

QUESTION

Determine the equation of the straight line passing through the point (3; 1) and with an angle of inclination of 135° .

SOLUTION

Step 1: Use the angle of inclination to determine the gradient of the line

$$\begin{aligned}m &= \tan \theta \\&= \tan 135^\circ \\ \therefore m &= -1\end{aligned}$$

Step 2: Write down the gradient–point form of the straight line equation

$$y - y_1 = m(x - x_1)$$

Substitute $m = -1$

$$y - y_1 = -(x - x_1)$$

Substitute the given point (3; 1)

$$\begin{aligned}y - 1 &= -(x - 3) \\y &= -x + 3 + 1 \\&= -x + 4\end{aligned}$$

Step 3: Write the final answer

The equation of the straight line is $y = -x + 4$.

Worked example 10: Inclination of a straight line

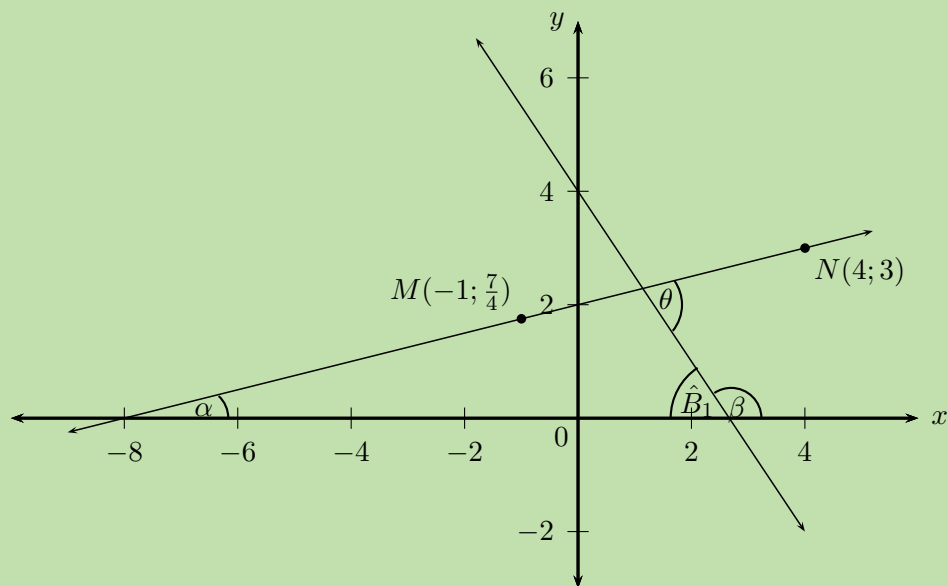
QUESTION

Determine the acute angle (correct to 1 decimal place) between the line passing through the points $M(-1; 1\frac{3}{4})$ and $N(4; 3)$ and the straight line $y = -\frac{3}{2}x + 4$.

SOLUTION

Step 1: Draw a sketch

Draw the line through points $M(-1; 1\frac{3}{4})$ and $N(4; 3)$ and the line $y = -\frac{3}{2}x + 4$ on a suitable system of axes. Label α and β , the angles of inclination of the two lines. Label θ , the acute angle between the two straight lines.



Notice that α and θ are acute angles and β is an obtuse angle.

$$\hat{B}_1 = 180^\circ - \beta \quad (\angle \text{ on str. line})$$

$$\text{and } \theta = \alpha + \hat{B}_1 \quad (\text{ext. } \angle \text{ of } \triangle = \text{sum int. opp})$$

$$\therefore \theta = \alpha + (180^\circ - \beta)$$

$$= 180^\circ + \alpha - \beta$$

Step 2: Use the gradient to determine the angle of inclination β

From the equation $y = -\frac{3}{2}x + 4$ we see that $m < 0$, therefore β is an obtuse angle such that $90^\circ < \beta < 180^\circ$.

$$\begin{aligned} \tan \beta &= m \\ &= -\frac{3}{2} \\ \tan^{-1} \left(-\frac{3}{2} \right) &= -56,3^\circ \end{aligned}$$

This negative angle lies in the fourth quadrant. We know that the angle of inclination β is an obtuse angle that lies in the second quadrant, therefore

$$\begin{aligned} \beta &= -56,3^\circ + 180^\circ \\ &= 123,7^\circ \end{aligned}$$

Step 3: Determine the gradient and angle of inclination of the line through M and N

Determine the gradient

$$\begin{aligned}m &= \frac{y_2 - y_1}{x_2 - x_1} \\&= \frac{3 - \frac{7}{4}}{4 - (-1)} \\&= \frac{\frac{5}{4}}{5} \\&= \frac{1}{4}\end{aligned}$$

Determine the angle of inclination

$$\begin{aligned}\tan \alpha &= m \\&= \frac{1}{4} \\\therefore \alpha &= \tan^{-1} \left(\frac{1}{4} \right) \\&= 14,0^\circ\end{aligned}$$

Step 4: Write the final answer

$$\begin{aligned}\theta &= 180^\circ + \alpha - \beta \\&= 180^\circ + 14,0^\circ - 123,7^\circ \\&= 70,3^\circ\end{aligned}$$

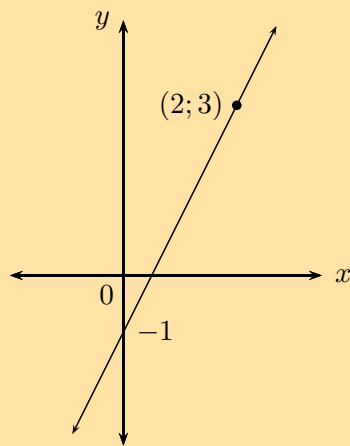
The acute angle between the two straight lines is $70,3^\circ$.

Exercise 4 – 6: Inclination of a straight line

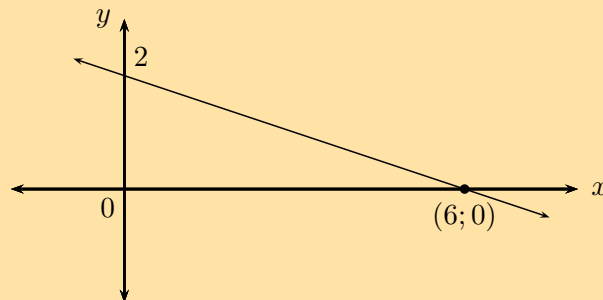
1. Determine the angle of inclination for each of the following:

- a) a line with $m = \frac{4}{5}$
- b) $x + y + 1 = 0$
- c) a line with $m = 5,69$
- d) the line that passes through $(1; 1)$ and $(-2; 7)$
- e) $3 - 2y = 9x$
- f) the line that passes through $(-1; -6)$ and $(-\frac{1}{2}; -\frac{11}{2})$
- g) $5 = 10y - 15x$

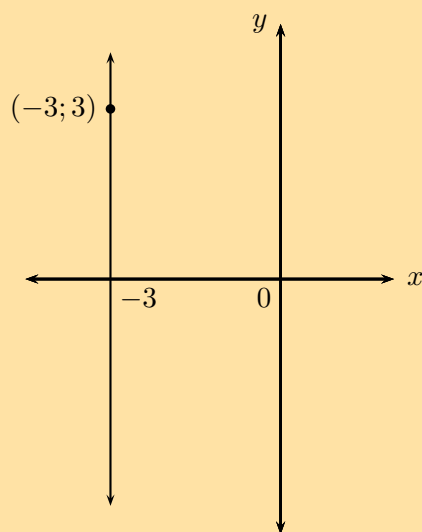
h)



i)



j)



2. Determine the acute angle between the line passing through the points $A(-2; \frac{1}{5})$ and $B(0; 1)$ and the line passing through the points $C(1; 0)$ and $D(-2; 6)$.
3. Determine the angle between the line $y + x = 3$ and the line $x = y + \frac{1}{2}$.
4. Find the angle between the line $y = 2x$ and the line passing through the points $(-1; \frac{7}{3})$ and $(0; 2)$.

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- 1a. [22MM](#) 1b. [22MN](#) 1c. [22MP](#) 1d. [22MQ](#) 1e. [22MR](#) 1f. [22MS](#)
 1g. [22MT](#) 1h. [22MV](#) 1i. [22MW](#) 1j. [22MX](#) 2. [22MY](#) 3. [22MZ](#)
 4. [22N2](#)



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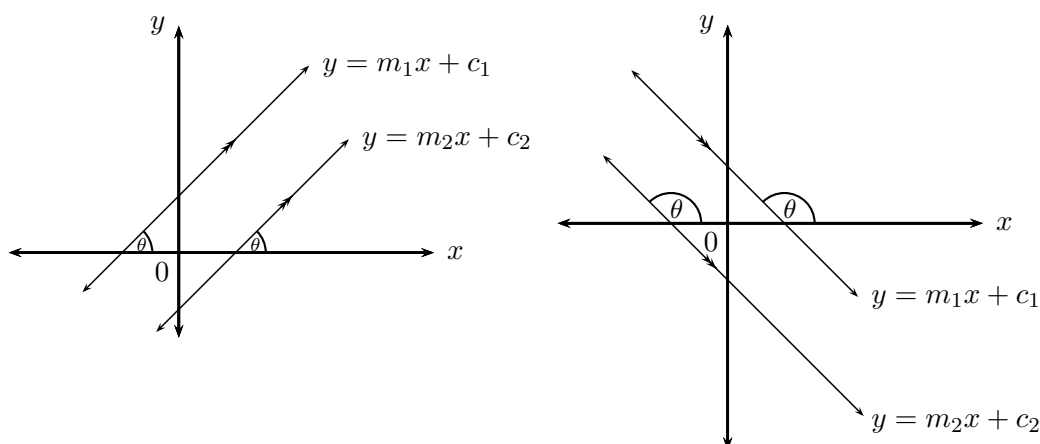
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Investigation: Parallel lines

1. Draw a sketch of the line passing through the points $P(-1; 0)$ and $Q(1; 4)$ and the line passing through the points $R(1; 2)$ and $S(2; 4)$.
2. Label and measure α and β , the angles of inclination of straight lines PQ and RS respectively.
3. Describe the relationship between α and β .
4. " α and β are alternate angles, therefore $PQ \parallel RS$." Is this a true statement? If not, provide a correct statement.
5. Use your calculator to determine $\tan \alpha$ and $\tan \beta$.
6. Complete the sentence: lines have angles of inclination.
7. Determine the equations of the straight lines PQ and RS .
8. What do you notice about m_{PQ} and m_{RS} ?
9. Complete the sentence: lines have gradients.

Another method of determining the equation of a straight line is to be given a point on the unknown line, $(x_1; y_1)$, and the equation of a line which is parallel to the unknown line.

Let the equation of the unknown line be $y = m_1x + c_1$ and the equation of the given line be $y = m_2x + c_2$.



If the two lines are parallel then

$$m_1 = m_2$$

Important: when determining the gradient of a line using the coefficient of x , make sure the given equation is written in the gradient–intercept (standard) form. $y = mx + c$

Substitute the value of m_2 and the given point $(x_1; y_1)$, into the gradient–intercept form of a straight line equation

$$y - y_1 = m(x - x_1)$$

and determine the equation of the unknown line.

Worked example 11: Parallel lines

QUESTION

Determine the equation of the line that passes through the point $(-1; 1)$ and is parallel to the line $y - 2x + 1 = 0$.

SOLUTION

Step 1: Write the equation in gradient–intercept form

We write the given equation in gradient–intercept form and determine the value of m .

$$y = 2x - 1$$

We know that the two lines are parallel, therefore $m_1 = m_2 = 2$.

Step 2: Write down the gradient–point form of the straight line equation

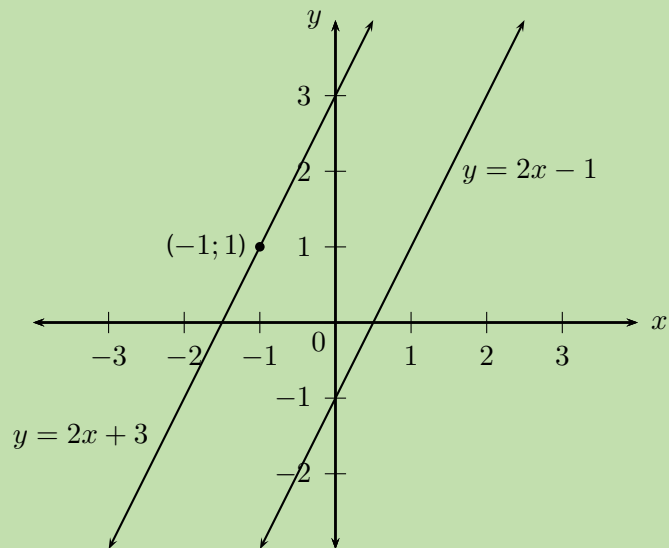
$$y - y_1 = m(x - x_1)$$

Substitute $m = 2$

$$y - y_1 = 2(x - x_1)$$

Substitute the given point $(-1; 1)$

$$\begin{aligned} y - 1 &= 2(x - (-1)) \\ y - 1 &= 2x + 2 \\ y &= 2x + 2 + 1 \\ &= 2x + 3 \end{aligned}$$



A sketch was not required, but it is always helpful and can be used to check answers.

Step 3: Write the final answer

The equation of the straight line is $y = 2x + 3$.

Worked example 12: Parallel lines

QUESTION

Line AB passes through the point $A(0; 3)$ and has an angle of inclination of $153,4^\circ$. Determine the equation of the line CD which passes through the point $C(2; -3)$ and is parallel to AB .

SOLUTION

Step 1: Use the given angle of inclination to determine the gradient

$$\begin{aligned} m_{AB} &= \tan \theta \\ &= \tan 153,4^\circ \\ &= -0,5 \end{aligned}$$

Step 2: Parallel lines have equal gradients

Since we are given $AB \parallel CD$,

$$m_{CD} = m_{AB} = -0,5$$

Step 3: Write down the gradient–point form of a straight line equation

$$y - y_1 = m(x - x_1)$$

Substitute the gradient $m_{CD} = -0,5$.

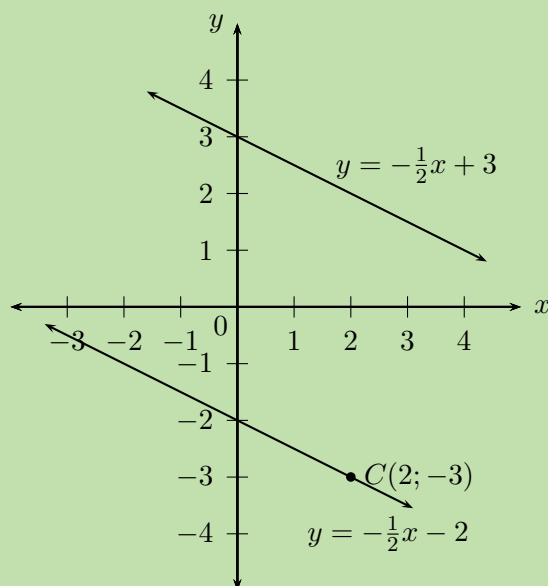
$$y - y_1 = -\frac{1}{2}(x - x_1)$$

Substitute the given point $(2; -3)$.

$$y - (-3) = -\frac{1}{2}(x - 2)$$

$$y + 3 = -\frac{1}{2}x + 1$$

$$y = -\frac{1}{2}x - 2$$



A sketch was not required, but it is always useful.

Step 4: Write the final answer

The equation of the straight line is $y = -\frac{1}{2}x - 2$.

► See video: 22N3 at www.everythingmaths.co.za

Exercise 4 – 7: Parallel lines

- Determine whether or not the following two lines are parallel:
 - $y + 2x = 1$ and $-2x + 3 = y$
 - $\frac{y}{3} + x + 5 = 0$ and $2y + 6x = 1$
 - $y = 2x - 7$ and the line passing through $(1; -2)$ and $(\frac{1}{2}; -1)$
 - $y + 1 = x$ and $x + y = 3$
 - The line passing through points $(-2; -1)$ and $(-4; -3)$ and the line $-y + x - 4 = 0$
 - $y - 1 = \frac{1}{3}x$ and the line passing through points $(-2; 4)$ and $(1; 5)$
- Determine the equation of the straight line that passes through the point $(1; -5)$ and is parallel to the line $y + 2x - 1 = 0$.
- Determine the equation of the straight line that passes through the point $(-2; -6)$ and is parallel to the line $2y + 1 = 6x$.
- Determine the equation of the straight line that passes through the point $(-2; -2)$ and is parallel to the line with angle of inclination $\theta = 56,31^\circ$.
- Determine the equation of the straight line that passes through the point $(-2; \frac{2}{5})$ and is parallel to the line with angle of inclination $\theta = 145^\circ$.

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- 1a. [22N4](#) 1b. [22N5](#) 1c. [22N6](#) 1d. [22N7](#) 1e. [22N8](#) 1f. [22N9](#)
2. [22NB](#) 3. [22NC](#) 4. [22ND](#) 5. [22NF](#)



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4.5 Perpendicular lines

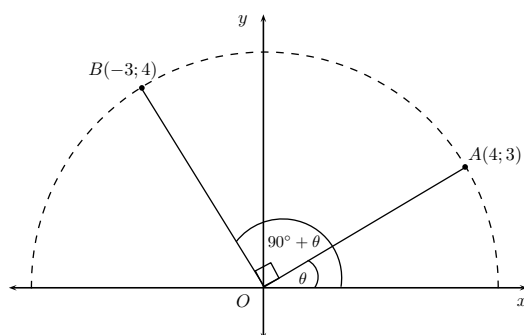
EMBGG

Investigation: Perpendicular lines

- Draw a sketch of the line passing through the points $A(-2; -3)$ and $B(2; 5)$ and the line passing through the points $C(-1; \frac{1}{2})$ and $D(4; -2)$.
- Label and measure α and β , the angles of inclination of straight lines AB and CD respectively.
- Label and measure θ , the angle between the lines AB and CD .
- Describe the relationship between the lines AB and CD .
- " θ is a reflex angle, therefore $AB \perp CD$." Is this a true statement? If not, provide a correct statement.

6. Determine the equation of the straight line AB and the line CD .
7. Use your calculator to determine $\tan \alpha \times \tan \beta$.
8. Determine $m_{AB} \times m_{CD}$.
9. What do you notice about these products?
10. Complete the sentence: if two lines are to each other, then the product of their is equal
11. Complete the sentence: if the gradient of a straight line is equal to the negative of the gradient of another straight line, then the two lines are

Deriving the formula: $m_1 \times m_2 = -1$



Consider the point $A(4; 3)$ on the Cartesian plane with an angle of inclination $\hat{AOX} = \theta$. Rotate through an angle of 90° and place point B at $(-3; 4)$ so that we have the angle of inclination $\hat{BOX} = 90^\circ + \theta$.

We determine the gradient of OA :

$$\begin{aligned} m_{OA} &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{3 - 0}{4 - 0} \\ &= \frac{3}{4} \end{aligned}$$

And determine the gradient of OB :

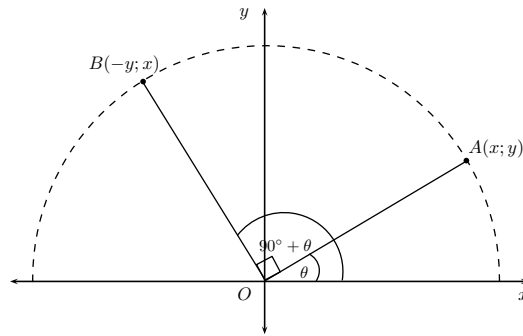
$$\begin{aligned} m_{OB} &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{4 - 0}{-3 - 0} \\ &= \frac{4}{-3} \end{aligned}$$

By rotating through an angle of 90° we know that $OB \perp OA$:

$$\begin{aligned} m_{OA} \times m_{OB} &= \frac{3}{4} \times \frac{4}{-3} \\ &= -1 \end{aligned}$$

We can also write that

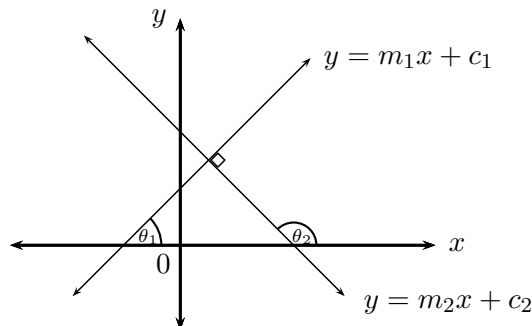
$$m_{OA} = -\frac{1}{m_{OB}}$$



If we have the general point $A(x; y)$ with an angle of inclination $\angle AOX = \theta$ and point $B(-y; x)$ such that $\angle BOX = 90^\circ + \theta$, then we know that

$$\begin{aligned} m_{OA} &= \frac{y}{x} \\ m_{OB} &= -\frac{x}{y} \\ \therefore m_{OA} \times m_{OB} &= \frac{y}{x} \times -\frac{x}{y} \\ &= -1 \end{aligned}$$

Another method of determining the equation of a straight line is to be given a point on the line, $(x_1; y_1)$, and the equation of a line which is perpendicular to the unknown line. Let the equation of the unknown line be $y = m_1x + c_1$ and the equation of the given line be $y = m_2x + c_2$.



If the two lines are perpendicular then

$$m_1 \times m_2 = -1$$

Note: this rule does not apply to vertical or horizontal lines.

When determining the gradient of a line using the coefficient of x , make sure the given equation is written in the gradient–intercept (standard) form $y = mx + c$. Then we know that

$$m_1 = -\frac{1}{m_2}$$

Substitute the value of m_1 and the given point $(x_1; y_1)$, into the gradient–intercept form of the straight line equation $y - y_1 = m(x - x_1)$ and determine the equation of the unknown line.

Worked example 13: Perpendicular lines

QUESTION

Determine the equation of the straight line passing through the point $T(2; 2)$ and perpendicular to the line $3y + 2x - 6 = 0$.

SOLUTION

Step 1: Write the equation in standard form

Let the gradient of the unknown line be m_1 and the given gradient be m_2 . We write the given equation in gradient–intercept form and determine the value of m_2 .

$$\begin{aligned} 3y + 2x - 6 &= 0 \\ 3y &= -2x + 6 \\ y &= -\frac{2}{3}x + 2 \\ \therefore m_2 &= -\frac{2}{3} \end{aligned}$$

We know that the two lines are perpendicular, therefore $m_1 \times m_2 = -1$. Therefore $m_1 = \frac{3}{2}$.

Step 2: Write down the gradient–point form of the straight line equation

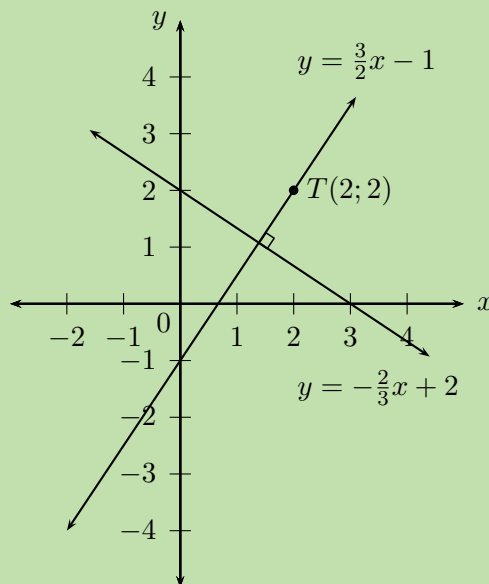
$$y - y_1 = m(x - x_1)$$

Substitute $m_1 = \frac{3}{2}$.

$$y - y_1 = \frac{3}{2}(x - x_1)$$

Substitute the given point $T(2; 2)$.

$$\begin{aligned} y - 2 &= \frac{3}{2}(x - 2) \\ y - 2 &= \frac{3}{2}x - 3 \\ y &= \frac{3}{2}x - 1 \end{aligned}$$



A sketch was not required, but it is useful for checking the answer.

Step 3: Write the final answer

The equation of the straight line is $y = \frac{3}{2}x - 1$.

Worked example 14: Perpendicular lines

QUESTION

Determine the equation of the straight line passing through the point $(2; \frac{1}{3})$ and perpendicular to the line with an angle of inclination of $71,57^\circ$.

SOLUTION

Step 1: Use the given angle of inclination to determine gradient

Let the gradient of the unknown line be m_1 and let the given gradient be m_2 .

$$\begin{aligned} m_2 &= \tan \theta \\ &= \tan 71,57^\circ \\ &= 3,0 \end{aligned}$$

Step 2: Determine the unknown gradient

Since we are given that the two lines are perpendicular,

$$\begin{aligned} m_1 \times m_2 &= -1 \\ \therefore m_1 &= -\frac{1}{3} \end{aligned}$$

Step 3: Write down the gradient–point form of the straight line equation

$$y - y_1 = m(x - x_1)$$

Substitute the gradient $m_1 = -\frac{1}{3}$.

$$y - y_1 = -\frac{1}{3}(x - x_1)$$

Substitute the given point $(2; \frac{1}{3})$.

$$y - \left(\frac{1}{3}\right) = -\frac{1}{3}(x - 2)$$

$$y - \frac{1}{3} = -\frac{1}{3}x + \frac{2}{3}$$

$$y = -\frac{1}{3}x + 1$$

Step 4: Write the final answer

The equation of the straight line is $y = -\frac{1}{3}x + 1$.

► See video: 22NG at www.everythingmaths.co.za

Exercise 4 – 8: Perpendicular lines

1. Calculate whether or not the following two lines are perpendicular:

a) $y - 1 = 4x$ and $4y + x + 2 = 0$

b) $10x = 5y - 1$ and $5y - x - 10 = 0$

c) $x = y - 5$ and the line passing through $(-1; \frac{5}{4})$ and $(3; -\frac{11}{4})$

d) $y = 2$ and $x = 1$

e) $\frac{y}{3} = x$ and $3y + x = 9$

f) $1 - 2x = y$ and the line passing through $(2; -1)$ and $(-1; 5)$

g) $y = x + 2$ and $2y + 1 = 2x$

2. Determine the equation of the straight line that passes through the point $(-2; -4)$ and is perpendicular to the line $y + 2x = 1$.

3. Determine the equation of the straight line that passes through the point $(2; -7)$ and is perpendicular to the line $5y - x = 0$.

4. Determine the equation of the straight line that passes through the point $(3; -1)$ and is perpendicular to the line with angle of inclination $\theta = 135^\circ$.
5. Determine the equation of the straight line that passes through the point $(-2; \frac{2}{5})$ and is perpendicular to the line $y = \frac{4}{3}$.

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1a. [22NH](#) 1b. [22NJ](#) 1c. [22NK](#) 1d. [22NM](#) 1e. [22NN](#) 1f. [22NP](#)
 1g. [22NQ](#) 2. [22NR](#) 3. [22NS](#) 4. [22NT](#) 5. [22NV](#)



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4.6 Summary

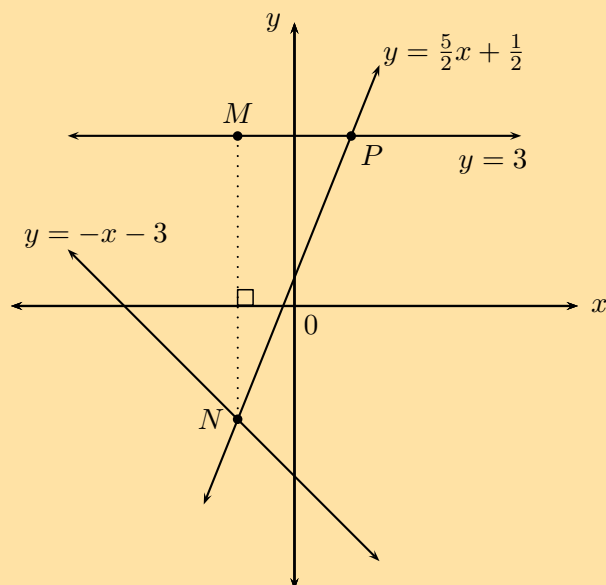
EMBGH

▶ See presentation: [22NW](#) at www.everythingmaths.co.za

- Distance between two points: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- Gradient of a line between two points: $m = \frac{y_2 - y_1}{x_2 - x_1}$
- Mid-point of a line: $M(x; y) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
- Parallel lines: $m_1 = m_2$
- Perpendicular lines: $m_1 \times m_2 = -1$
- General form of a straight line equation: $ax + by + c = 0$
- Two-point form of a straight line equation: $\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$
- Gradient–point form of a straight line equation: $y - y_1 = m(x - x_1)$
- Gradient–intercept form of a straight line equation (standard form): $y = mx + c$
- Angle of inclination of a straight line: θ , the angle formed between the line and the positive x -axis; $m = \tan \theta$

Exercise 4 – 9: End of chapter exercises

1. Determine the equation of the line:
 - a) through points $(-1; 3)$ and $(1; 4)$
 - b) through points $(7; -3)$ and $(0; 4)$
 - c) parallel to $y = \frac{1}{2}x + 3$ and passing through $(-2; 3)$
 - d) perpendicular to $y = -\frac{1}{2}x + 3$ and passing through $(-1; 2)$
 - e) perpendicular to $3y + x = 6$ and passing through the origin
2. Determine the angle of inclination of the following lines:
 - a) $y = 2x - 3$
 - b) $y = \frac{1}{3}x - 7$
 - c) $4y = 3x + 8$
 - d) $y = -\frac{2}{3}x + 3$
 - e) $3y + x - 3 = 0$
3. $P(2; 3)$, $Q(-4; 0)$ and $R(5; -3)$ are the vertices of $\triangle PQR$ in the Cartesian plane. PR intersects the x -axis at S . Determine the following:
 - a) the equation of the line PR
 - b) the coordinates of point S
 - c) the angle of inclination of PR (correct to two decimal places)
 - d) the gradient of line PQ
 - e) $\hat{QPR}$
 - f) the equation of the line perpendicular to PQ and passing through the origin
 - g) the mid-point M of QR
 - h) the equation of the line parallel to PR and passing through point M
4. Points $A(-3; 5)$, $B(-7; -4)$ and $C(2; 0)$ are given.
 - a) Plot the points on the Cartesian plane.
 - b) Determine the coordinates of D if $ABCD$ is a parallelogram.
 - c) Prove that $ABCD$ is a rhombus.
- 5.



Consider the sketch above, with the following lines shown:

$$y = -x - 3$$

$$y = 3$$

$$y = \frac{5}{2}x + \frac{1}{2}$$

- a) Determine the coordinates of the point N .
 - b) Determine the coordinates of the point P .
 - c) Determine the equation of the vertical line MN .
 - d) Determine the length of the vertical line MN .
 - e) Find $M\hat{N}P$.
 - f) Determine the equation of the line parallel to NP and passing through the point M .
6. The following points are given: $A(-2; 3)$, $B(2; 4)$, $C(3; 0)$.
- a) Plot the points on the Cartesian plane.
 - b) Prove that $\triangle ABC$ is a right-angled isosceles triangle.
 - c) Determine the equation of the line AB .
 - d) Determine the coordinates of D if $ABCD$ is a square.
 - e) Determine the coordinates of E , the mid-point of BC .
7. Given points $S(2; 5)$, $T(-3; -4)$ and $V(4; -2)$.
- a) Determine the equation of the line ST .
 - b) Determine the size of $T\hat{S}V$.
8. Consider triangle FGH with vertices $F(-1; 3)$, $G(2; 1)$ and $H(4; 4)$.
- a) Sketch $\triangle FGH$ on the Cartesian plane.
 - b) Show that $\triangle FGH$ is an isosceles triangle.
 - c) Determine the equation of the line PQ , perpendicular bisector of FH .
 - d) Does G lie on the line PQ ?
 - e) Determine the equation of the line parallel to GH and passing through point F .
9. Given the points $A(-1; 5)$, $B(5; -3)$ and $C(0; -6)$. M is the mid-point of AB and N is the mid-point of AC .
- a) Draw a sketch on the Cartesian plane.
 - b) Show that the coordinates of M and N are $(2; 1)$ and $(-\frac{1}{2}; -\frac{1}{2})$ respectively.
 - c) Use analytical geometry methods to prove the mid-point theorem. (Prove that $NM \parallel CB$ and $NM = \frac{1}{2}CB$.)

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- | | | | | | |
|----------|----------|----------|----------|----------|----------|
| 1a. 22NX | 1b. 22NY | 1c. 22NZ | 1d. 22P2 | 1e. 22P3 | 2a. 22P4 |
| 2b. 22P5 | 2c. 22P6 | 2d. 22P7 | 2e. 22P8 | 3. 22P9 | 4. 22PB |
| 5. 22PC | 6. 22PD | 7. 22PF | 8. 22PG | 9. 22PH | |



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Functions

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A function describes a specific relationship between two variables; where an independent (input) variable has exactly one dependent (output) variable. Every element in the domain maps to only one element in the range. Functions can be one-to-one relations or many-to-one relations. A many-to-one relation associates two or more values of the independent variable with a single value of the dependent variable. Functions allow us to visualise relationships in the form of graphs, which are much easier to read and interpret than lists of numbers.

5.1 Quadratic functions

EMBGJ

Revision

EMBGK

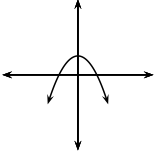
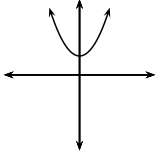
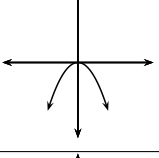
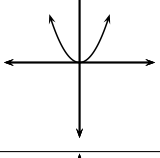
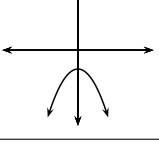
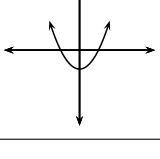
Functions of the form $y = ax^2 + q$

Functions of the general form $y = ax^2 + q$ are called parabolic functions, where a and q are constants.

The effects of a and q on $f(x) = ax^2 + q$:

- The effect of q on vertical shift

- For $q > 0$, $f(x)$ is shifted vertically upwards by q units.
The turning point of $f(x)$ is above the x -axis.
- For $q < 0$, $f(x)$ is shifted vertically downwards by q units.
The turning point of $f(x)$ is below the x -axis.
- q is also the y -intercept of the parabola.

	$a < 0$	$a > 0$
$q > 0$		
$q = 0$		
$q < 0$		

- The effect of a on shape

- For $a > 0$; the graph of $f(x)$ is a “smile” and has a minimum turning point $(0; q)$. As the value of a becomes larger, the graph becomes narrower.
As a gets closer to 0, $f(x)$ becomes wider.
- For $a < 0$; the graph of $f(x)$ is a “frown” and has a maximum turning point $(0; q)$. As the value of a becomes smaller, the graph becomes narrower.
As a gets closer to 0, $f(x)$ becomes wider.

Exercise 5 – 1: Revision

1. On separate axes, accurately draw each of the following functions.

- Use tables of values if necessary.
- Use graph paper if available.

a) $y_1 = x^2$

b) $y_2 = \frac{1}{2}x^2$

c) $y_3 = -x^2 - 1$

d) $y_4 = -2x^2 + 4$

2. Use your sketches of the functions given above to complete the following table (the first column has been completed as an example):

	y_1	y_2	y_3	y_4
value of q	$q = 0$			
effect of q	$y_{\text{int}} = 0$			
value of a	$a = 1$			
effect of a	standard parabola			
turning point	$(0; 0)$			
axis of symmetry	$x = 0$ (y -axis)			
domain	$\{x : x \in \mathbb{R}\}$			
range	$\{y : y \geq 0\}$			

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1a. [22PJ](#) 1b. [22PK](#) 1c. [22PM](#) 1d. [22PN](#) 2. [22PP](#)



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▶ See video: [22PQ](#) at www.everythingmaths.co.za

We now consider parabolic functions of the form $y = a(x + p)^2 + q$ and the effects of parameter p .

Investigation: The effects of a , p and q on a parabolic graph

1. On the same system of axes, plot the following graphs:

- | | |
|----------------------|----------------------|
| a) $y_1 = x^2$ | d) $y_4 = (x + 1)^2$ |
| b) $y_2 = (x - 2)^2$ | |
| c) $y_3 = (x - 1)^2$ | e) $y_5 = (x + 2)^2$ |

Use your sketches of the functions above to complete the following table:

	y_1	y_2	y_3	y_4	y_5
x-intercept(s)					
y-intercept					
turning point					
axis of symmetry					
domain					
range					
effect of p					

2. On the same system of axes, plot the following graphs:

- | | |
|--------------------------|--------------------------|
| a) $y_1 = x^2 + 2$ | d) $y_4 = (x + 1)^2 + 1$ |
| b) $y_2 = (x - 2)^2 - 1$ | |
| c) $y_3 = (x - 1)^2 + 1$ | e) $y_5 = (x + 2)^2 - 1$ |

Use your sketches of the functions above to complete the following table:

	y_1	y_2	y_3	y_4	y_5
x-intercept(s)					
y-intercept					
turning point					
axis of symmetry					
domain					
range					
effect of q					

3. Consider the three functions given below and answer the questions that follow:

- $y_1 = (x - 2)^2 + 1$
- $y_2 = 2(x - 2)^2 + 1$
- $y_3 = -\frac{1}{2}(x - 2)^2 + 1$

- What is the value of a for y_2 ?
- Does y_1 have a minimum or maximum turning point?

- c) What are the coordinates of the turning point of y_2 ?
- d) Compare the graphs of y_1 and y_2 . Discuss the similarities and differences.
- e) What is the value of a for y_3 ?
- f) Will the graph of y_3 be narrower or wider than the graph of y_1 ?
- g) Determine the coordinates of the turning point of y_3 .
- h) Compare the graphs of y_1 and y_3 . Describe any differences.

► See video: 22PR at www.everythingmaths.co.za

The effect of the parameters on $y = a(x + p)^2 + q$

The effect of p is a horizontal shift because all points are moved the same distance in the same direction (the entire graph slides to the left or to the right).

- For $p > 0$, the graph is shifted to the left by p units.
- For $p < 0$, the graph is shifted to the right by p units.

The value of p also affects whether the turning point is to the left of the y -axis ($p > 0$) or to the right of the y -axis ($p < 0$). The axis of symmetry is the line $x = -p$.

The effect of q is a vertical shift. The value of q affects whether the turning point of the graph is above the x -axis ($q > 0$) or below the x -axis ($q < 0$).

The value of a affects the shape of the graph. If $a < 0$, the graph is a “frown” and has a maximum turning point. If $a > 0$ then the graph is a “smile” and has a minimum turning point. When $a = 0$, the graph is a horizontal line $y = q$.

	$p > 0$		$p < 0$	
	$a < 0$	$a > 0$	$a < 0$	$a > 0$
$q > 0$				
$q < 0$				

► See simulation: 22PS at www.everythingmaths.co.za

Discovering the characteristics

For functions of the general form $f(x) = y = a(x + p)^2 + q$:

Domain and range

The domain is $\{x : x \in \mathbb{R}\}$ because there is no value of x for which $f(x)$ is undefined.

The range of $f(x)$ depends on whether the value for a is positive or negative. If $a > 0$ we have:

$$\begin{aligned}(x + p)^2 &\geq 0 && \text{(perfect square is always positive)} \\ \therefore a(x + p)^2 &\geq 0 && (a \text{ is positive}) \\ \therefore a(x + p)^2 + q &\geq q \\ \therefore f(x) &\geq q\end{aligned}$$

The range is therefore $\{y : y \geq q, y \in \mathbb{R}\}$ if $a > 0$. Similarly, if $a < 0$, the range is $\{y : y \leq q, y \in \mathbb{R}\}$.

Worked example 1: Domain and range

QUESTION

State the domain and range for $g(x) = -2(x - 1)^2 + 3$.

SOLUTION

Step 1: Determine the domain

The domain is $\{x : x \in \mathbb{R}\}$ because there is no value of x for which $g(x)$ is undefined.

Step 2: Determine the range

The range of $g(x)$ can be calculated from:

$$\begin{aligned}(x - 1)^2 &\geq 0 \\ -2(x - 1)^2 &\leq 0 \\ -2(x - 1)^2 + 3 &\leq 3 \\ g(x) &\leq 3\end{aligned}$$

Therefore the range is $\{g(x) : g(x) \leq 3\}$ or in interval notation $(-\infty; 3]$.

Notice in the example above that it helps to have the function in the form $y = a(x + p)^2 + q$.

We use the method of **completing the square** to write a quadratic function of the general form $y = ax^2 + bx + c$ in the form $y = a(x + p)^2 + q$ (see Chapter 2).

Exercise 5 – 2: Domain and range

Give the domain and range for each of the following functions:

1. $f(x) = (x - 4)^2 - 1$

4. $j(x) = -2(x + 1)^2$

2. $g(x) = -(x - 5)^2 + 4$

3. $h(x) = x^2 - 6x + 9$

5. $k(x) = -x^2 + 2x - 3$

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1. [22PT](#) 2. [22PV](#) 3. [22PW](#) 4. [22PX](#) 5. [22PY](#)



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Intercepts

The y -intercept:

Every point on the y -axis has an x -coordinate of 0, therefore to calculate the y -intercept we let $x = 0$.

For example, the y -intercept of $g(x) = (x - 1)^2 + 5$ is determined by setting $x = 0$:

$$\begin{aligned} g(x) &= (x - 1)^2 + 5 \\ g(0) &= (0 - 1)^2 + 5 \\ &= 6 \end{aligned}$$

This gives the point $(0; 6)$.

The x -intercept:

Every point on the x -axis has a y -coordinate of 0, therefore to calculate the x -intercept we let $y = 0$.

For example, the x -intercept of $g(x) = (x - 1)^2 + 5$ is determined by setting $y = 0$:

$$\begin{aligned} g(x) &= (x - 1)^2 + 5 \\ 0 &= (x - 1)^2 + 5 \\ -5 &= (x - 1)^2 \end{aligned}$$

which has no real solutions. Therefore, the graph of $g(x)$ lies above the x -axis and does not have any x -intercepts.

Exercise 5 – 3: Intercepts

Determine the x - and y -intercepts for each of the following functions:

1. $f(x) = (x + 4)^2 - 1$

4. $j(x) = 4(x - 3)^2 - 1$

2. $g(x) = 16 - 8x + x^2$

5. $k(x) = 4(x - 3)^2 + 1$

3. $h(x) = -x^2 + 4x - 3$

6. $l(x) = 2x^2 - 3x - 4$

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. [22PZ](#) 2. [22Q2](#) 3. [22Q3](#) 4. [22Q4](#) 5. [22Q5](#) 6. [22Q6](#)



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Turning point

The turning point of the function $f(x) = a(x + p)^2 + q$ is determined by examining the range of the function:

- If $a > 0$, $f(x)$ has a minimum turning point and the range is $[q; \infty)$:

The minimum value of $f(x)$ is q .

If $f(x) = q$, then $a(x + p)^2 = 0$, and therefore $x = -p$.

This gives the turning point $(-p; q)$.

- If $a < 0$, $f(x)$ has a maximum turning point and the range is $(-\infty; q]$:

The maximum value of $f(x)$ is q .

If $f(x) = q$, then $a(x + p)^2 = 0$, and therefore $x = -p$.

This gives the turning point $(-p; q)$.

Therefore the turning point of the quadratic function $f(x) = a(x + p)^2 + q$ is $(-p; q)$.

Alternative form for quadratic equations:

We can also write the quadratic equation in the form

$$y = a(x - p)^2 + q$$

The effect of p is still a horizontal shift, however notice that:

- For $p > 0$, the graph is shifted to the **right** by p units.
- For $p < 0$, the graph is shifted to the **left** by p units.

The turning point is $(p; q)$ and the axis of symmetry is the line $x = p$.

Worked example 2: Turning point

QUESTION

Determine the turning point of $g(x) = 3x^2 - 6x - 1$.

SOLUTION

Step 1: Write the equation in the form $y = a(x + p)^2 + q$

We use the method of completing the square:

$$\begin{aligned}g(x) &= 3x^2 - 6x - 1 \\&= 3(x^2 - 2x) - 1 \\&= 3((x - 1)^2 - 1) - 1 \\&= 3(x - 1)^2 - 3 - 1 \\&= 3(x - 1)^2 - 4\end{aligned}$$

Step 2: Determine turning point $(-p; q)$

From the equation $g(x) = 3(x - 1)^2 - 4$ we know that the turning point for $g(x)$ is $(1; -4)$.

Worked example 3: Turning point

QUESTION

1. Show that the x -value for the turning point of $h(x) = ax^2 + bx + c$ is given by $x = -\frac{b}{2a}$.
2. Hence, determine the turning point of $k(x) = 2 - 10x + 5x^2$.

SOLUTION

Step 1: Write the equation in the form $y = a(x + p)^2 + q$ and show that $p = -\frac{b}{2a}$

We use the method of completing the square:

$$\begin{aligned}h(x) &= ax^2 + bx + c \\&= a\left(x^2 + \frac{b}{a}x + \frac{c}{a}\right)\end{aligned}$$

Take half the coefficient of the x term and square it; then add and subtract it from the

expression.

$$\begin{aligned}h(x) &= a \left(x^2 + \frac{b}{a}x + \left(\frac{b}{2a} \right)^2 - \left(\frac{b}{2a} \right)^2 + \frac{c}{a} \right) \\&= a \left(\left(x + \frac{b}{2a} \right)^2 - \frac{b^2}{4a^2} + \frac{c}{a} \right) \\&= a \left(\left(x + \frac{b}{2a} \right)^2 - \frac{b^2 - 4ac}{4a^2} \right) \\&= a \left(x + \frac{b}{2a} \right)^2 - \frac{b^2 - 4ac}{4a}\end{aligned}$$

From the above we have that the turning point is at $x = -p = -\frac{b}{2a}$ and $y = q = -\frac{b^2 - 4ac}{4a}$.

Step 2: Determine the turning point of $k(x)$

Write the equation in the general form $y = ax^2 + bx + c$.

$$k(x) = 5x^2 - 10x + 2$$

Therefore $a = 5$; $b = -10$; $c = 2$.

Use the results obtained above to determine $x = -\frac{b}{2a}$:

$$\begin{aligned}x &= -\left(\frac{-10}{2(5)} \right) \\&= 1\end{aligned}$$

Substitute $x = 1$ to obtain the corresponding y -value :

$$\begin{aligned}y &= 5x^2 - 10x + 2 \\&= 5(1)^2 - 10(1) + 2 \\&= 5 - 10 + 2 \\&= -3\end{aligned}$$

The turning point of $k(x)$ is $(1; -3)$.

Exercise 5 – 4: Turning points

Determine the turning point of each of the following:

1. $y = x^2 - 6x + 8$
2. $y = -x^2 + 4x - 3$
3. $y = \frac{1}{2}(x + 2)^2 - 1$
4. $y = 2x^2 + 2x + 1$

5. $y = 18 + 6x - 3x^2$

6. $y = -2[(x + 1)^2 + 3]$

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1. 22Q7 2. 22Q8 3. 22Q9 4. 22QB 5. 22QC 6. 22QD



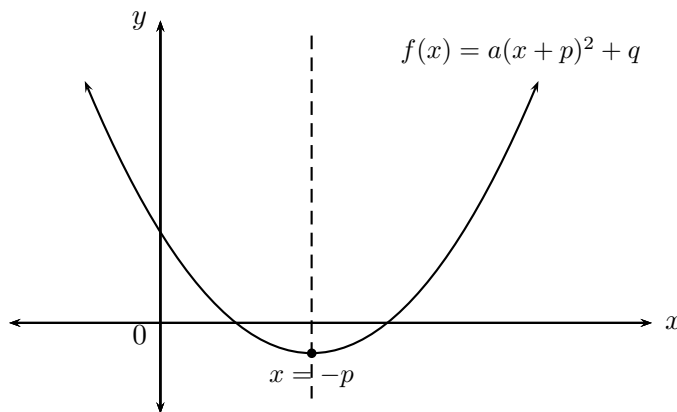
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Axis of symmetry

The axis of symmetry for $f(x) = a(x + p)^2 + q$ is the vertical line $x = -p$. The axis of symmetry passes through the turning point $(-p; q)$ and is parallel to the y -axis.



Exercise 5 – 5: Axis of symmetry

1. Determine the axis of symmetry of each of the following:

a) $y = 2x^2 - 5x - 18$

b) $y = 3(x - 2)^2 + 1$

c) $y = 4x - x^2$

2. Write down the equation of a parabola where the y -axis is the axis of symmetry.

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1a. 22QF 1b. 22QG 1c. 22QH 2. 22QJ



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Sketching graphs of the form $f(x) = a(x + p)^2 + q$

In order to sketch graphs of the form $f(x) = a(x + p)^2 + q$, we need to determine five characteristics:

- sign of a
- turning point
- y -intercept
- x -intercept(s) (if they exist)
- domain and range

▶ See video: [22QK](https://www.everythingmaths.co.za) at www.everythingmaths.co.za

Worked example 4: Sketching a parabola

QUESTION

Sketch the graph of $y = -\frac{1}{2}(x + 1)^2 - 3$.

Mark the intercepts, turning point and the axis of symmetry. State the domain and range of the function.

SOLUTION

Step 1: Examine the equation of the form $y = a(x + p)^2 + q$

We notice that $a < 0$, therefore the graph is a “frown” and has a maximum turning point.

Step 2: Determine the turning point $(-p; q)$

From the equation we know that the turning point is $(-1; -3)$.

Step 3: Determine the axis of symmetry $x = -p$

From the equation we know that the axis of symmetry is $x = -1$.

Step 4: Determine the y -intercept

The y -intercept is obtained by letting $x = 0$:

$$\begin{aligned}y &= -\frac{1}{2}((0) + 1)^2 - 3 \\&= -\frac{1}{2} - 3 \\&= -3\frac{1}{2}\end{aligned}$$

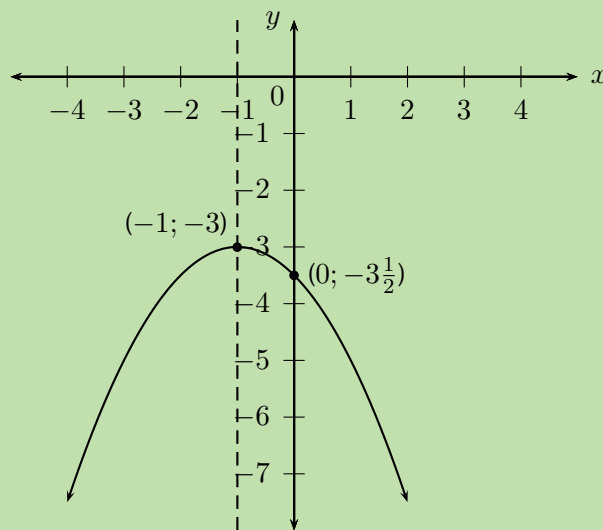
This gives the point $(0; -3\frac{1}{2})$.

Step 5: Determine the x -intercepts

The x -intercepts are obtained by letting $y = 0$:

$$\begin{aligned}0 &= -\frac{1}{2}(x+1)^2 - 3 \\3 &= -\frac{1}{2}(x+1)^2 \\-6 &= (x+1)^2\end{aligned}$$

which has no real solutions. Therefore, there are no x -intercepts and the graph lies below the x -axis.

Step 6: Plot the points and sketch the graph**Step 7: State the domain and range**

Domain: $\{x : x \in \mathbb{R}\}$

Range: $\{y : y \leq -3, y \in \mathbb{R}\}$

▶ See video: [22QM](#) at www.everythingmaths.co.za

Worked example 5: Sketching a parabola**QUESTION**

Sketch the graph of $y = \frac{1}{2}x^2 - 4x + \frac{7}{2}$.

Determine the intercepts, turning point and the axis of symmetry. Give the domain and range of the function.

SOLUTION

Step 1: Examine the equation of the form $y = ax^2 + bx + c$

We notice that $a > 0$, therefore the graph is a “smile” and has a minimum turning point.

Step 2: Determine the turning point and the axis of symmetry

Check that the equation is in standard form and identify the coefficients.

$$a = \frac{1}{2}; \quad b = -4; \quad c = \frac{7}{2}$$

Calculate the x -value of the turning point using

$$\begin{aligned} x &= -\frac{b}{2a} \\ &= -\left(\frac{-4}{2\left(\frac{1}{2}\right)}\right) \\ &= 4 \end{aligned}$$

Therefore the axis of symmetry is $x = 4$.

Substitute $x = 4$ into the original equation to obtain the corresponding y -value.

$$\begin{aligned} y &= \frac{1}{2}x^2 - 4x + \frac{7}{2} \\ &= \frac{1}{2}(4)^2 - 4(4) + \frac{7}{2} \\ &= 8 - 16 + \frac{7}{2} \\ &= -4\frac{1}{2} \end{aligned}$$

This gives the point $(4; -4\frac{1}{2})$.

Step 3: Determine the y -intercept

The y -intercept is obtained by letting $x = 0$:

$$\begin{aligned} y &= \frac{1}{2}(0)^2 - 4(0) + \frac{7}{2} \\ &= \frac{7}{2} \end{aligned}$$

This gives the point $(0; \frac{7}{2})$.

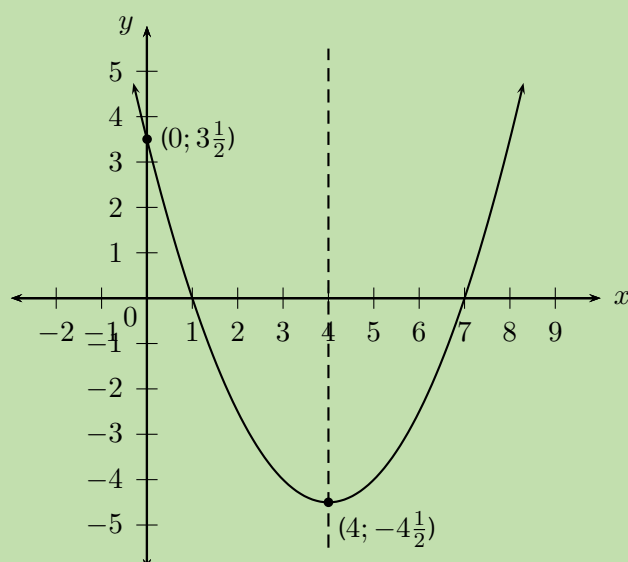
Step 4: Determine the x -intercepts

The x -intercepts are obtained by letting $y = 0$:

$$\begin{aligned} 0 &= \frac{1}{2}x^2 - 4x + \frac{7}{2} \\ &= x^2 - 8x + 7 \\ &= (x - 1)(x - 7) \end{aligned}$$

Therefore $x = 1$ or $x = 7$. This gives the points $(1; 0)$ and $(7; 0)$.

Step 5: Plot the points and sketch the graph



Step 6: State the domain and range

Domain: $\{x : x \in \mathbb{R}\}$

Range: $\{y : y \geq -4\frac{1}{2}, y \in \mathbb{R}\}$

► See video: [22QN](https://www.everythingmaths.co.za/22QN) at www.everythingmaths.co.za

Investigation: Shifting the equation of a parabola

Carl and Eric are doing their Mathematics homework and decide to check each others answers.

Homework question:

If the parabola $y = 3x^2 + 1$ is shifted 2 units to the right, determine the equation of the new parabola.

- Carl's answer:

A shift to the right means moving in the positive x direction, therefore x is replaced with $x + 2$ and the new equation is $y = 3(x + 2)^2 + 1$.

- Eric's answer:

We replace x with $x - 2$, therefore the new equation is $y = 3(x - 2)^2 + 1$.

Work together in pairs. Discuss the two different answers and decide which one is correct. Use calculations and sketches to help explain your reasoning.

Writing an equation of a shifted parabola

The parabola is shifted horizontally:

- If the parabola is shifted m units to the right, x is replaced by $(x - m)$.
- If the parabola is shifted m units to the left, x is replaced by $(x + m)$.

The parabola is shifted vertically:

- If the parabola is shifted n units down, y is replaced by $(y + n)$.
- If the parabola is shifted n units up, y is replaced by $(y - n)$.

Worked example 6: Shifting a parabola

QUESTION

Given $y = x^2 - 2x - 3$.

1. If the parabola is shifted 1 unit to the right, determine the new equation of the parabola.
2. If the parabola is shifted 3 units down, determine the new equation of the parabola.

SOLUTION

Step 1: Determine the new equation of the shifted parabola

1. The parabola is shifted 1 unit to the right, so x must be replaced by $(x - 1)$.

$$\begin{aligned}y &= x^2 - 2x - 3 \\&= (x - 1)^2 - 2(x - 1) - 3 \\&= x^2 - 2x + 1 - 2x + 2 - 3 \\&= x^2 - 4x\end{aligned}$$

Be careful not to make a common error: replacing x with $x + 1$ for a shift to the right.

2. The parabola is shifted 3 units down, so y must be replaced by $(y + 3)$.

$$\begin{aligned}y + 3 &= x^2 - 2x - 3 \\y &= x^2 - 2x - 3 - 3 \\&= x^2 - 2x - 6\end{aligned}$$

Exercise 5 – 6: Sketching parabolas

1. Sketch graphs of the following functions and determine:

- intercepts
- turning point
- axes of symmetry
- domain and range

a) $y = -x^2 + 4x + 5$

b) $y = 2(x + 1)^2$

c) $y = 3x^2 - 2(x + 2)$

d) $y = 3(x - 2)^2 + 1$

2. Draw the following graphs on the same system of axes:

$$f(x) = -x^2 + 7$$

$$g(x) = -(x - 2)^2 + 7$$

$$h(x) = (x - 2)^2 - 7$$

3. Draw a sketch of each of the following graphs:

a) $y = ax^2 + bx + c$ if $a > 0, b > 0, c < 0$.

b) $y = ax^2 + bx + c$ if $a < 0, b = 0, c > 0$.

c) $y = ax^2 + bx + c$ if $a < 0, b < 0, b^2 - 4ac < 0$.

d) $y = (x + p)^2 + q$ if $p < 0, q < 0$ and the x -intercepts have different signs.

e) $y = a(x + p)^2 + q$ if $a < 0, p < 0, q > 0$ and one root is zero.

f) $y = a(x + p)^2 + q$ if $a > 0, p = 0, b^2 - 4ac > 0$.

4. Determine the new equation (in the form $y = ax^2 + bx + c$) if:

a) $y = 2x^2 + 4x + 2$ is shifted 3 units to the left.

b) $y = -(x + 1)^2$ is shifted 1 unit up.

c) $y = 3(x - 1)^2 + 2(x - \frac{1}{2})$ is shifted 2 units to the right.

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1a. [22QP](#) 1b. [22QQ](#) 1c. [22QR](#) 1d. [22QS](#) 2. [22QT](#) 3a. [22QV](#)
3b. [22QW](#) 3c. [22QX](#) 3d. [22QY](#) 3e. [22QZ](#) 3f. [22R2](#) 4a. [22R3](#)
4b. [22R4](#) 4c. [22R5](#)



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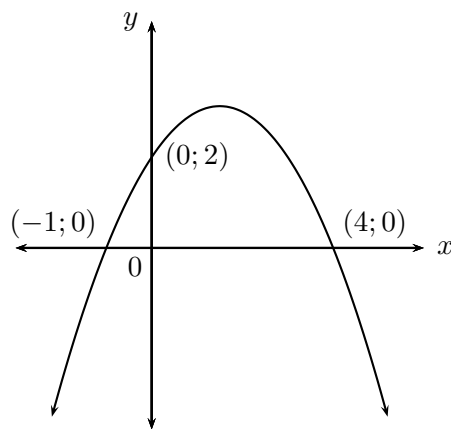


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Finding the equation of a parabola from the graph

If the intercepts are given, use $y = a(x - x_1)(x - x_2)$.

Example:



x -intercepts: $(-1; 0)$ and $(4; 0)$

$$y = a(x - x_1)(x - x_2)$$

$$= a(x + 1)(x - 4)$$

$$= ax^2 - 3ax - 4a$$

y -intercept: $(0; 2)$

$$-4a = 2$$

$$a = -\frac{1}{2}$$

Equation of the parabola:

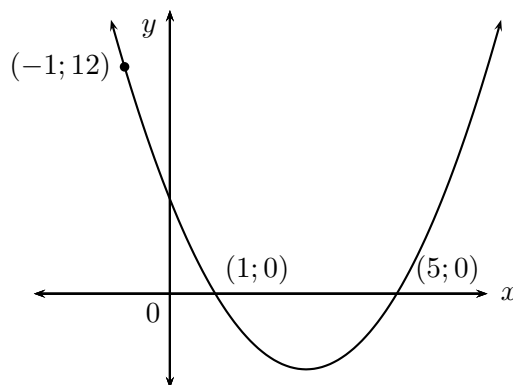
$$y = ax^2 - 3ax - 4a$$

$$= -\frac{1}{2}x^2 - 3\left(-\frac{1}{2}\right)x - 4\left(-\frac{1}{2}\right)$$

$$= -\frac{1}{2}x^2 + \frac{3}{2}x + 2$$

If the x -intercepts and another point are given, use $y = a(x - x_1)(x - x_2)$.

Example:



x -intercepts: $(1; 0)$ and $(5; 0)$

$$y = a(x - x_1)(x - x_2)$$

$$= a(x - 1)(x - 5)$$

$$= ax^2 - 6ax + 5a$$

Substitute the point: $(-1; 12)$

$$12 = a(-1)^2 - 6a(-1) + 5a$$

$$12 = a + 6a + 5a$$

$$12 = 12a$$

$$1 = a$$

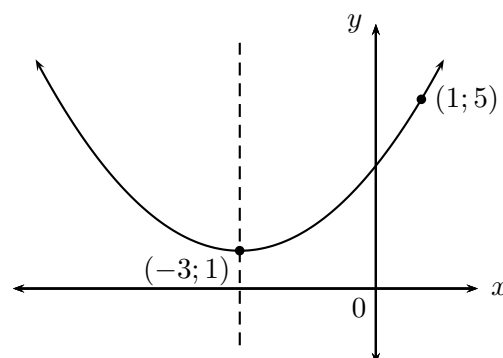
Equation of the parabola:

$$y = ax^2 - 6ax + 5a$$

$$= x^2 - 6x + 5$$

If the turning point and another point are given, use $y = a(x + p)^2 + q$.

Example:



Turning point: $(-3; 1)$

$$y = a(x + p)^2 + q$$

$$= a(x + 3)^2 + 1$$

$$= ax^2 + 6ax + 9a + 1$$

Substitute the point: $(1; 5)$

$$5 = a(1)^2 + 6a(1) + 9a + 1$$

$$4 = 16a$$

$$\frac{1}{4} = a$$

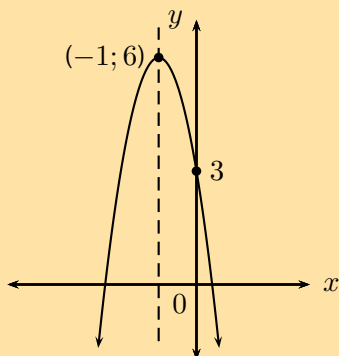
Equation of the parabola:

$$y = \frac{1}{4}(x + 3)^2 + 1$$

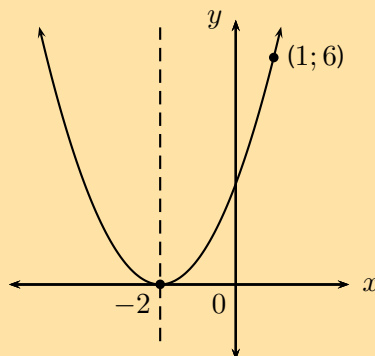
Exercise 5 – 7: Finding the equation

Determine the equations of the following graphs. Write your answers in the form $y = a(x + p)^2 + q$.

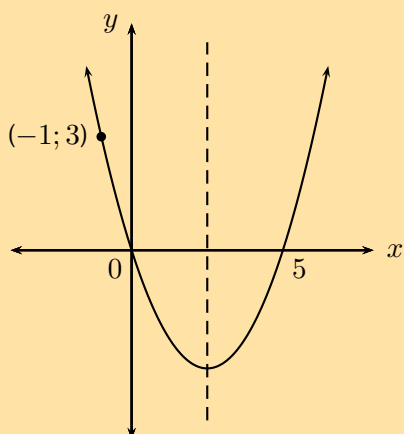
1.



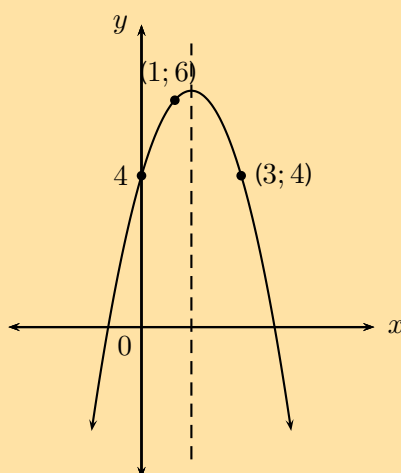
3.



2.



4.



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1. 22R6 2. 22R7 3. 22R8 4. 22R9

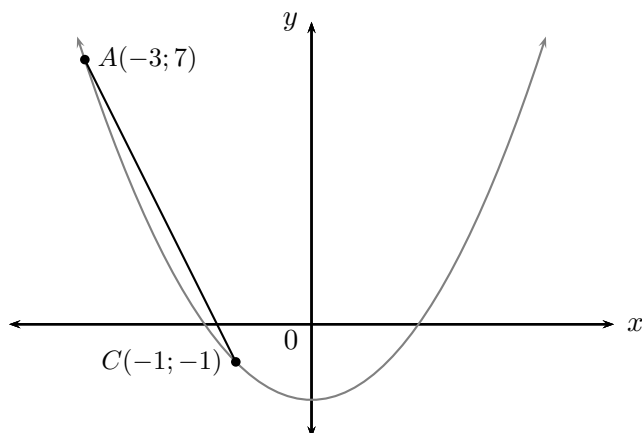


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We notice that the gradient of a curve changes at every point on the curve, therefore we need to work with the average gradient. The average gradient between any two points on a curve is the gradient of the straight line passing through the two points.



For the diagram above, the gradient of the line AC is

$$\begin{aligned}\text{Gradient} &= \frac{y_A - y_C}{x_A - x_C} \\ &= \frac{7 - (-1)}{-3 - (-1)} \\ &= \frac{8}{-2} \\ &= -4\end{aligned}$$

This is the average gradient of the curve between the points A and C .

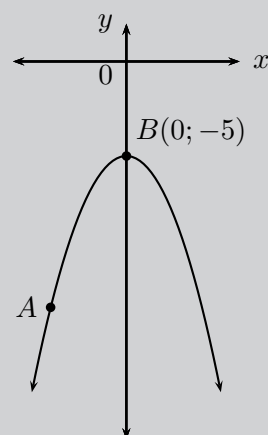
What happens to the gradient if we fix the position of one point and move the second point closer to the fixed point?

► See video: [22RB](https://www.everythingmaths.co.za) at www.everythingmaths.co.za

Investigation: Gradient at a single point on a curve

The curve shown here is defined by $y = -2x^2 - 5$. Point B is fixed at $(0; -5)$ and the position of point A varies.

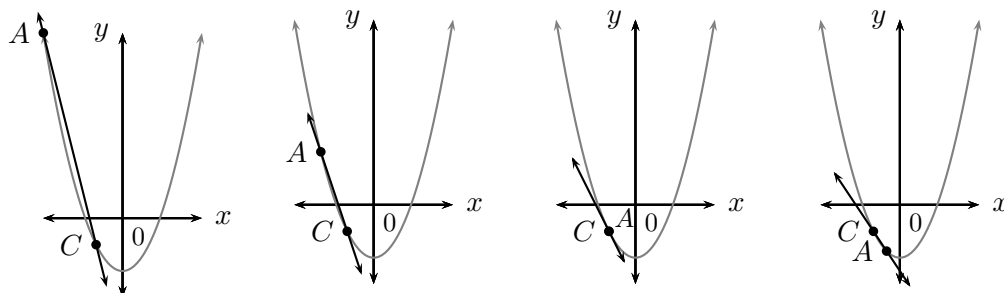
Complete the table below by calculating the y -coordinates of point A for the given x -coordinates and then calculating the average gradient between points A and B .



x_A	y_A	Average gradient
-2		
-1,5		
-1		
-0,5		
0		
0,5		
1		
1,5		
2		

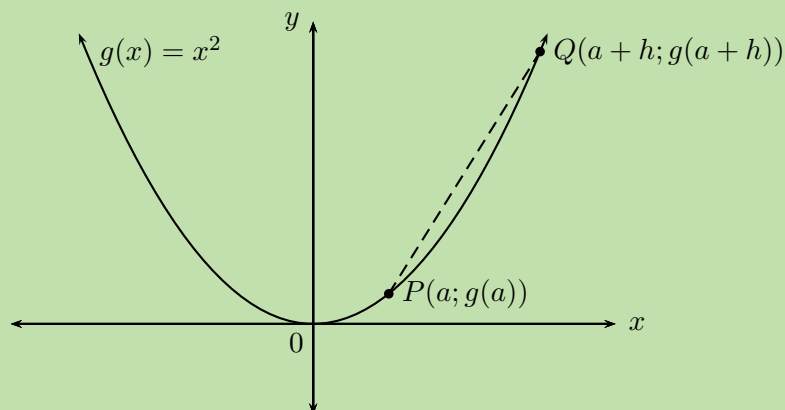
1. What happens to the average gradient as A moves towards B ?
2. What happens to the average gradient as A moves away from B ?
3. What is the average gradient when A overlaps with B ?

In the example above, the gradient of the straight line that passes through points A and C changes as A moves closer to C . At the point where A and C overlap, the straight line only passes through one point on the curve. This line is known as a tangent to the curve.



We therefore introduce the idea of the gradient at a single point on a curve. The gradient at a point on a curve is the gradient of the tangent to the curve at the given point.

QUESTION



1. Find the average gradient between two points $P(a; g(a))$ and $Q(a+h; g(a+h))$ on a curve $g(x) = x^2$.
2. Determine the average gradient between $P(2; g(2))$ and $Q(5; g(5))$.
3. Explain what happens to the average gradient if Q moves closer to P .

SOLUTION

Step 1: Assign labels to the x -values for the given points

$$x_1 = a$$

$$x_2 = a + h$$

Step 2: Determine the corresponding y -coordinates

Using the function $g(x) = x^2$, we can determine:

$$\begin{aligned} y_1 &= g(a) \\ &= a^2 \end{aligned}$$

$$\begin{aligned} y_2 &= g(a+h) \\ &= (a+h)^2 \\ &= a^2 + 2ah + h^2 \end{aligned}$$

Step 3: Calculate the average gradient

$$\begin{aligned}\frac{y_2 - y_1}{x_2 - x_1} &= \frac{(a^2 + 2ah + h^2) - (a^2)}{(a + h) - (a)} \\&= \frac{a^2 + 2ah + h^2 - a^2}{a + h - a} \\&= \frac{2ah + h^2}{h} \\&= \frac{h(2a + h)}{h} \\&= 2a + h\end{aligned}$$

The average gradient between $P(a; g(a))$ and $Q(a + h; g(a + h))$ on the curve $g(x) = x^2$ is $2a + h$.

Step 4: Calculate the average gradient between $P(2; g(2))$ and $Q(5; g(5))$

The x -coordinate of P is a and the x -coordinate of Q is $a + h$ therefore if we know that $a = 2$ and $a + h = 5$, then $h = 3$.

The average gradient is therefore $2a + h = 2(2) + (3) = 7$

Step 5: When Q moves closer to P

When point Q moves closer to point P , h gets smaller.

When the point Q overlaps with the point P , $h = 0$ and the gradient is given by $2a$.

We can write the equation for average gradient in another form. Given a curve $f(x)$ with two points P and Q with $P(a; f(a))$ and $Q(a + h; f(a + h))$. The average gradient between P and Q is:

$$\begin{aligned}\text{Average gradient} &= \frac{y_Q - y_P}{x_Q - x_P} \\&= \frac{f(a + h) - f(a)}{(a + h) - (a)} \\&= \frac{f(a + h) - f(a)}{h}\end{aligned}$$

This result is important for calculating the gradient at a point on a curve and will be explored in greater detail in Grade 12.

Worked example 8: Average gradient

QUESTION

Given $f(x) = -2x^2$.

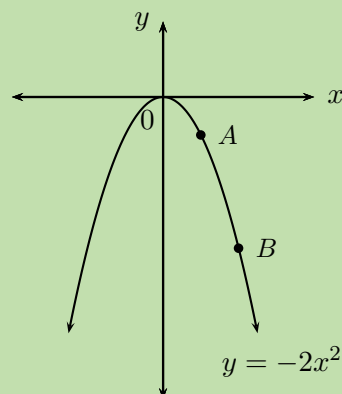
1. Draw a sketch of the function and determine the average gradient between the points A , where $x = 1$, and B , where $x = 3$.
2. Determine the gradient of the curve at point A .

SOLUTION

Step 1: Examine the form of the equation

From the equation we see that $a < 0$, therefore the graph is a “frown” and has a maximum turning point. We also see that when $x = 0$, $y = 0$, therefore the graph passes through the origin.

Step 2: Draw a rough sketch



Step 3: Calculate the average gradient between A and B

$$\begin{aligned}\text{Average gradient} &= \frac{f(3) - f(1)}{3 - 1} \\ &= \frac{-2(3)^2 - (-2(1)^2)}{2} \\ &= \frac{-18 + 2}{2} \\ &= \frac{-16}{2} \\ &= -8\end{aligned}$$

Step 4: Calculate the average gradient for $f(x)$

$$\begin{aligned}\text{Average gradient} &= \frac{f(a+h) - f(a)}{(a+h) - a} \\&= \frac{-2(a+h)^2 - (-2a^2)}{h} \\&= \frac{-2a^2 - 4ah - 2h^2 + 2a^2}{h} \\&= \frac{-4ah - 2h^2}{h} \\&= \frac{h(-4a - 2h)}{h} \\&= -4a - 2h\end{aligned}$$

At point A , $h = 0$ and $a = 1$. Therefore

$$\begin{aligned}\text{Average gradient} &= -4a - 2h \\&= -4(1) - 2(0) \\&= -4\end{aligned}$$

Exercise 5 – 8:

1.
 - a) Determine the average gradient of the curve $f(x) = x(x+3)$ between $x = 5$ and $x = 3$.
 - b) Hence, state what you can deduce about the function f between $x = 5$ and $x = 3$.
2. $A(1; 3)$ is a point on $f(x) = 3x^2$.
 - a) Draw a sketch of $f(x)$ and label point A .
 - b) Determine the gradient of the curve at point A .
 - c) Determine the equation of the tangent line at A .
3. Given: $g(x) = -x^2 + 1$.
 - a) Draw a sketch of $g(x)$.
 - b) Determine the average gradient of the curve between $x = -2$ and $x = 1$.
 - c) Determine the gradient of g at $x = 2$.
 - d) Determine the gradient of g at $x = 0$.

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1. 22RC 2. 22RD 3. 22RF



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Functions of the form $y = \frac{a}{x} + q$

Functions of the general form $y = \frac{a}{x} + q$ are called hyperbolic functions, where a and q are constants.

The effects of a and q on $f(x) = \frac{a}{x} + q$:

- **The effect of q on vertical shift**

- For $q > 0$, $f(x)$ is shifted vertically upwards by q units.
- For $q < 0$, $f(x)$ is shifted vertically downwards by q units.
- The horizontal asymptote is the line $y = q$.
- The vertical asymptote is the y -axis, the line $x = 0$.

- **The effect of a on shape and quadrants**

- For $a > 0$, $f(x)$ lies in the first and third quadrants.
- For $a > 1$, $f(x)$ will be further away from both axes than $y = \frac{1}{x}$.
- For $0 < a < 1$, as a tends to 0, $f(x)$ moves closer to the axes than $y = \frac{1}{x}$.
- For $a < 0$, $f(x)$ lies in the second and fourth quadrants.
- For $a < -1$, $f(x)$ will be further away from both axes than $y = -\frac{1}{x}$.
- For $-1 < a < 0$, as a tends to 0, $f(x)$ moves closer to the axes than $y = -\frac{1}{x}$.

	$a < 0$	$a > 0$
$q > 0$		
$q = 0$		
$q < 0$		

Exercise 5 – 9: Revision

1. Consider the following hyperbolic functions:

- $y_1 = \frac{1}{x}$
- $y_2 = -\frac{4}{x}$
- $y_3 = \frac{4}{x} - 2$
- $y_4 = -\frac{4}{x} + 1$

Complete the table to summarise the properties of the hyperbolic function:

	y_1	y_2	y_3	y_4
value of q	$q = 0$			
effect of q	no vertical shift			
value of a	$a = 1$			
effect of a	lies in I and III quad			
asymptotes	y -axis, $x = 0$ x -axis, $y = 0$			
axes of symmetry	$y = x$ $y = -x$			
domain	$\{x : x \in \mathbb{R}, x \neq 0\}$			
range	$\{y : y \in \mathbb{R}, y \neq 0\}$			

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1. [22RG](#)



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Functions of the form $y = \frac{a}{x+p} + q$

EMBGR

We now consider hyperbolic functions of the form $y = \frac{a}{x+p} + q$ and the effects of parameter p .

Investigation: The effects of a , p and q on a hyperbolic graph

1. On the same system of axes, plot the following graphs:

- a) $y_1 = \frac{1}{x}$
- b) $y_2 = \frac{1}{x-2}$
- c) $y_3 = \frac{1}{x-1}$
- d) $y_4 = \frac{1}{x+1}$

Use your sketches of the functions above to complete the following table:

	y_1	y_2	y_3	y_4
intercept(s)				
asymptotes				
axes of symmetry				
domain				
range				
effect of p				

2. Complete the following sentences for functions of the form $y = \frac{a}{x+p} + q$:

- A change in p causes a shift.
- If the value of p increases, the graph and the vertical asymptote
- If the value of q changes, then the asymptote of the hyperbola will shift.
- If the value of p decreases, the graph and the vertical asymptote

The effect of the parameters on $y = \frac{a}{x+p} + q$

The effect of p is a horizontal shift because all points are moved the same distance in the same direction (the entire graph slides to the left or to the right).

- For $p > 0$, the graph is shifted to the left by p units.
- For $p < 0$, the graph is shifted to the right by p units.

The value of p also affects the vertical asymptote, the line $x = -p$.

The effect of q is a vertical shift. The value of q also affects the horizontal asymptotes, the line $y = q$.

The value of a affects the shape of the graph and its position on the Cartesian plane.

	$p > 0$		$p < 0$	
	$a < 0$	$a > 0$	$a < 0$	$a > 0$
$q > 0$				
$q < 0$				

Discovering the characteristics

For functions of the general form: $f(x) = y = \frac{a}{x+p} + q$:

Domain and range

The domain is $\{x : x \in \mathbb{R}, x \neq -p\}$. If $x = -p$, the denominator is equal to zero and the function is undefined.

We see that

$$y = \frac{a}{x+p} + q$$

can be re-written as:

$$y - q = \frac{a}{x+p}$$

If $x \neq -p$ then:

$$\begin{aligned}(y - q)(x + p) &= a \\ x + p &= \frac{a}{y - q}\end{aligned}$$

The range is therefore $\{y : y \in \mathbb{R}, y \neq q\}$.

These restrictions on the domain and range determine the vertical asymptote $x = -p$ and the horizontal asymptote $y = q$.

Worked example 9: Domain and range

QUESTION

Determine the domain and range for $g(x) = \frac{2}{x+1} + 2$.

SOLUTION

Step 1: Determine the domain

The domain is $\{x : x \in \mathbb{R}, x \neq -1\}$ since $g(x)$ is undefined for $x = -1$.

Step 2: Determine the range

Let $g(x) = y$:

$$\begin{aligned}y &= \frac{2}{x+1} + 2 \\ y - 2 &= \frac{2}{x+1} \\ (y - 2)(x + 1) &= 2 \\ x + 1 &= \frac{2}{y - 2}\end{aligned}$$

Therefore the range is $\{g(x) : g(x) \in \mathbb{R}, g(x) \neq 2\}$.

Exercise 5 – 10: Domain and range

Determine the domain and range for each of the following functions:

1. $y = \frac{1}{x} + 1$

4. $x = \frac{2}{3-y} + 5$

2. $g(x) = \frac{8}{x-8} + 4$

3. $y = -\frac{4}{x+1} - 3$

5. $(y-2)(x+2) = 3$

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1. [22RJ](#) 2. [22RK](#) 3. [22RM](#) 4. [22RN](#) 5. [22RP](#)



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Intercepts

The y -intercept:

To calculate the y -intercept we let $x = 0$. For example, the y -intercept of $g(x) = \frac{2}{x+1} + 2$ is determined by setting $x = 0$:

$$\begin{aligned} g(x) &= \frac{2}{x+1} + 2 \\ g(0) &= \frac{2}{0+1} + 2 \\ &= 2 + 2 \\ &= 4 \end{aligned}$$

This gives the point $(0; 4)$.

The x -intercept:

To calculate the x -intercept we let $y = 0$. For example, the x -intercept of $g(x) = \frac{2}{x+1} + 2$ is determined by setting $y = 0$:

$$\begin{aligned} g(x) &= \frac{2}{x+1} + 2 \\ 0 &= \frac{2}{x+1} + 2 \\ -2 &= \frac{2}{x+1} \\ -2(x+1) &= 2 \\ -2x - 2 &= 2 \\ -2x &= 4 \\ x &= -2 \end{aligned}$$

This gives the point $(-2; 0)$.

Exercise 5 – 11: Intercepts

Determine the x - and y -intercepts for each of the following functions:

1. $f(x) = \frac{1}{x+4} - 2$

4. $h(x) = \frac{3}{6-x} + 1$

2. $g(x) = -\frac{5}{x} + 2$

3. $j(x) = \frac{2}{x-1} + 3$

5. $k(x) = \frac{5}{x+2} - \frac{1}{2}$

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1. [22RQ](#) 2. [22RR](#) 3. [22RS](#) 4. [22RT](#) 5. [22RV](#)



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Asymptotes

There are two asymptotes for functions of the form $y = \frac{a}{x+p} + q$. The asymptotes indicate the values of x for which the function does not exist. In other words, the values that are excluded from the domain and the range. The horizontal asymptote is the line $y = q$ and the vertical asymptote is the line $x = -p$.

Exercise 5 – 12: Asymptotes

Determine the asymptotes for each of the following functions:

1. $y = \frac{1}{x+4} - 2$

4. $y = \frac{1}{x} - 8$

2. $y = -\frac{5}{x}$

3. $y = \frac{3}{2-x} + 1$

5. $y = -\frac{2}{x-2}$

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1. [22RW](#) 2. [22RX](#) 3. [22RY](#) 4. [22RZ](#) 5. [22S2](#)



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Axes of symmetry

There are two lines about which a hyperbola is symmetrical.

For the standard hyperbola $y = \frac{1}{x}$, we see that if we replace $x \Rightarrow y$ and $y \Rightarrow x$, we get $y = \frac{1}{x}$. Similarly, if we replace $x \Rightarrow -y$ and $y \Rightarrow -x$, the function remains the same. Therefore the function is symmetrical about the lines $y = x$ and $y = -x$.

For the shifted hyperbola $y = \frac{a}{x+p} + q$, the axes of symmetry intersect at the point $(-p; q)$.

To determine the axes of symmetry we define the two straight lines $y_1 = m_1x + c_1$ and $y_2 = m_2x + c_2$. For the standard and shifted hyperbolic function, the gradient of one of the lines of symmetry is 1 and the gradient of the other line of symmetry is -1 . The axes of symmetry are perpendicular to each other and the product of their gradients equals -1 . Therefore we let $y_1 = x + c_1$ and $y_2 = -x + c_2$. We then substitute $(-p; q)$, the point of intersection of the axes of symmetry, into both equations to determine the values of c_1 and c_2 .

Worked example 10: Axes of symmetry

QUESTION

Determine the axes of symmetry for $y = \frac{2}{x+1} - 2$.

SOLUTION

Step 1: Determine the point of intersection $(-p; q)$

From the equation we see that $p = 1$ and $q = -2$. So the axes of symmetry will intersect at $(-1; -2)$.

Step 2: Define two straight line equations

$$\begin{aligned}y_1 &= x + c_1 \\y_2 &= -x + c_2\end{aligned}$$

Step 3: Solve for c_1 and c_2

Use $(-1; -2)$ to solve for c_1 :

$$\begin{aligned}y_1 &= x + c_1 \\-2 &= -1 + c_1 \\-1 &= c_1\end{aligned}$$

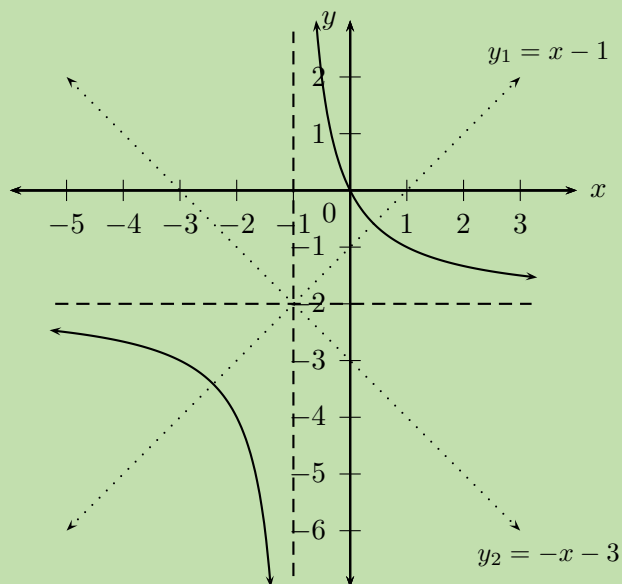
Use $(-1; -2)$ to solve for c_2 :

$$\begin{aligned}y_2 &= -x + c_2 \\-2 &= -(-1) + c_2 \\-3 &= c_2\end{aligned}$$

Step 4: Write the final answer

The axes of symmetry for $y = \frac{2}{x+1} - 2$ are the lines

$$\begin{aligned}y_1 &= x - 1 \\y_2 &= -x - 3\end{aligned}$$



Exercise 5 – 13: Axes of symmetry

1. Complete the following for $f(x)$ and $g(x)$:

- Sketch the graph.
- Determine $(-p; q)$.
- Find the axes of symmetry.

Compare $f(x)$ and $g(x)$ and also their axes of symmetry. What do you notice?

a) $f(x) = \frac{2}{x}$
 $g(x) = \frac{2}{x} + 1$

b) $f(x) = -\frac{3}{x}$
 $g(x) = -\frac{3}{x+1}$

c) $f(x) = \frac{5}{x}$
 $g(x) = \frac{5}{x-1} - 1$

2. A hyperbola of the form $k(x) = \frac{a}{x+p} + q$ passes through the point $(4; 3)$. If the axes of symmetry intersect at $(-1; 2)$, determine the equation of $k(x)$.

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1a. 22S3 1b. 22S4 1c. 22S5 2. 22S6



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Sketching graphs of the form $f(x) = \frac{a}{x+p} + q$

In order to sketch graphs of functions of the form, $f(x) = \frac{a}{x+p} + q$, we need to calculate five characteristics:

- quadrants
- asymptotes
- y -intercept
- x -intercept
- domain and range

Worked example 11: Sketching a hyperbola

QUESTION

Sketch the graph of $y = \frac{2}{x+1} + 2$. Determine the intercepts, asymptotes and axes of symmetry. State the domain and range of the function.

SOLUTION

Step 1: Examine the equation of the form $y = \frac{a}{x+p} + q$

We notice that $a > 0$, therefore the graph will lie in the first and third quadrants.

Step 2: Determine the asymptotes

From the equation we know that $p = 1$ and $q = 2$.

Therefore the horizontal asymptote is the line $y = 2$ and the vertical asymptote is the line $x = -1$.

Step 3: Determine the y -intercept

The y -intercept is obtained by letting $x = 0$:

$$\begin{aligned}y &= \frac{2}{0+1} + 2 \\ &= 4\end{aligned}$$

This gives the point $(0; 4)$.

Step 4: Determine the x -intercept

The x -intercept is obtained by letting $y = 0$:

$$\begin{aligned}0 &= \frac{2}{x+1} + 2 \\ -2 &= \frac{2}{x+1}\end{aligned}$$

$$\begin{aligned}
 -2(x+1) &= 2 \\
 -2x - 2 &= 2 \\
 -2x &= 4 \\
 x &= -2
 \end{aligned}$$

This gives the point $(-2; 0)$.

Step 5: Determine the axes of symmetry

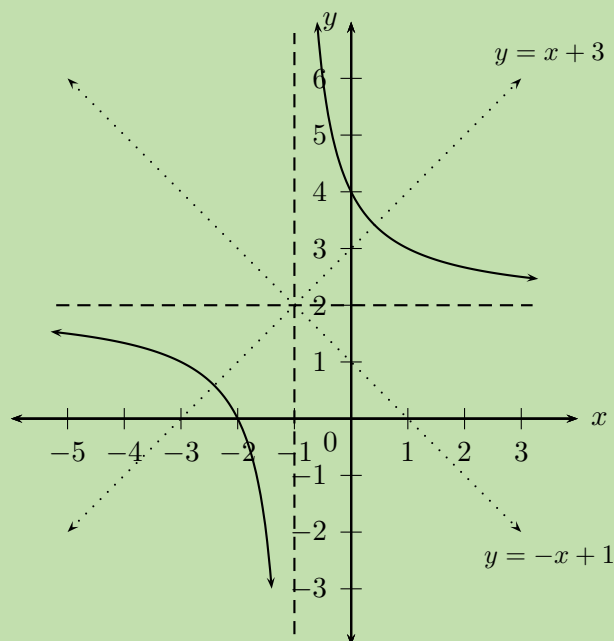
Using $(-1; 2)$ to solve for c_1 :

$$\begin{aligned}
 y_1 &= x + c_1 \\
 2 &= -1 + c_1 \\
 3 &= c_1
 \end{aligned}$$

$$\begin{aligned}
 y_2 &= -x + c_2 \\
 2 &= -(-1) + c_2 \\
 1 &= c_2
 \end{aligned}$$

Therefore the axes of symmetry are $y = x + 3$ and $y = -x + 1$.

Step 6: Plot the points and sketch the graph



Step 7: State the domain and range

Domain: $\{x : x \in \mathbb{R}, x \neq -1\}$

Range: $\{y : y \in \mathbb{R}, y \neq 2\}$

Worked example 12: Sketching a hyperbola

QUESTION

Use horizontal and vertical shifts to sketch the graph of $f(x) = \frac{1}{x-2} + 3$.

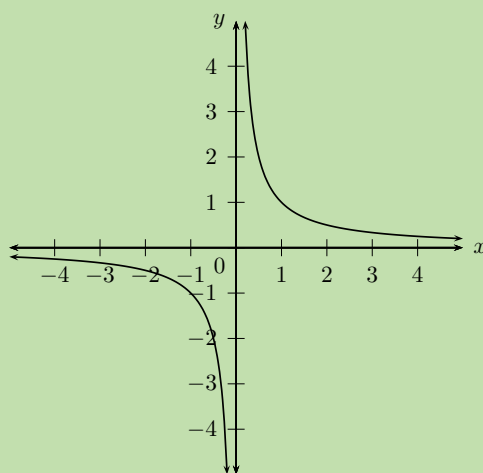
SOLUTION

Step 1: Examine the equation of the form $y = \frac{a}{x+p} + q$

We notice that $a > 0$, therefore the graph will lie in the first and third quadrants.

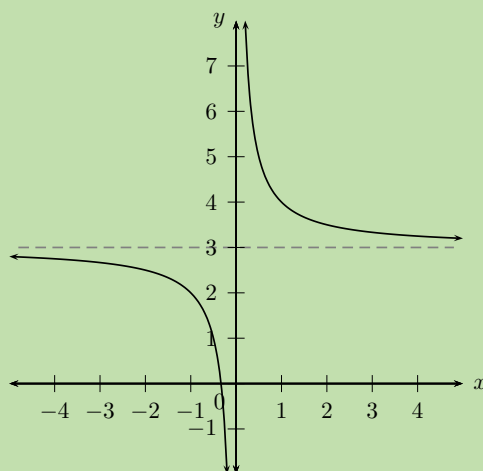
Step 2: Sketch the standard hyperbola $y = \frac{1}{x}$

Start with a sketch of the standard hyperbola $g(x) = \frac{1}{x}$. The vertical asymptote is $x = 0$ and the horizontal asymptote is $y = 0$.



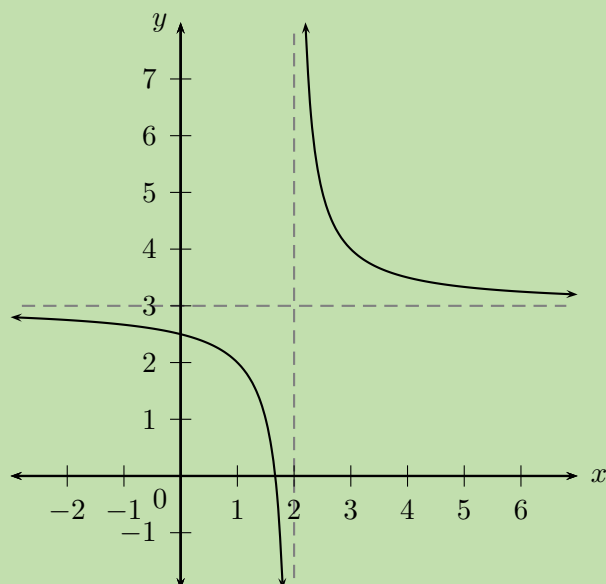
Step 3: Determine the vertical shift

From the equation we see that $q = 3$, which means $g(x)$ must be shifted 3 units up. The horizontal asymptote is also shifted 3 units up to $y = 3$.



Step 4: Determine the horizontal shift

From the equation we see that $p = -2$, which means $g(x)$ must be shifted 2 units to the right. The vertical asymptote is also shifted 2 units to the right.



Step 5: Determine the y -intercept

The y -intercept is obtained by letting $x = 0$:

$$\begin{aligned} y &= \frac{1}{0-2} + 3 \\ &= 2\frac{1}{2} \end{aligned}$$

This gives the point $(0; 2\frac{1}{2})$.

Step 6: Determine the x -intercept

The x -intercept is obtained by letting $y = 0$:

$$\begin{aligned} 0 &= \frac{1}{x-2} + 3 \\ -3 &= \frac{1}{x-2} \\ -3(x-2) &= 1 \\ -3x + 6 &= 1 \\ -3x &= -5 \\ x &= \frac{5}{3} \end{aligned}$$

This gives the point $(\frac{5}{3}; 0)$.

Step 7: Determine the domain and range

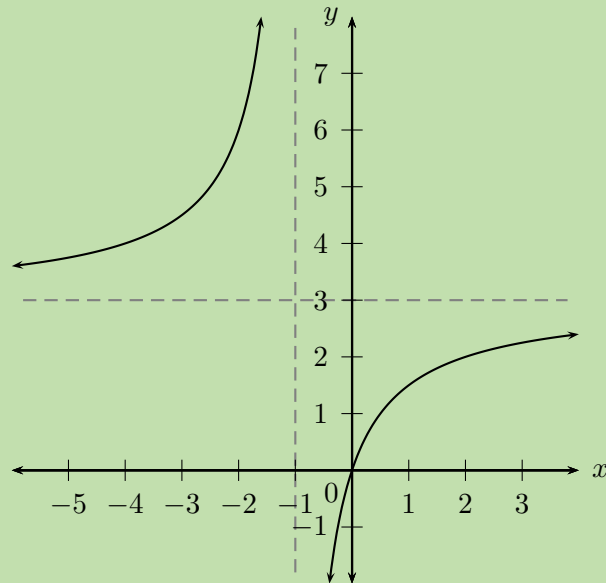
Domain: $\{x : x \in \mathbb{R}, x \neq 2\}$

Range: $\{y : y \in \mathbb{R}, y \neq 3\}$

Worked example 13: Finding the equation of a hyperbola from the graph

QUESTION

Use the graph below to determine the values of a , p and q for $y = \frac{a}{x+p} + q$.



SOLUTION

Step 1: Examine the graph and deduce the sign of a

We notice that the graph lies in the second and fourth quadrants, therefore $a < 0$.

Step 2: Determine the asymptotes

From the graph we see that the vertical asymptote is $x = -1$, therefore $p = 1$. The horizontal asymptote is $y = 3$, and therefore $q = 3$.

$$y = \frac{a}{x+1} + 3$$

Step 3: Determine the value of a

To determine the value of a we substitute a point on the graph, namely $(0; 0)$:

$$y = \frac{a}{x+1} + 3$$

$$0 = \frac{a}{0+1} + 3$$

$$\therefore -3 = a$$

Step 4: Write the final answer

$$y = -\frac{3}{x+1} + 3$$

Exercise 5 – 14: Sketching graphs

1. Draw the graphs of the following functions and indicate:

- asymptotes
- intercepts, where applicable
- axes of symmetry
- domain and range

a) $y = \frac{1}{x} + 2$

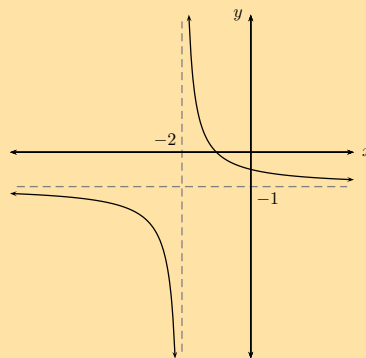
d) $y = -\frac{5}{x-2\frac{1}{2}} - 2$

b) $y = \frac{1}{x+4} - 2$

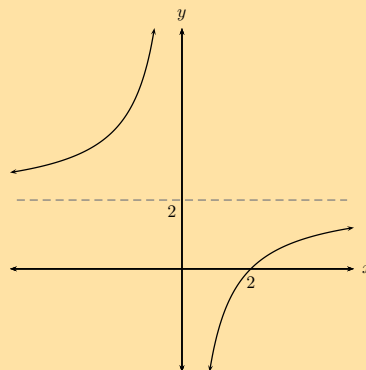
e) $y = \frac{8}{x-8} + 4$

c) $y = -\frac{1}{x+1} + 3$

2. Given the graph of the hyperbola of the form $y = \frac{1}{x+p} + q$, determine the values of p and q .



3. Given a sketch of the function of the form $y = \frac{a}{x+p} + q$, determine the values of a , p and q .



4. a) Draw the graph of $f(x) = -\frac{3}{x}$, $x > 0$.
 b) Determine the average gradient of the graph between $x = 1$ and $x = 3$.
 c) Is the gradient at $(\frac{1}{2}; -6)$ less than or greater than the average gradient between $x = 1$ and $x = 3$? Illustrate this on your graph.

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- 1a. 22S7 1b. 22S8 1c. 22S9 1d. 22SB 1e. 22SC 2. 22SD
 3. 22SF 4. 22SG



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Functions of the form $y = ab^x + q$

Functions of the general form $y = ab^x + q$, for $b > 0$, are called exponential functions, where a , b and q are constants.

The effects of a , b and q on $f(x) = ab^x + q$:

- **The effect of q on vertical shift**

- For $q > 0$, $f(x)$ is shifted vertically upwards by q units.
- For $q < 0$, $f(x)$ is shifted vertically downwards by q units.
- The horizontal asymptote is the line $y = q$.

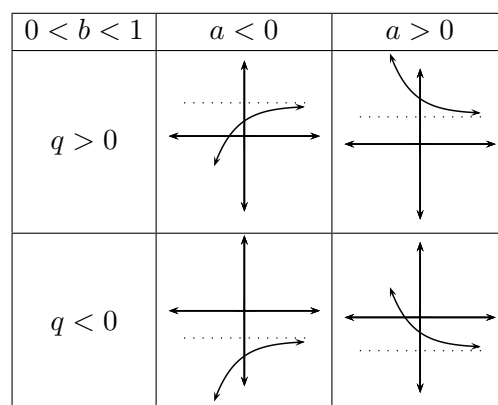
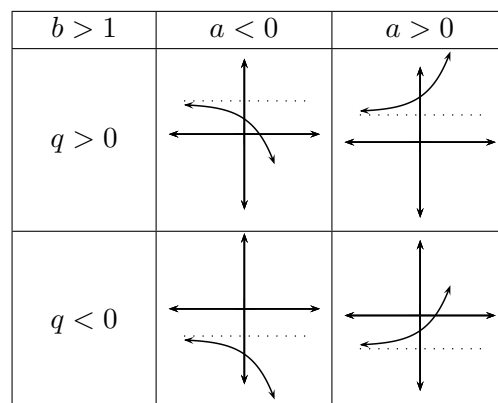
- **The effect of a on shape**

- For $a > 0$, $f(x)$ is increasing.
- For $a < 0$, $f(x)$ is decreasing. The graph is reflected about the horizontal asymptote.

- **The effect of b on direction**

Assuming $a > 0$:

- If $b > 1$, $f(x)$ is an increasing function.
- If $0 < b < 1$, $f(x)$ is a decreasing function.
- If $b \leq 0$, $f(x)$ is not defined.

**Exercise 5 – 15: Revision**

1. On separate axes, accurately draw each of the following functions:

- Use tables of values if necessary.
- Use graph paper if available.

- $y_1 = 3^x$
- $y_2 = -2 \times 3^x$
- $y_3 = 2 \times 3^x + 1$
- $y_4 = 3^x - 2$

2. Use your sketches of the functions given above to complete the following table (the first column has been completed as an example):

	y_1	y_2	y_3	y_4
value of q	$q = 0$			
effect of q	no vertical shift			
value of a	$a = 1$			
effect of a	increasing			
asymptote	x -axis, $y = 0$			
domain	$\{x : x \in \mathbb{R}\}$			
range	$\{y : y \in \mathbb{R}, y > 0\}$			

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1a. [22SH](#) 1b. [22SJ](#) 1c. [22SK](#) 1d. [22SM](#) 2. [22SN](#)



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Functions of the form $y = ab^{(x+p)} + q$

EMBGV

We now consider exponential functions of the form $y = ab^{(x+p)} + q$ and the effects of parameter p .

▶ See video: [22SP](#) at www.everythingmaths.co.za

Investigation: The effects of a , p and q on an exponential graph

1. On the same system of axes, plot the following graphs:

- a) $y_1 = 2^x$
- b) $y_2 = 2^{(x-2)}$
- c) $y_3 = 2^{(x-1)}$
- d) $y_4 = 2^{(x+1)}$
- e) $y_5 = 2^{(x+2)}$

Use your sketches of the functions above to complete the following table:

	y_1	y_2	y_3	y_4	y_5
intercept(s)					
asymptote					
domain					
range					
effect of p					

2. On the same system of axes, plot the following graphs:

- a) $y_1 = 2^{(x-1)} + 2$

- b) $y_2 = 3 \times 2^{(x-1)} + 2$
 c) $y_3 = \frac{1}{2} \times 2^{(x-1)} + 2$
 d) $y_4 = 0 \times 2^{(x-1)} + 2$
 e) $y_5 = -3 \times 2^{(x-1)} + 2$

Use your sketches of the functions above to complete the following table:

	y_1	y_2	y_3	y_4	y_5
intercept(s)					
asymptotes					
domain					
range					
effect of a					

The effect of the parameters on $y = ab^{x+p} + q$

The effect of p is a horizontal shift because all points are moved the same distance in the same direction (the entire graph slides to the left or to the right).

- For $p > 0$, the graph is shifted to the left by p units.
- For $p < 0$, the graph is shifted to the right by p units.

The effect of q is a vertical shift. The value of q also affects the horizontal asymptotes, the line $y = q$.

The value of a affects the shape of the graph and its position relative to the horizontal asymptote.

- For $a > 0$, the graph lies above the horizontal asymptote, $y = q$.
- For $a < 0$, the graph lies below the horizontal asymptote, $y = q$.

	$p > 0$		$p < 0$	
	$a < 0$	$a > 0$	$a < 0$	$a > 0$
$q > 0$				
$q < 0$				

Discovering the characteristics

For functions of the general form: $f(x) = y = ab^{(x+p)} + q$:

Domain and range

The domain is $\{x : x \in \mathbb{R}\}$ because there is no value of x for which $f(x)$ is undefined.

The range of $f(x)$ depends on whether the value for a is positive or negative.

If $a > 0$ we have:

$$\begin{aligned}b^{(x+p)} &> 0 \\ab^{(x+p)} &> 0 \\ab^{(x+p)} + q &> q \\f(x) &> q\end{aligned}$$

The range is therefore $\{y : y > q, y \in \mathbb{R}\}$.

Similarly, if $a < 0$, the range is $\{y : y < q, y \in \mathbb{R}\}$.

Worked example 14: Domain and range

QUESTION

State the domain and range for $g(x) = 5 \times 3^{(x+1)} - 1$.

SOLUTION

Step 1: Determine the domain

The domain is $\{x : x \in \mathbb{R}\}$ because there is no value of x for which $g(x)$ is undefined.

Step 2: Determine the range

The range of $g(x)$ can be calculated from:

$$\begin{aligned}3^{(x+1)} &> 0 \\5 \times 3^{(x+1)} &> 0 \\5 \times 3^{(x+1)} - 1 &> -1 \\\therefore g(x) &> -1\end{aligned}$$

Therefore the range is $\{g(x) : g(x) > -1\}$ or in interval notation $(-1; \infty)$.

Exercise 5 – 16: Domain and range

Give the domain and range for each of the following functions:

1. $y = \left(\frac{3}{2}\right)^{(x+3)}$

4. $y = n + 3^{(x-m)}$

2. $f(x) = -5^{(x-2)} + 1$

3. $y + 3 = 2^{(x+1)}$

5. $\frac{y}{2} = 3^{(x-1)} - 1$

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1. 22SQ 2. 22SR 3. 22SS 4. 22ST 5. 22SV



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Intercepts

The y -intercept:

To calculate the y -intercept we let $x = 0$. For example, the y -intercept of $g(x) = 3 \times 2^{(x+1)} + 2$ is determined by setting $x = 0$:

$$\begin{aligned} g(0) &= 3 \times 2^{(0+1)} + 2 \\ &= 3 \times 2 + 2 \\ &= 8 \end{aligned}$$

This gives the point $(0; 8)$.

The x -intercept:

To calculate the x -intercept we let $y = 0$. For example, the x -intercept of $g(x) = 3 \times 2^{(x+1)} + 2$ is determined by setting $y = 0$:

$$\begin{aligned} 0 &= 3 \times 2^{(x+1)} + 2 \\ -2 &= 3 \times 2^{(x+1)} \\ -\frac{2}{3} &= 2^{(x+1)} \end{aligned}$$

which has no real solutions. Therefore, the graph of $g(x)$ lies above the x -axis and does not have any x -intercepts.

Exercise 5 – 17: Intercepts

Determine the x - and y -intercepts for each of the following functions:

1. $f(x) = 2^{(x+1)} - 8$

2. $y = 2 \times 3^{(x-1)} - 18$

3. $y + 5^{(x+2)} = 5$

4. $y = \frac{1}{2} \left(\frac{3}{2}\right)^{(x+3)} - 0,75$

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1. [22SW](#) 2. [22SX](#) 3. [22SY](#) 4. [22SZ](#)



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Asymptote

Exponential functions of the form $y = ab^{(x+p)} + q$ have a horizontal asymptote, the line $y = q$.

Worked example 15: Asymptote

QUESTION

Determine the asymptote for $y = 5 \times 3^{(x+1)} - 1$.

SOLUTION

Step 1: Determine the asymptote

The asymptote of $g(x)$ can be calculated as:

$$\begin{aligned} 3^{(x+1)} &\neq 0 \\ 5 \times 3^{(x+1)} &\neq 0 \\ 5 \times 3^{(x+1)} - 1 &\neq -1 \\ \therefore y &\neq -1 \end{aligned}$$

Therefore the asymptote is the line $y = -1$.

Exercise 5 – 18: Asymptote

Give the asymptote for each of the following functions:

1. $y = -5^{(x+1)}$

4. $y = 7^{(x+1)} - 2$

2. $y = 3^{(x-2)} + 1$

3. $\left(\frac{3y}{2}\right) = 5^{(x+3)} - 1$

5. $\frac{y}{2} + 1 = 3^{(x+2)}$

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. 22T2 2. 22T3 3. 22T4 4. 22T5 5. 22T6



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Sketching graphs of the form $f(x) = ab^{(x+p)} + q$

In order to sketch graphs of functions of the form, $f(x) = ab^{(x+p)} + q$, we need to determine five characteristics:

- shape
- y -intercept
- x -intercept
- asymptote
- domain and range

Worked example 16: Sketching an exponential graph

QUESTION

Sketch the graph of $2y = 10 \times 2^{(x+1)} - 5$.

Mark the intercept(s) and asymptote. State the domain and range of the function.

SOLUTION

Step 1: Examine the equation of the form $y = ab^{(x+p)} + q$

We notice that $a > 0$ and $b > 1$, therefore the function is increasing.

Step 2: Determine the y -intercept

The y -intercept is obtained by letting $x = 0$:

$$\begin{aligned}2y &= 10 \times 2^{(0+1)} - 5 \\&= 10 \times 2 - 5 \\&= 15 \\ \therefore y &= 7\frac{1}{2}\end{aligned}$$

This gives the point $(0; 7\frac{1}{2})$.

Step 3: Determine the x -intercept

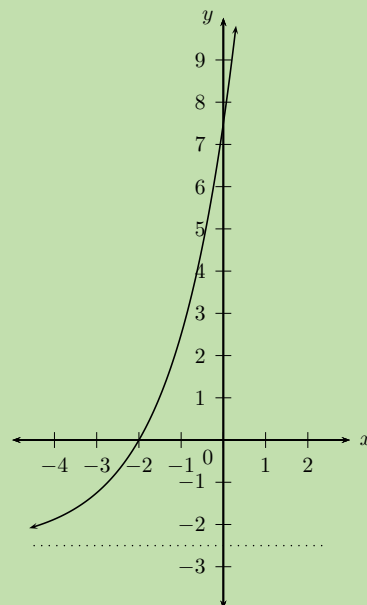
The x -intercept is obtained by letting $y = 0$:

$$\begin{aligned}0 &= 10 \times 2^{(x+1)} - 5 \\5 &= 10 \times 2^{(x+1)} \\ \frac{1}{2} &= 2^{(x+1)} \\ 2^{-1} &= 2^{(x+1)} \\ \therefore -1 &= x + 1 \quad (\text{same base}) \\ -2 &= x\end{aligned}$$

This gives the point $(-2; 0)$.

Step 4: Determine the asymptote

The horizontal asymptote is the line $y = -\frac{5}{2}$.

Step 5: Plot the points and sketch the graph**Step 6: State the domain and range**

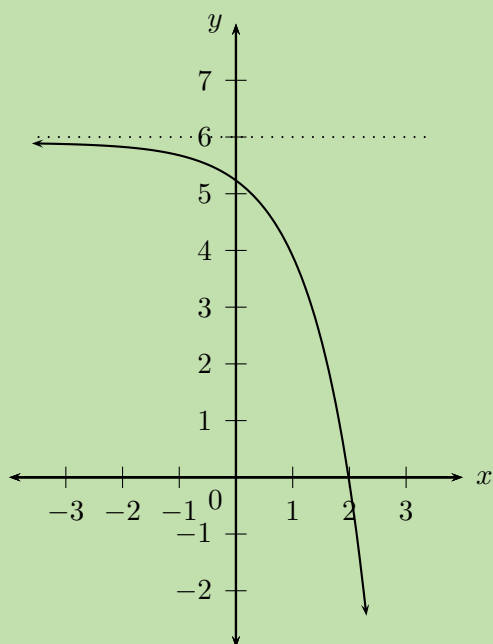
Domain: $\{x : x \in \mathbb{R}\}$

Range: $\{y : y > -\frac{5}{2}, y \in \mathbb{R}\}$

Worked example 17: Finding the equation of an exponential function from a graph

QUESTION

Use the given graph of $y = -2 \times 3^{(x+p)} + q$ to determine the values of p and q .



SOLUTION

Step 1: Examine the equation of the form $y = ab^{(x+p)} + q$

From the graph we see that the function is decreasing. We also note that $a = -2$ and $b = 3$. We need to solve for p and q .

Step 2: Use the asymptote to determine q

The horizontal asymptote $y = 6$ is given, therefore we know that $q = 6$.

$$y = -2 \times 3^{(x+p)} + 6$$

Step 3: Use the x -intercept to determine p

Substitute $(2; 0)$ into the equation and solve for p :

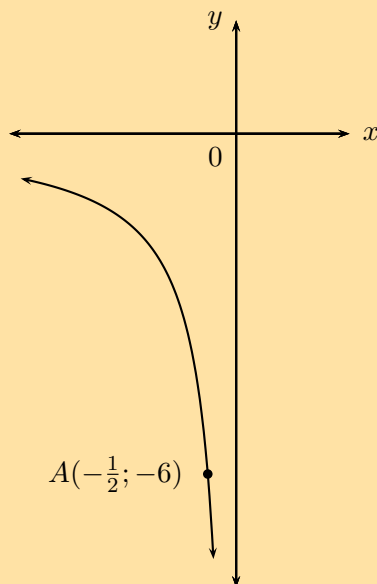
$$\begin{aligned} y &= -2 \times 3^{(x+p)} + 6 \\ 0 &= -2 \times 3^{(2+p)} + 6 \\ -6 &= -2 \times 3^{(2+p)} \\ 3 &= 3^{(2+p)} \\ \therefore 1 &= 2 + p \quad (\text{same base}) \\ \therefore p &= -1 \end{aligned}$$

Step 4: Write the final answer

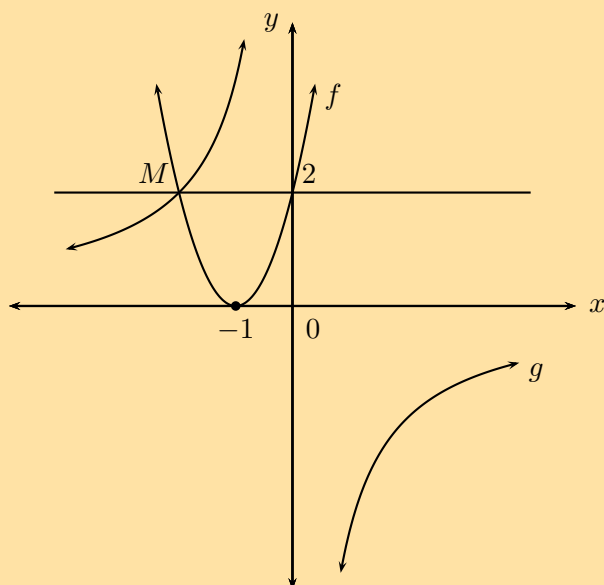
$$y = -2 \times 3^{(x-1)} + 6$$

Exercise 5 – 19: Mixed exercises

1. Given the graph of the hyperbola of the form $h(x) = \frac{k}{x}$, $x < 0$, which passes through the point $A(-\frac{1}{2}; -6)$.



- Show that $k = 3$.
 - Write down the equation for the new function formed if $h(x)$:
 - is shifted 3 units vertically upwards
 - is shifted to the right by 3 units
 - is reflected about the y -axis
 - is shifted so that the asymptotes are $x = 0$ and $y = -\frac{1}{4}$
 - is shifted upwards to pass through the point $(-1; 1)$
 - is shifted to the left by 2 units and 1 unit vertically downwards (for $x < 0$)
2. Given the graphs of $f(x) = a(x + p)^2$ and $g(x) = \frac{a}{x}$.
- The axis of symmetry for $f(x)$ is $x = -1$ and $f(x)$ and $g(x)$ intersect at point M .
The line $y = 2$ also passes through M .



Determine:

- the coordinates of M
- the equation of $g(x)$
- the equation of $f(x)$
- the values for which $f(x) < g(x)$
- the range of $f(x)$

3. On the same system of axes, sketch:

- the graphs of $k(x) = 2(x + \frac{1}{2})^2 - 4\frac{1}{2}$ and $h(x) = 2^{(x+\frac{1}{2})}$. Determine all intercepts, turning point(s) and asymptotes.
- the reflection of $h(x)$ about the x -axis. Label this function as $j(x)$.

4. Sketch the graphs of $y = ax^2 + bx + c$ for:

- $a < 0, b > 0, b^2 < 4ac$
- $a > 0, b > 0$, one root $= 0$

5. On separate systems of axes, sketch the graphs:

$$y = \frac{2}{x-2}$$

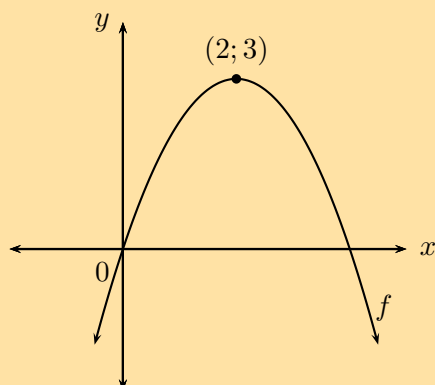
$$y = \frac{2}{x} - 2$$

$$y = -2^{(x-2)}$$

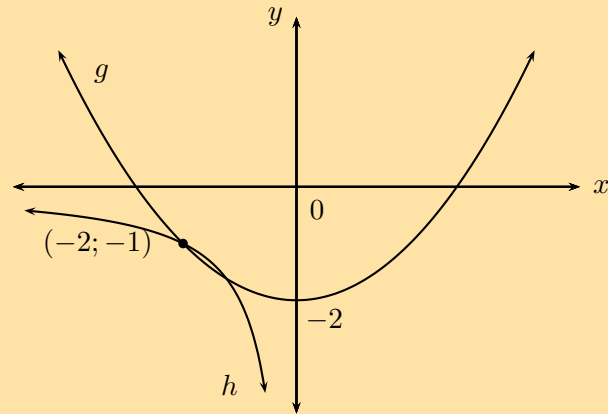
6. For the diagrams shown below, determine:

- the equations of the functions; $f(x) = a(x + p)^2 + q$, $g(x) = ax^2 + q$, $h(x) = \frac{a}{x}, x < 0$ and $k(x) = b^x + q$
- the axes of symmetry of each function
- the domain and range of each function

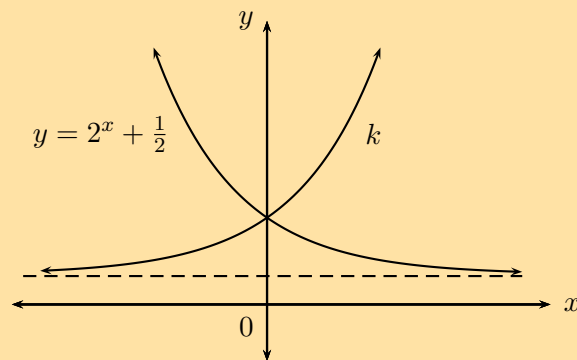
a)



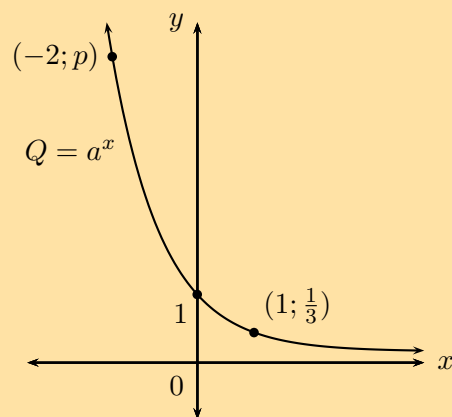
b)



c)



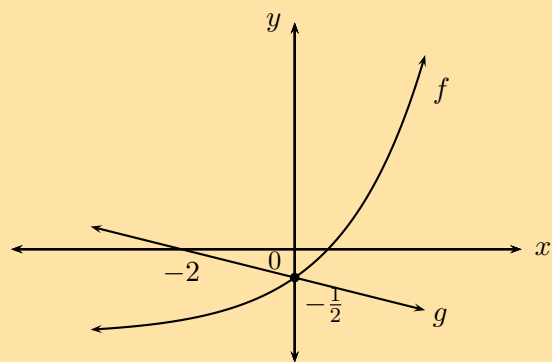
7. Given the graph of the function $Q(x) = a^x$.



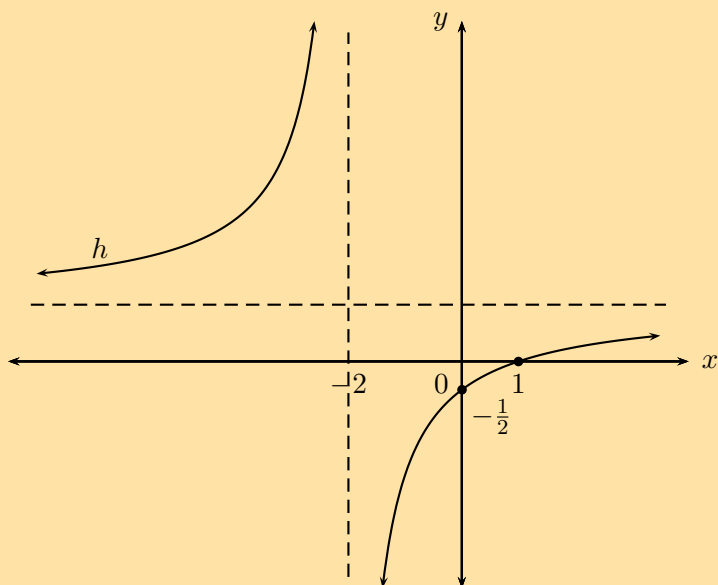
- Show that $a = \frac{1}{3}$.
- Find the value of p if the point $(-2; p)$ is on Q .
- Calculate the average gradient of the curve between $x = -2$ and $x = 1$.
- Determine the equation of the new function formed if Q is shifted 2 units vertically downwards and 2 units to the left.

8. Find the equation for each of the functions shown below:

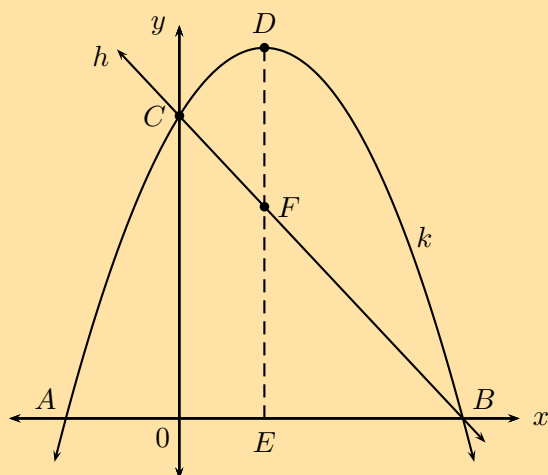
- $f(x) = 2^x + q$
 $g(x) = mx + c$



b) $h(x) = \frac{k}{x+p} + q$



9. Given: the graph of $k(x) = -x^2 + 3x + 10$ with turning point at D . The graph of the straight line $h(x) = mx + c$ passing through points B and C is also shown.



Determine:

- the lengths AO , OB , OC and DE
- the equation of DE
- the equation of $h(x)$
- the x -values for which $k(x) < 0$

- e) the x -values for which $k(x) \geq h(x)$
 f) the length of DF

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1. 22T7 2. 22T8 3. 22T9 4. 22TB 5. 22TC 6a. 22TD
 6b. 22TF 6c. 22TG 7. 22TH 8a. 22TJ 8b. 22TK 9. 22TM



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IMPORTANT: Trigonometric functions are examined in PAPER 2.

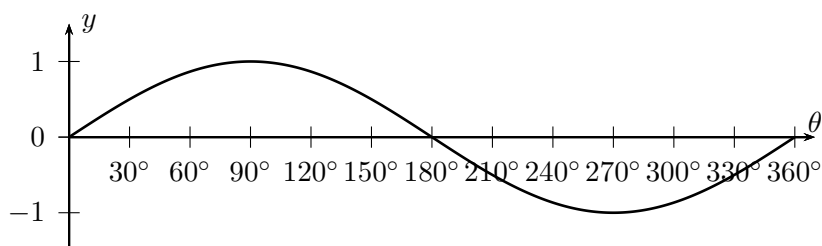
5.5 The sine function

EMBGW

Revision

EMBGX

Functions of the form $y = \sin \theta$ for $0^\circ \leq \theta \leq 360^\circ$



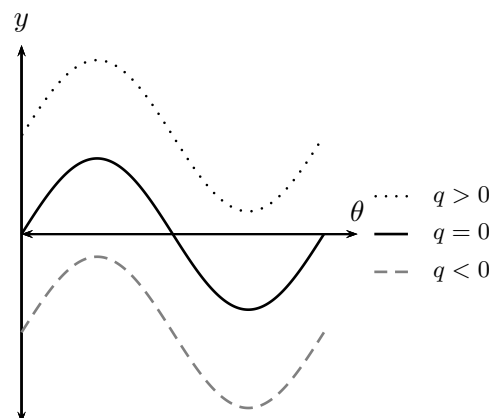
- Period of one complete wave is 360° .
- Amplitude is the maximum height of the wave above and below the x -axis and is always positive. Amplitude = 1.
- Domain: $[0^\circ; 360^\circ]$
 For $y = \sin \theta$, the domain is $\{\theta : \theta \in \mathbb{R}\}$, however in this case, the domain has been restricted to the interval $0^\circ \leq \theta \leq 360^\circ$.
- Range: $[-1; 1]$
- x -intercepts: $(0^\circ; 0)$, $(180^\circ; 0)$, $(360^\circ; 0)$
- y -intercept: $(0^\circ; 0)$
- Maximum turning point: $(90^\circ; 1)$
- Minimum turning point: $(270^\circ; -1)$

Functions of the form $y = a \sin \theta + q$

The effects of a and q on $f(\theta) = a \sin \theta + q$:

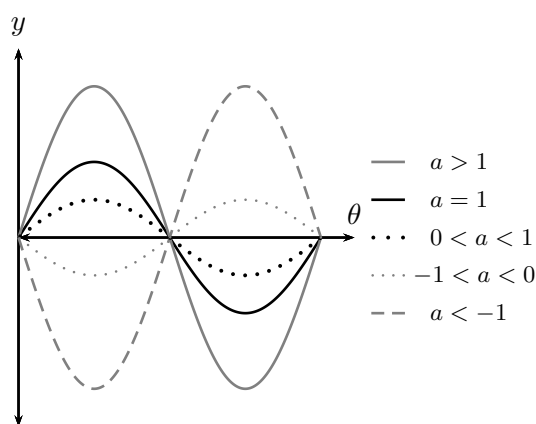
- The effect of q on vertical shift

- For $q > 0$, $f(\theta)$ is shifted vertically upwards by q units.
- For $q < 0$, $f(\theta)$ is shifted vertically downwards by q units.



- The effect of a on shape

- For $a > 1$, the amplitude of $f(\theta)$ increases.
- For $0 < a < 1$, the amplitude of $f(\theta)$ decreases.
- For $a < 0$, there is a reflection about the x -axis.
- For $-1 < a < 0$, there is a reflection about the x -axis and the amplitude decreases.
- For $a < -1$, there is a reflection about the x -axis and the amplitude increases.



Exercise 5 – 20: Revision

On separate axes, accurately draw each of the following functions for $0^\circ \leq \theta \leq 360^\circ$.

- Use tables of values if necessary.
- Use graph paper if available.

For each function also determine the following:

- Period
- Amplitude
- Domain and range
- x - and y -intercepts
- Maximum and minimum turning points

1. $y_1 = \sin \theta$
2. $y_2 = -2 \sin \theta$
3. $y_3 = \sin \theta + 1$
4. $y_4 = \frac{1}{2} \sin \theta - 1$

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1. 22TN 2. 22TP 3. 22TQ 4. 22TR



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Functions of the form $y = \sin k\theta$

EMBGY

We now consider cosine functions of the form $y = \sin k\theta$ and the effects of k .

Investigation: The effects of k on a sine graph

1. Complete the following table for $y_1 = \sin \theta$ for $-360^\circ \leq \theta \leq 360^\circ$:

θ	-360°	-270°	-180°	-90°	0°	90°	180°	270°	360°
$\sin \theta$									

2. Use the table of values to plot the graph of $y_1 = \sin \theta$ for $-360^\circ \leq \theta \leq 360^\circ$.
3. On the same system of axes, plot the following graphs:
 - a) $y_2 = \sin(-\theta)$
 - b) $y_3 = \sin 2\theta$
 - c) $y_4 = \sin \frac{\theta}{2}$
4. Use your sketches of the functions above to complete the following table:

	y_1	y_2	y_3	y_4
period				
amplitude				
domain				
range				
maximum turning points				
minimum turning points				
y-intercept(s)				
x-intercept(s)				
effect of k				

5. What do you notice about $y_1 = \sin \theta$ and $y_2 = \sin(-\theta)$?
6. Is $\sin(-\theta) = -\sin \theta$ a true statement? Explain your answer.
7. Can you deduce a formula for determining the period of $y = \sin k\theta$?

The effect of the parameter on $y = \sin k\theta$

The value of k affects the period of the sine function. If k is negative, then the graph is reflected about the y -axis.

- **For $k > 0$:**

For $k > 1$, the period of the sine function decreases.

For $0 < k < 1$, the period of the sine function increases.

- **For $k < 0$:**

For $-1 < k < 0$, the graph is reflected about the y -axis and the period increases.

For $k < -1$, the graph is reflected about the y -axis and the period decreases.

Negative angles:

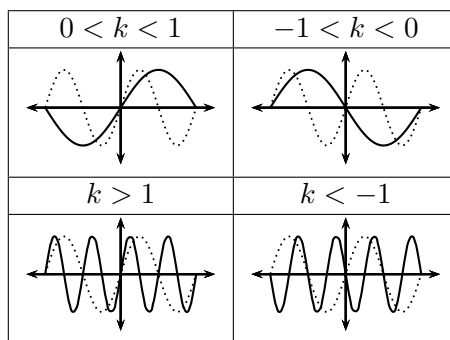
$$\sin(-\theta) = -\sin \theta$$

Calculating the period:

To determine the period of $y = \sin k\theta$ we use,

$$\text{Period} = \frac{360^\circ}{|k|}$$

where $|k|$ is the absolute value of k (this means that k is always considered to be positive).



Worked example 18: Sine function

QUESTION

1. Sketch the following functions on the same set of axes for $-180^\circ \leq \theta \leq 180^\circ$.

a) $y_1 = \sin \theta$

b) $y_2 = \sin \frac{3\theta}{2}$

2. For each function determine the following:

a) Period

b) Amplitude

c) Domain and range

d) x - and y -intercepts

e) Maximum and minimum turning points

SOLUTION

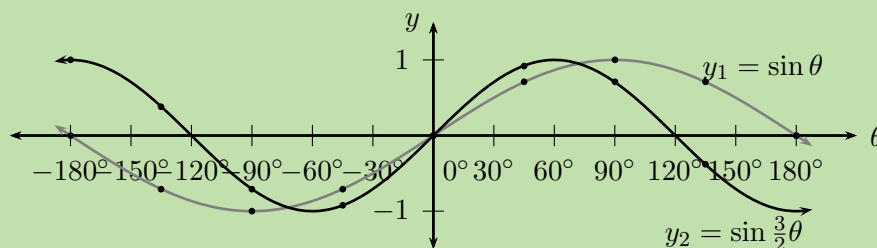
Step 1: Examine the equations of the form $y = \sin k\theta$

Notice that $k > 1$ for $y_2 = \sin \frac{3\theta}{2}$, therefore the period of the graph decreases.

Step 2: Complete a table of values

θ	-180°	-135°	-90°	-45°	0°	45°	90°	135°	180°
$\sin \theta$	0	-0,71	-1	-0,71	0	0,71	1	0,71	0
$\sin \frac{3\theta}{2}$	1	0,38	-0,71	-0,92	0	0,92	0,71	-0,38	-1

Step 3: Sketch the sine graphs



Step 4: Complete the table

	$y_1 = \sin \theta$	$y_2 = \sin \frac{3\theta}{2}$
period	360°	240°
amplitude	1	1
domain	$[-180^\circ; 180^\circ]$	$[-180^\circ; 180^\circ]$
range	$[-1; 1]$	$[-1; 1]$
maximum turning points	$(90^\circ; 1)$	$(-180^\circ; 1)$ and $(60^\circ; 1)$
minimum turning points	$(-90^\circ; -1)$	$(-60^\circ; -1)$ and $(180^\circ; -1)$
y-intercept(s)	$(0^\circ; 0)$	$(0^\circ; 0)$
x-intercept(s)	$(-180^\circ; 0)$, $(0^\circ; 0)$ and $(180^\circ; 0)$	$(-120^\circ; 0)$, $(0^\circ; 0)$ and $(120^\circ; 0)$

Discovering the characteristics

For functions of the general form: $f(\theta) = y = \sin k\theta$:

Domain and range

The domain is $\{\theta : \theta \in \mathbb{R}\}$ because there is no value for θ for which $f(\theta)$ is undefined.

The range is $\{f(\theta) : -1 \leq f(\theta) \leq 1, f(\theta) \in \mathbb{R}\}$ or $[-1; 1]$.

Intercepts

The x -intercepts are determined by letting $f(\theta) = 0$ and solving for θ .

The y -intercept is calculated by letting $\theta = 0^\circ$ and solving for $f(\theta)$.

$$\begin{aligned}y &= \sin k\theta \\&= \sin 0^\circ \\&= 0\end{aligned}$$

This gives the point $(0^\circ; 0)$.

Exercise 5 – 21: Sine functions of the form $y = \sin k\theta$

1. Sketch the following functions for $-180^\circ \leq \theta \leq 180^\circ$ and for each graph determine:

- Period
- Amplitude
- Domain and range
- x - and y -intercepts
- Maximum and minimum turning points

a) $f(\theta) = \sin 3\theta$

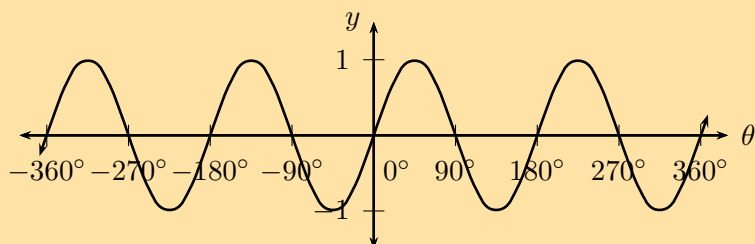
b) $g(\theta) = \sin \frac{\theta}{3}$

c) $h(\theta) = \sin(-2\theta)$

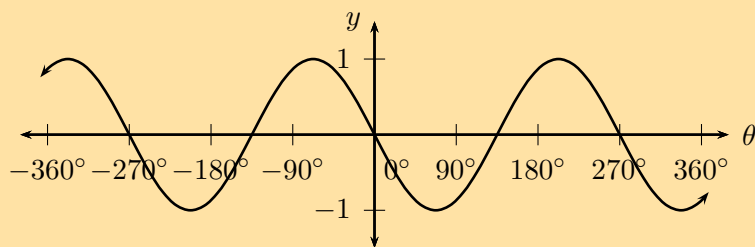
d) $k(\theta) = \sin \frac{3\theta}{4}$

2. For each graph of the form $f(\theta) = \sin k\theta$, determine the value of k :

a)



b)



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1a. 22TS 1b. 22TT 1c. 22TV 1d. 22TW 2a. 22TX 2b. 22TY



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Functions of the form $y = \sin(\theta + p)$

EMBGZ

Investigation: The effects of p on a sine graph

1. On the same system of axes, plot the following graphs for $-360^\circ \leq \theta \leq 360^\circ$:

- a) $y_1 = \sin \theta$
- b) $y_2 = \sin(\theta - 90^\circ)$
- c) $y_3 = \sin(\theta - 60^\circ)$
- d) $y_4 = \sin(\theta + 90^\circ)$
- e) $y_5 = \sin(\theta + 180^\circ)$

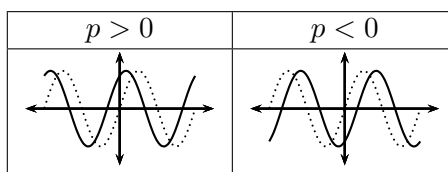
2. Use your sketches of the functions above to complete the following table:

	y_1	y_2	y_3	y_4	y_5
period					
amplitude					
domain					
range					
maximum turning points					
minimum turning points					
y-intercept(s)					
x-intercept(s)					
effect of p					

The effect of the parameter on $y = \sin(\theta + p)$

The effect of p on the sine function is a horizontal shift, also called a phase shift; the entire graph slides to the left or to the right.

- For $p > 0$, the graph of the sine function shifts to the left by p .
- For $p < 0$, the graph of the sine function shifts to the right by p .



Worked example 19: Sine function

QUESTION

1. Sketch the following functions on the same set of axes for $-360^\circ \leq \theta \leq 360^\circ$.
 - a) $y_1 = \sin \theta$
 - b) $y_2 = \sin(\theta - 30^\circ)$
2. For each function determine the following:
 - a) Period
 - b) Amplitude
 - c) Domain and range
 - d) x - and y -intercepts
 - e) Maximum and minimum turning points

SOLUTION

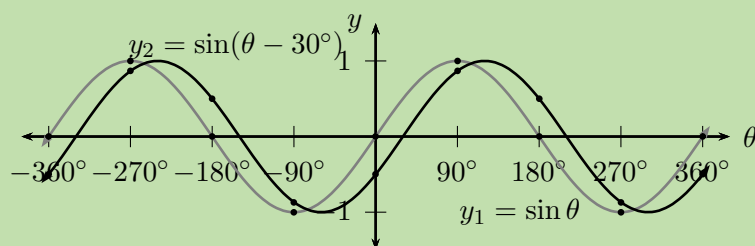
Step 1: Examine the equations of the form $y = \sin(\theta + p)$

Notice that for $y_1 = \sin \theta$ we have $p = 0$ (no phase shift) and for $y_2 = \sin(\theta - 30^\circ)$, $p < 0$ therefore the graph shifts to the right by 30° .

Step 2: Complete a table of values

θ	-360°	-270°	-180°	-90°	0°	90°	180°	270°	360°
$\sin \theta$	0	1	0	-1	0	1	0	-1	0
$\sin(\theta - 30^\circ)$	-0,5	0,87	0,5	-0,87	-0,5	0,87	0,5	-0,87	-0,5

Step 3: Sketch the sine graphs



Step 4: Complete the table

	$y_1 = \sin \theta$	$y_2 = \sin(\theta - 30^\circ)$
period	360°	360°
amplitude	1	1
domain	$[-360^\circ; 360^\circ]$	$[-360^\circ; 360^\circ]$
range	$[-1; 1]$	$[-1; 1]$
maximum turning points	$(-270^\circ; 1)$ and $(90^\circ; 1)$	$(-240^\circ; 1)$ and $(120^\circ; 1)$
minimum turning points	$(-90^\circ; -1)$ and $(270^\circ; -1)$	$(-60^\circ; -1)$ and $(300^\circ; -1)$
y-intercept(s)	$(0^\circ; 0)$	$(0^\circ; -\frac{1}{2})$
x-intercept(s)	$(-360^\circ; 0)$, $(-180^\circ; 0)$, $(0^\circ; 0)$, $(180^\circ; 0)$ and $(360^\circ; 0)$	$(-330^\circ; 0)$, $(-150^\circ; 0)$, $(30^\circ; 0)$ and $(210^\circ; 0)$

Discovering the characteristics

For functions of the general form: $f(\theta) = y = \sin(\theta + p)$:

Domain and range

The domain is $\{\theta : \theta \in \mathbb{R}\}$ because there is no value for θ for which $f(\theta)$ is undefined.

The range is $\{f(\theta) : -1 \leq f(\theta) \leq 1, f(\theta) \in \mathbb{R}\}$.

Intercepts

The x -intercepts are determined by letting $f(\theta) = 0$ and solving for θ .

The y -intercept is calculated by letting $\theta = 0^\circ$ and solving for $f(\theta)$.

Exercise 5 – 22: Sine functions of the form $y = \sin(\theta + p)$

Sketch the following functions for $-360^\circ \leq \theta \leq 360^\circ$.

For each function, determine the following:

- Period
- Amplitude
- Domain and range
- x - and y -intercepts
- Maximum and minimum turning points

1. $f(\theta) = \sin(\theta + 30^\circ)$

2. $g(\theta) = \sin(\theta - 45^\circ)$

3. $h(\theta) = \sin(\theta + 60^\circ)$

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1. [22TZ](#) 2. [22V2](#) 3. [22V3](#)



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Sketching sine graphs

EMBH2

Worked example 20: Sketching a sine graph

QUESTION

Sketch the graph of $f(\theta) = \sin(45^\circ - \theta)$ for $0^\circ \leq \theta \leq 360^\circ$.

SOLUTION

Step 1: Examine the form of the equation

Write the equation in the form $y = \sin(\theta + p)$.

$$\begin{aligned} f(\theta) &= \sin(45^\circ - \theta) \\ &= \sin(-\theta + 45^\circ) \\ &= \sin(-(\theta - 45^\circ)) \\ &= -\sin(\theta - 45^\circ) \end{aligned}$$

To draw a graph of the above function, we know that the standard sine graph, $y = \sin \theta$,

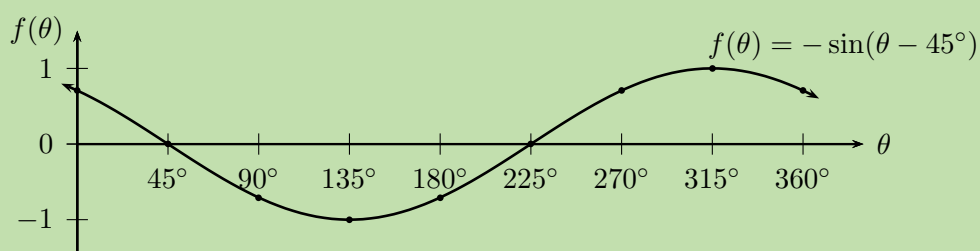
must:

- be reflected about the x -axis
- be shifted to the right by 45°

Step 2: Complete a table of values

θ	0°	45°	90°	135°	180°	225°	270°	315°	360°
$f(\theta)$	0,71	0	-0,71	-1	-0,71	0	0,71	1	0,71

Step 3: Plot the points and join with a smooth curve



Period: 360°

Amplitude: 1

Domain: $[-360^\circ; 360^\circ]$

Range: $[-1; 1]$

Maximum turning point: $(315^\circ; 1)$

Minimum turning point: $(135^\circ; -1)$

y -intercepts: $(0^\circ; 0,71)$

x -intercept: $(45^\circ; 0)$ and $(225^\circ; 0)$

Worked example 21: Sketching a sine graph

QUESTION

Sketch the graph of $f(\theta) = \sin(3\theta + 60^\circ)$ for $0^\circ \leq \theta \leq 180^\circ$.

SOLUTION

Step 1: Examine the form of the equation

Write the equation in the form $y = \sin k(\theta + p)$.

$$\begin{aligned}f(\theta) &= \sin(3\theta + 60^\circ) \\&= \sin 3(\theta + 20^\circ)\end{aligned}$$

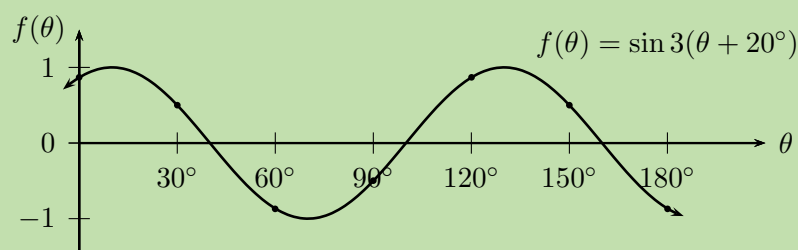
To draw a graph of the above equation, the standard sine graph, $y = \sin \theta$, must be changed in the following ways:

- decrease the period by a factor of 3;
- shift to the left by 20° .

Step 2: Complete a table of values

θ	0°	30°	60°	90°	120°	150°	180°
$f(\theta)$	0,87	0,5	-0,87	-0,5	0,87	0,5	-0,87

Step 3: Plot the points and join with a smooth curve



Period: 120°

Amplitude: 1

Domain: $[0^\circ; 180^\circ]$

Range: $[-1; 1]$

Maximum turning point: $(10^\circ; 1)$ and $(130^\circ; 1)$

Minimum turning point: $(70^\circ; -1)$

y -intercept: $(0^\circ; 0,87)$

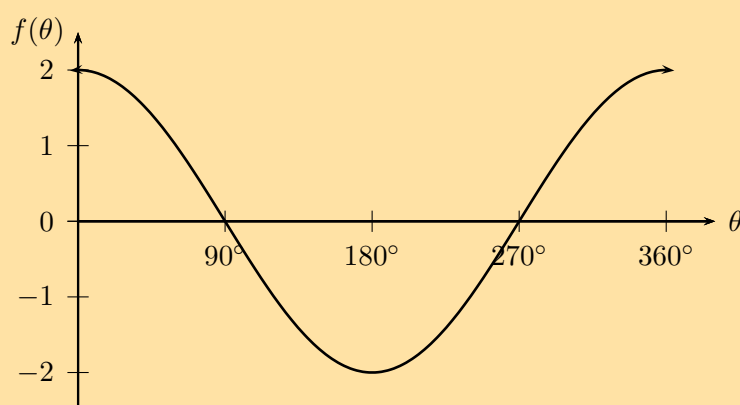
x -intercepts: $(40^\circ; 0)$, $(100^\circ; 0)$ and $(160^\circ; 0)$

Exercise 5 – 23: The sine function

1. Sketch the following graphs on separate axes:

- $y = 2 \sin \frac{\theta}{2}$ for $-360^\circ \leq \theta \leq 360^\circ$
- $f(\theta) = \frac{1}{2} \sin(\theta - 45^\circ)$ for $-90^\circ \leq \theta \leq 90^\circ$
- $y = \sin(\theta + 90^\circ) + 1$ for $0^\circ \leq \theta \leq 360^\circ$
- $y = \sin(-\frac{3\theta}{2})$ for $-180^\circ \leq \theta \leq 180^\circ$
- $y = \sin(30^\circ - \theta)$ for $-360^\circ \leq \theta \leq 360^\circ$

2. Given the graph of the function $y = a \sin(\theta + p)$, determine the values of a and p .



Can you describe this graph in terms of $\cos \theta$?

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1a. 22V4 1b. 22V5 1c. 22V6 1d. 22V7 1e. 22V8 2. 22V9



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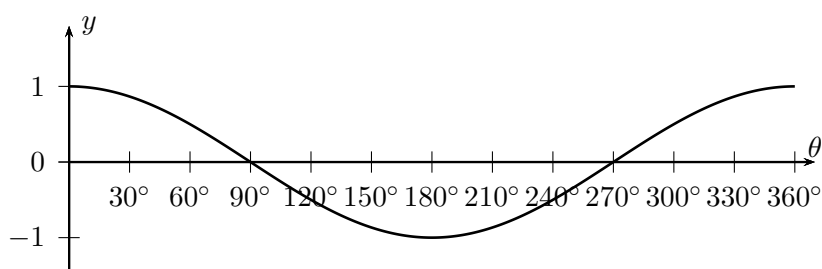
5.6 The cosine function

EMBH3

Revision

EMBH4

Functions of the form $y = \cos \theta$ for $0^\circ \leq \theta \leq 360^\circ$



- The period is 360° and the amplitude is 1.
- Domain: $[0^\circ; 360^\circ]$
For $y = \cos \theta$, the domain is $\{\theta : \theta \in \mathbb{R}\}$, however in this case, the domain has been restricted to the interval $0^\circ \leq \theta \leq 360^\circ$.
- Range: $[-1; 1]$
- x -intercepts: $(90^\circ; 0)$, $(270^\circ; 0)$
- y -intercept: $(0^\circ; 1)$
- Maximum turning points: $(0^\circ; 1)$, $(360^\circ; 1)$
- Minimum turning point: $(180^\circ; -1)$

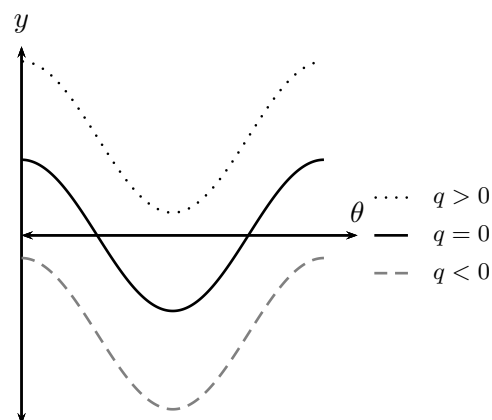
Functions of the form $y = a \cos \theta + q$

Cosine functions of the general form $y = a \cos \theta + q$, where a and q are constants.

The effects of a and q on $f(\theta) = a \cos \theta + q$:

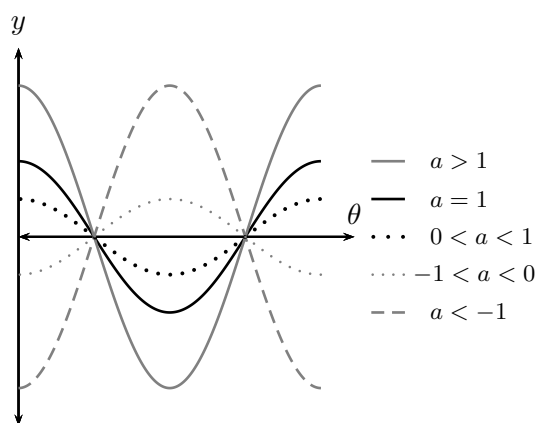
- **The effect of q on vertical shift**

- For $q > 0$, $f(\theta)$ is shifted vertically upwards by q units.
- For $q < 0$, $f(\theta)$ is shifted vertically downwards by q units.



- **The effect of a on shape**

- For $a > 1$, the amplitude of $f(\theta)$ increases.
- For $0 < a < 1$, the amplitude of $f(\theta)$ decreases.
- For $a < 0$, there is a reflection about the x -axis.
- For $-1 < a < 0$, there is a reflection about the x -axis and the amplitude decreases.
- For $a < -1$, there is a reflection about the x -axis and the amplitude increases.



Exercise 5 – 24: Revision

On separate axes, accurately draw each of the following functions for $0^\circ \leq \theta \leq 360^\circ$:

- Use tables of values if necessary.
- Use graph paper if available.

For each function in the previous problem determine the following:

- Period
- Amplitude
- Domain and range
- x - and y -intercepts
- Maximum and minimum turning points

1. $y_1 = \cos \theta$
2. $y_2 = -3 \cos \theta$
3. $y_3 = \cos \theta + 2$
4. $y_4 = \frac{1}{2} \cos \theta - 1$

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1. 22VB 2. 22VC 3. 22VD 4. 22VF



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Functions of the form $y = \cos(k\theta)$

EMBH5

We now consider cosine functions of the form $y = \cos k\theta$ and the effects of k .

Investigation: The effects of k on a cosine graph

1. Complete the following table for $y_1 = \cos \theta$ for $-360^\circ \leq \theta \leq 360^\circ$:

θ	-360°	-300°	-240°	-180°	-120°	-60°	0°
$\cos \theta$							
θ	60°	120°	180°	240°	300°	360°	
$\cos \theta$							

2. Use the table of values to plot the graph of $y_1 = \cos \theta$ for $-360^\circ \leq \theta \leq 360^\circ$.
3. On the same system of axes, plot the following graphs:
 - a) $y_2 = \cos(-\theta)$
 - b) $y_3 = \cos 3\theta$
 - c) $y_4 = \cos \frac{3\theta}{4}$

4. Use your sketches of the functions above to complete the following table:

	y_1	y_2	y_3	y_4
period				
amplitude				
domain				
range				
maximum turning points				
minimum turning points				
y-intercept(s)				
x-intercept(s)				
effect of k				

5. What do you notice about $y_1 = \cos \theta$ and $y_2 = \cos(-\theta)$?
6. Is $\cos(-\theta) = -\cos \theta$ a true statement? Explain your answer.
7. Can you deduce a formula for determining the period of $y = \cos k\theta$?

The effect of the parameter k on $y = \cos k\theta$

The value of k affects the period of the cosine function.

- **For $k > 0$:**
 For $k > 1$, the period of the cosine function decreases.
 For $0 < k < 1$, the period of the cosine function increases.
- **For $k < 0$:**
 For $-1 < k < 0$, the period increases.
 For $k < -1$, the period decreases.

Negative angles:

$$\cos(-\theta) = \cos \theta$$

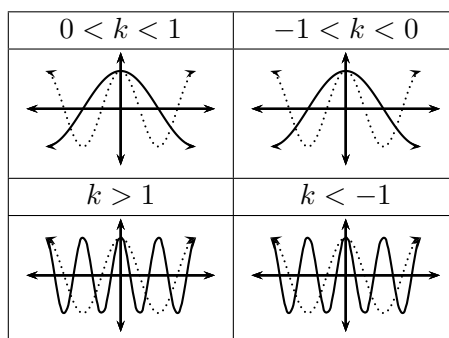
Notice that for negative values of θ , the graph is **not** reflected about the x -axis.

Calculating the period:

To determine the period of $y = \cos k\theta$ we use,

$$\text{Period} = \frac{360^\circ}{|k|}$$

where $|k|$ is the absolute value of k .



Worked example 22: Cosine function

QUESTION

1. Sketch the following functions on the same set of axes for $-180^\circ \leq \theta \leq 180^\circ$.

a) $y_1 = \cos \theta$

b) $y_2 = \cos \frac{\theta}{2}$

2. For each function determine the following:

- a) Period
- b) Amplitude
- c) Domain and range
- d) x - and y -intercepts
- e) Maximum and minimum turning points

SOLUTION

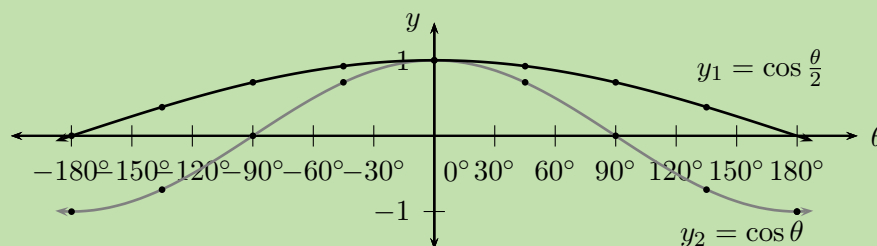
Step 1: Examine the equations of the form $y = \cos k\theta$

Notice that for $y_2 = \cos \frac{\theta}{2}$, $k < 1$ therefore the period of the graph increases.

Step 2: Complete a table of values

θ	-180°	-135°	-90°	-45°	0°	45°	90°	135°	180°
$\cos \theta$	-1	-0,71	0	0,71	1	0,71	0	-0,71	-1
$\cos \frac{\theta}{2}$	0	0,38	0,71	0,92	1	0,92	0,71	0,38	0

Step 3: Sketch the cosine graphs



Step 4: Complete the table

	$y_1 = \cos \theta$	$y_2 = \cos \frac{\theta}{2}$
period	360°	720°
amplitude	1	1
domain	$[-180^\circ; 180^\circ]$	$[-180^\circ; 180^\circ]$
range	$[-1; 1]$	$[0; 1]$
maximum turning points	$(0^\circ; 1)$	$(0^\circ; 1)$
minimum turning points	$(-180^\circ; -1)$ and $(180^\circ; -1)$	none
y-intercept(s)	$(0^\circ; 1)$	$(0^\circ; 1)$
x-intercept(s)	$(-90^\circ; 0)$ and $(90^\circ; 0)$	$(-180^\circ; 0)$ and $(180^\circ; 0)$

Discovering the characteristics

For functions of the general form: $f(\theta) = y = \cos k\theta$:

Domain and range

The domain is $\{\theta : \theta \in \mathbb{R}\}$ because there is no value for θ for which $f(\theta)$ is undefined.

The range is $\{f(\theta) : -1 \leq f(\theta) \leq 1, f(\theta) \in \mathbb{R}\}$ or $[-1; 1]$.

Intercepts

The x -intercepts are determined by letting $f(\theta) = 0$ and solving for θ .

The y -intercept is calculated by letting $\theta = 0^\circ$ and solving for $f(\theta)$.

$$\begin{aligned}y &= \cos k\theta \\&= \cos 0^\circ \\&= 1\end{aligned}$$

This gives the point $(0^\circ; 1)$.

Exercise 5 – 25: Cosine functions of the form $y = \cos k\theta$

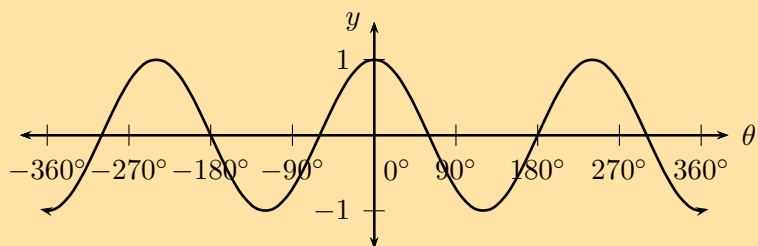
1. Sketch the following functions for $-180^\circ \leq \theta \leq 180^\circ$. For each graph determine:

- Period
- Amplitude
- Domain and range
- x - and y -intercepts
- Maximum and minimum turning points

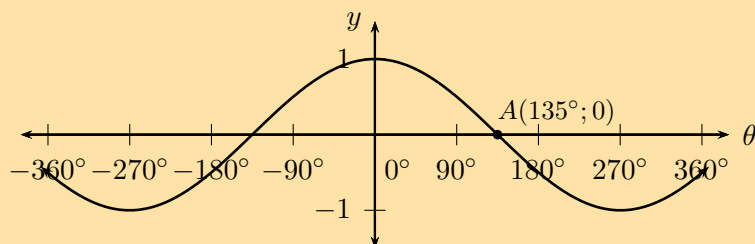
- $f(\theta) = \cos 2\theta$
- $g(\theta) = \cos \frac{\theta}{3}$
- $h(\theta) = \cos(-2\theta)$
- $k(\theta) = \cos \frac{3\theta}{4}$

2. For each graph of the form $f(\theta) = \cos k\theta$, determine the value of k :

a)



b)



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1a. [22VG](#) 1b. [22VH](#) 1c. [22VJ](#) 1d. [22VK](#) 2a. [22VM](#) 2b. [22VN](#)



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Functions of the form $y = \cos(\theta + p)$

EMBH6

We now consider cosine functions of the form $y = \cos(\theta + p)$ and the effects of parameter p .

Investigation: The effects of p on a cosine graph

1. On the same system of axes, plot the following graphs for $-360^\circ \leq \theta \leq 360^\circ$:

- a) $y_1 = \cos \theta$
- b) $y_2 = \cos(\theta - 90^\circ)$
- c) $y_3 = \cos(\theta - 60^\circ)$
- d) $y_4 = \cos(\theta + 90^\circ)$
- e) $y_5 = \cos(\theta + 180^\circ)$

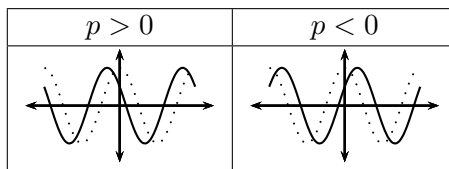
2. Use your sketches of the functions above to complete the following table:

	y_1	y_2	y_3	y_4	y_5
period					
amplitude					
domain					
range					
maximum turning points					
minimum turning points					
y-intercept(s)					
x-intercept(s)					
effect of p					

The effect of the parameter on $y = \cos(\theta + p)$

The effect of p on the cosine function is a horizontal shift (or phase shift); the entire graph slides to the left or to the right.

- For $p > 0$, the graph of the cosine function shifts to the left by p degrees.
- For $p < 0$, the graph of the cosine function shifts to the right by p degrees.



Worked example 23: Cosine function

QUESTION

- Sketch the following functions on the same set of axes for $-360^\circ \leq \theta \leq 360^\circ$.
 - $y_1 = \cos \theta$
 - $y_2 = \cos(\theta + 30^\circ)$
- For each function determine the following:
 - Period
 - Amplitude
 - Domain and range
 - x - and y -intercepts
 - Maximum and minimum turning points

SOLUTION

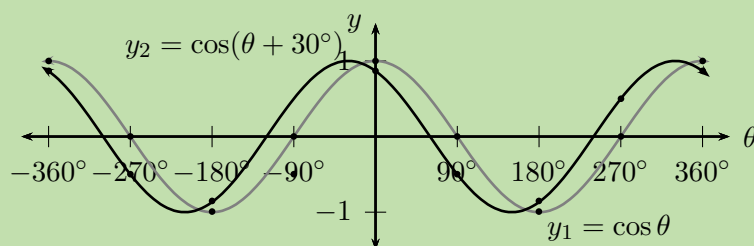
Step 1: Examine the equations of the form $y = \cos(\theta + p)$

Notice that for $y_1 = \cos \theta$ we have $p = 0$ (no phase shift) and for $y_2 = \cos(\theta + 30^\circ)$, $p < 0$ therefore the graph shifts to the left by 30° .

Step 2: Complete a table of values

θ	-360°	-270°	-180°	-90°	0°	90°	180°	270°	360°
$\cos \theta$	1	0	-1	0	1	0	-1	0	1
$\cos(\theta + 30^\circ)$	0,87	-0,5	-0,87	0,5	0,87	-0,5	-0,87	0,5	0,87

Step 3: Sketch the cosine graphs



Step 4: Complete the table

	y_1	y_2
period	360°	360°
amplitude	1	1
domain	$[-360^\circ; 360^\circ]$	$[-360^\circ; 360^\circ]$
range	$[-1; 1]$	$[-1; 1]$
maximum turning points	$(-360^\circ; 1)$, $(0^\circ; 1)$ and $(360^\circ; 1)$	$(-30^\circ; 1)$ and $(330^\circ; 1)$
minimum turning points	$(-180^\circ; -1)$ and $(180^\circ; -1)$	$(-210^\circ; -1)$ and $(150^\circ; -1)$
y-intercept(s)	$(0^\circ; 0)$	$(0^\circ; 0,87)$
x-intercept(s)	$(-270^\circ; 0)$, $(-90^\circ; 0)$, $(90^\circ; 0)$ and $(270^\circ; 0)$	$(-300^\circ; 0)$, $(-120^\circ; 0)$, $(60^\circ; 0)$ and $(240^\circ; 0)$

Discovering the characteristics

For functions of the general form: $f(\theta) = y = \cos(\theta + p)$:

Domain and range

The domain is $\{\theta : \theta \in \mathbb{R}\}$ because there is no value for θ for which $f(\theta)$ is undefined.

The range is $\{f(\theta) : -1 \leq f(\theta) \leq 1, f(\theta) \in \mathbb{R}\}$.

Intercepts

The x -intercepts are determined by letting $f(\theta) = 0$ and solving for θ .

The y -intercept is calculated by letting $\theta = 0^\circ$ and solving for $f(\theta)$.

$$\begin{aligned}
 y &= \cos(\theta + p) \\
 &= \cos(0^\circ + p) \\
 &= \cos p
 \end{aligned}$$

This gives the point $(0^\circ; \cos p)$.

Exercise 5 – 26: Cosine functions of the form $y = \cos(\theta + p)$

Sketch the following functions for $-360^\circ \leq \theta \leq 360^\circ$.

For each function, determine the following:

- Period
- Amplitude
- Domain and range
- x - and y -intercepts
- Maximum and minimum turning points

1. $f(\theta) = \cos(\theta + 45^\circ)$

2. $g(\theta) = \cos(\theta - 30^\circ)$

3. $h(\theta) = \cos(\theta + 60^\circ)$

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1. [22VP](#) 2. [22VQ](#) 3. [22VR](#)



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Sketching cosine graphs

EMBH7

Worked example 24: Sketching a cosine graph

QUESTION

Sketch the graph of $f(\theta) = \cos(180^\circ - 3\theta)$ for $0^\circ \leq \theta \leq 360^\circ$.

SOLUTION

Step 1: Examine the form of the equation

Write the equation in the form $y = \cos k(\theta + p)$.

$$\begin{aligned} f(\theta) &= \cos(180^\circ - 3\theta) \\ &= \cos(-3\theta + 180^\circ) \\ &= \cos(-3(\theta - 60^\circ)) \\ &= \cos 3(\theta - 60^\circ) \end{aligned}$$

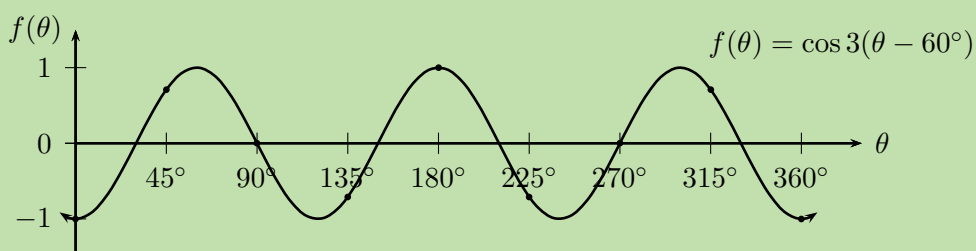
To draw a graph of the above function, the standard cosine graph, $y = \cos \theta$, must be changed in the following ways:

- decrease the period by a factor of 3
- shift to the right by 60° .

Step 2: Complete a table of values

θ	0°	45°	90°	135°	180°	225°	270°	315°	360°
$f(\theta)$	-1	0,71	0	-0,71	1	-0,71	0	0,71	-1

Step 3: Plot the points and join with a smooth curve



Period: 120°

Amplitude: 1

Domain: $[0^\circ; 360^\circ]$

Range: $[-1; 1]$

Maximum turning point: $(60^\circ; 1)$, $(180^\circ; 1)$ and $(300^\circ; 1)$

Minimum turning point: $(0^\circ; -1)$, $(120^\circ; -1)$, $(240^\circ; -1)$ and $(360^\circ; -1)$

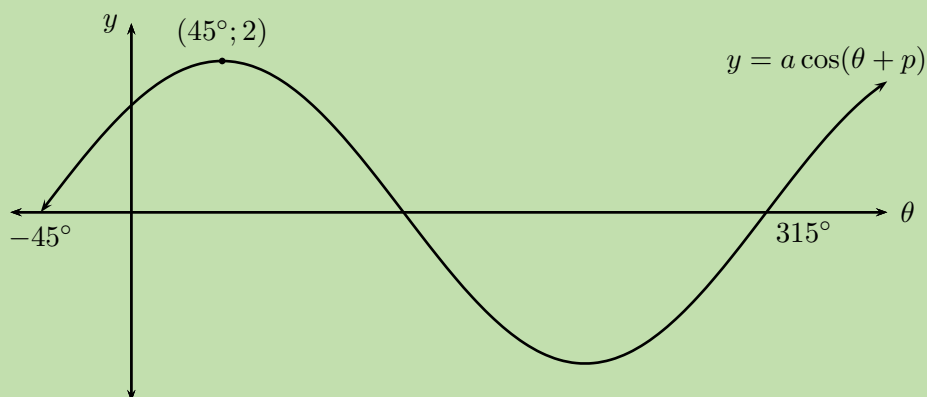
y -intercepts: $(0^\circ; -1)$

x -intercepts: $(30^\circ; 0)$, $(90^\circ; 0)$, $(150^\circ; 0)$, $(210^\circ; 0)$, $(270^\circ; 0)$ and $(330^\circ; 0)$

Worked example 25: Finding the equation of a cosine graph

QUESTION

Given the graph of $y = a \cos(k\theta + p)$, determine the values of a , k , p and the minimum turning point.



SOLUTION

Step 1: Determine the value of k

From the sketch we see that the period of the graph is 360° , therefore $k = 1$.

$$y = a \cos(\theta + p)$$

Step 2: Determine the value of a

From the sketch we see that the maximum turning point is $(45^\circ; 2)$, so we know that the amplitude of the graph is 2 and therefore $a = 2$.

$$y = 2 \cos(\theta + p)$$

Step 3: Determine the value of p

Compare the given graph with the standard cosine function $y = \cos \theta$ and notice the difference in the maximum turning points. We see that the given function has been shifted to the right by 45° , therefore $p = 45^\circ$.

$$y = 2 \cos(\theta - 45^\circ)$$

Step 4: Determine the minimum turning point

At the minimum turning point, $y = -2$:

$$\begin{aligned} y &= 2 \cos(\theta - 45^\circ) \\ -2 &= 2 \cos(\theta - 45^\circ) \\ -1 &= \cos(\theta - 45^\circ) \\ \cos^{-1}(-1) &= \theta - 45^\circ \\ 180^\circ &= \theta - 45^\circ \\ 225^\circ &= \theta \end{aligned}$$

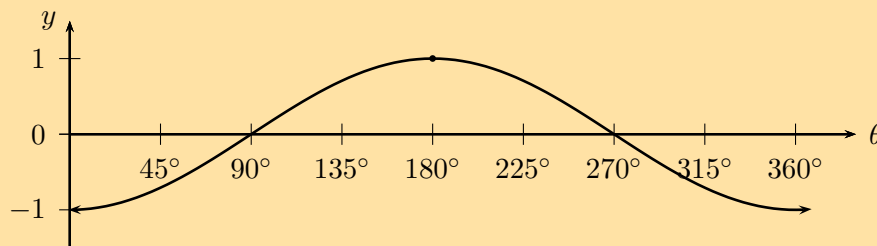
This gives the point $(225^\circ; -2)$.

Exercise 5 – 27: The cosine function

1. Sketch the following graphs on separate axes:

- a) $y = \cos(\theta + 15^\circ)$ for $-180^\circ \leq \theta \leq 180^\circ$
- b) $f(\theta) = \frac{1}{3} \cos(\theta - 60^\circ)$ for $-90^\circ \leq \theta \leq 90^\circ$
- c) $y = -2 \cos \theta$ for $0^\circ \leq \theta \leq 360^\circ$
- d) $y = \cos(30^\circ - \theta)$ for $-360^\circ \leq \theta \leq 360^\circ$
- e) $g(\theta) = 1 + \cos(\theta - 90^\circ)$ for $0^\circ \leq \theta \leq 360^\circ$
- f) $y = \cos(2\theta + 60^\circ)$ for $-360^\circ \leq \theta \leq 360^\circ$

2. Two girls are given the following graph:



- a) Audrey decides that the equation for the graph is a cosine function of the form $y = a \cos \theta$. Determine the value of a .
- b) Megan thinks that the equation for the graph is a cosine function of the form $y = \cos(\theta + p)$. Determine the value of p .
- c) What can they conclude?

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1a. [22VS](#) 1b. [22VT](#) 1c. [22VV](#) 1d. [22VW](#) 1e. [22VX](#) 1f. [22VY](#)
2. [22VZ](#)

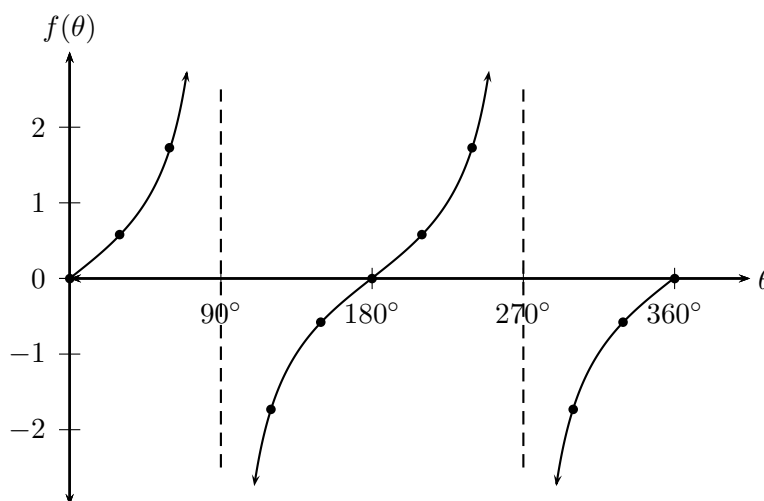


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Functions of the form $y = \tan \theta$ for $0^\circ \leq \theta \leq 360^\circ$



The dashed vertical lines are called the asymptotes. The asymptotes are at the values of θ where $\tan \theta$ is not defined.

- Period: 180°
- Domain: $\{\theta : 0^\circ \leq \theta \leq 360^\circ, \theta \neq 90^\circ; 270^\circ\}$
- Range: $\{f(\theta) : f(\theta) \in \mathbb{R}\}$
- x -intercepts: $(0^\circ; 0)$, $(180^\circ; 0)$, $(360^\circ; 0)$
- y -intercept: $(0^\circ; 0)$
- Asymptotes: the lines $\theta = 90^\circ$ and $\theta = 270^\circ$

Functions of the form $y = a \tan \theta + q$

Tangent functions of the general form $y = a \tan \theta + q$, where a and q are constants.

The effects of a and q on $f(\theta) = a \tan \theta + q$:

- **The effect of q on vertical shift**
 - For $q > 0$, $f(\theta)$ is shifted vertically upwards by q units.
 - For $q < 0$, $f(\theta)$ is shifted vertically downwards by q units.
- **The effect of a on shape**
 - For $a > 1$, branches of $f(\theta)$ are steeper.
 - For $0 < a < 1$, branches of $f(\theta)$ are less steep and curve more.

- For $a < 0$, there is a reflection about the x -axis.
- For $-1 < a < 0$, there is a reflection about the x -axis and the branches of the graph are less steep.
- For $a < -1$, there is a reflection about the x -axis and the branches of the graph are steeper.

	$a < 0$	$a > 0$
$q > 0$		
$q = 0$		
$q < 0$		

Exercise 5 – 28: Revision

On separate axes, accurately draw each of the following functions for $0^\circ \leq \theta \leq 360^\circ$:

- Use tables of values if necessary.
- Use graph paper if available.

For each function determine the following:

- Period
- Domain and range

- x - and y -intercepts
- Asymptotes

1. $y_1 = \tan \theta - \frac{1}{2}$
2. $y_2 = -3 \tan \theta$
3. $y_3 = \tan \theta + 2$
4. $y_4 = 2 \tan \theta - 1$

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. 22W2 2. 22W3 3. 22W4 4. 22W5



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Functions of the form $y = \tan(k\theta)$

EMBHB

Investigation: The effects of k on a tangent graph

1. Complete the following table for $y_1 = \tan \theta$ for $-360^\circ \leq \theta \leq 360^\circ$:

θ	-360°	-300°	-240°	-180°	-120°	-60°	0°
$\tan \theta$							
θ	60°	120°	180°	240°	300°	360°	
$\tan \theta$							

2. Use the table of values to plot the graph of $y_1 = \tan \theta$ for $-360^\circ \leq \theta \leq 360^\circ$.
3. On the same system of axes, plot the following graphs:
 - a) $y_2 = \tan(-\theta)$
 - b) $y_3 = \tan 3\theta$
 - c) $y_4 = \tan \frac{\theta}{2}$
4. Use your sketches of the functions above to complete the following table:

	y_1	y_2	y_3	y_4
period				
domain				
range				
y-intercept(s)				
x-intercept(s)				
asymptotes				
effect of k				

5. What do you notice about $y_1 = \tan \theta$ and $y_2 = \tan(-\theta)$?
6. Is $\tan(-\theta) = -\tan \theta$ a true statement? Explain your answer.
7. Can you deduce a formula for determining the period of $y = \tan k\theta$?

The effect of the parameter on $y = \tan k\theta$

The value of k affects the period of the tangent function. If k is negative, then the graph is reflected about the y -axis.

- **For $k > 0$:**

For $k > 1$, the period of the tangent function decreases.

For $0 < k < 1$, the period of the tangent function increases.

- **For $k < 0$:**

For $-1 < k < 0$, the graph is reflected about the y -axis and the period increases.

For $k < -1$, the graph is reflected about the y -axis and the period decreases.

Negative angles:

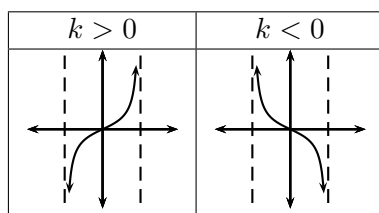
$$\tan(-\theta) = -\tan \theta$$

Calculating the period:

To determine the period of $y = \tan k\theta$ we use,

$$\text{Period} = \frac{180^\circ}{|k|}$$

where $|k|$ is the absolute value of k .



Worked example 26: Tangent function

QUESTION

1. Sketch the following functions on the same set of axes for $-180^\circ \leq \theta \leq 180^\circ$.

a) $y_1 = \tan \theta$

b) $y_2 = \tan \frac{3\theta}{2}$

2. For each function determine the following:

- Period
- Domain and range
- x - and y -intercepts
- Asymptotes

SOLUTION

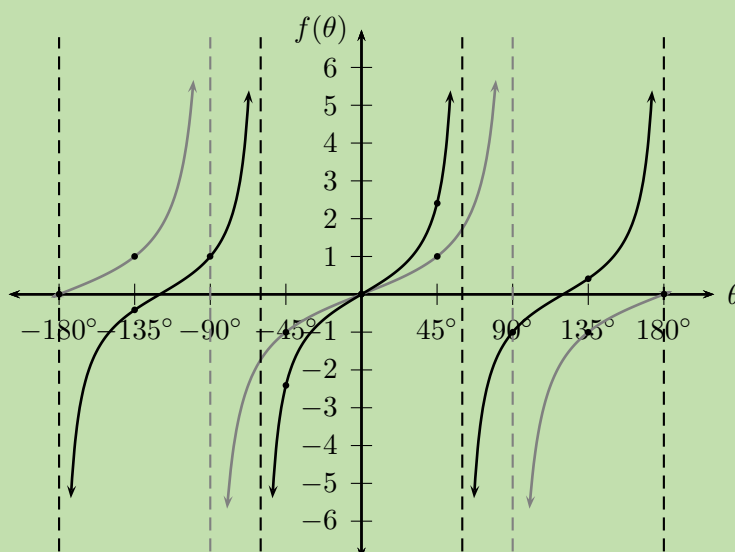
Step 1: Examine the equations of the form $y = \tan k\theta$

Notice that $k > 1$ for $y_2 = \tan \frac{3\theta}{2}$, therefore the period of the graph decreases.

Step 2: Complete a table of values

θ	-180°	-135°	-90°	-45°	0°	45°	90°	135°	180°
$\tan \theta$	0	1	UNDEF	-1	0	1	UNDEF	-1	0
$\tan \frac{3\theta}{2}$	UNDEF	-0,41	1	-2,41	0	2,41	-1	0,41	UNDEF

Step 3: Sketch the tangent graphs



Step 4: Complete the table

	$y_1 = \tan \theta$	$y_2 = \tan \frac{3\theta}{2}$
period	180°	120°
domain	$\{\theta : -180^\circ < \theta < 180^\circ, \theta \neq -90^\circ; 90^\circ\}$	$\{\theta : -180^\circ < \theta < 180^\circ, \theta \neq -60^\circ; 60^\circ\}$
range	$\{f(\theta) : f(\theta) \in \mathbb{R}\}$	$\{f(\theta) : f(\theta) \in \mathbb{R}\}$
y-intercept(s)	$(0^\circ; 0)$	$(0^\circ; 0)$
x-intercept(s)	$(-180^\circ; 0), (0^\circ; 0)$ and $(180^\circ; 0)$	$(-120^\circ; 0), (0^\circ; 0)$ and $(120^\circ; 0)$
asymptotes	$\theta = -90^\circ$ and $\theta = 90^\circ$	$\theta = -180^\circ; -60^\circ$ and 180°

Discovering the characteristics

For functions of the general form: $f(\theta) = y = \tan k\theta$:

Domain and range

The domain of one branch is $\{\theta : -\frac{90^\circ}{k} < \theta < \frac{90^\circ}{k}, \theta \in \mathbb{R}\}$ because $f(\theta)$ is undefined for $\theta = -\frac{90^\circ}{k}$ and $\theta = \frac{90^\circ}{k}$.

The range is $\{f(\theta) : f(\theta) \in \mathbb{R}\}$ or $(-\infty; \infty)$.

Intercepts

The x -intercepts are determined by letting $f(\theta) = 0$ and solving for θ .

The y -intercept is calculated by letting $\theta = 0$ and solving for $f(\theta)$.

$$\begin{aligned}y &= \tan k\theta \\&= \tan 0^\circ \\&= 0\end{aligned}$$

This gives the point $(0^\circ; 0)$.

Asymptotes

These are the values of $k\theta$ for which $\tan k\theta$ is undefined.

Exercise 5 – 29: Tangent functions of the form $y = \tan k\theta$

Sketch the following functions for $-180^\circ \leq \theta \leq 180^\circ$. For each graph determine:

- Period
- Domain and range
- x - and y -intercepts
- Asymptotes

1. $f(\theta) = \tan 2\theta$

2. $g(\theta) = \tan \frac{3\theta}{4}$

3. $h(\theta) = \tan(-2\theta)$

4. $k(\theta) = \tan \frac{2\theta}{3}$

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. 22W6 2. 22W7 3. 22W8 4. 22W9



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We now consider tangent functions of the form $y = \tan(\theta + p)$ and the effects of parameter p .

Investigation: The effects of p on a tangent graph

1. On the same system of axes, plot the following graphs for $-360^\circ \leq \theta \leq 360^\circ$:

- a) $y_1 = \tan \theta$
- b) $y_2 = \tan(\theta - 60^\circ)$
- c) $y_3 = \tan(\theta - 90^\circ)$
- d) $y_4 = \tan(\theta + 60^\circ)$
- e) $y_5 = \tan(\theta + 180^\circ)$

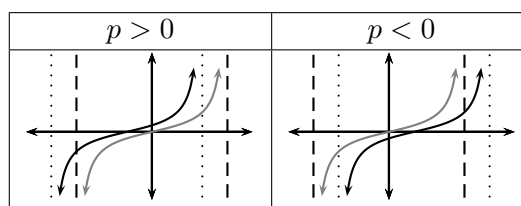
2. Use your sketches of the functions above to complete the following table:

	y_1	y_2	y_3	y_4	y_5
period					
domain					
range					
y-intercept(s)					
x-intercept(s)					
asymptotes					
effect of p					

The effect of the parameter on $y = \tan(\theta + p)$

The effect of p on the tangent function is a horizontal shift (or phase shift); the entire graph slides to the left or to the right.

- For $p > 0$, the graph of the tangent function shifts to the left by p .
- For $p < 0$, the graph of the tangent function shifts to the right by p .



Worked example 27: Tangent function

QUESTION

1. Sketch the following functions on the same set of axes for $-180^\circ \leq \theta \leq 180^\circ$.

a) $y_1 = \tan \theta$

b) $y_2 = \tan(\theta + 30^\circ)$

For each function determine the following:

2.
 - Period
 - Domain and range
 - x - and y -intercepts
 - Asymptotes

SOLUTION

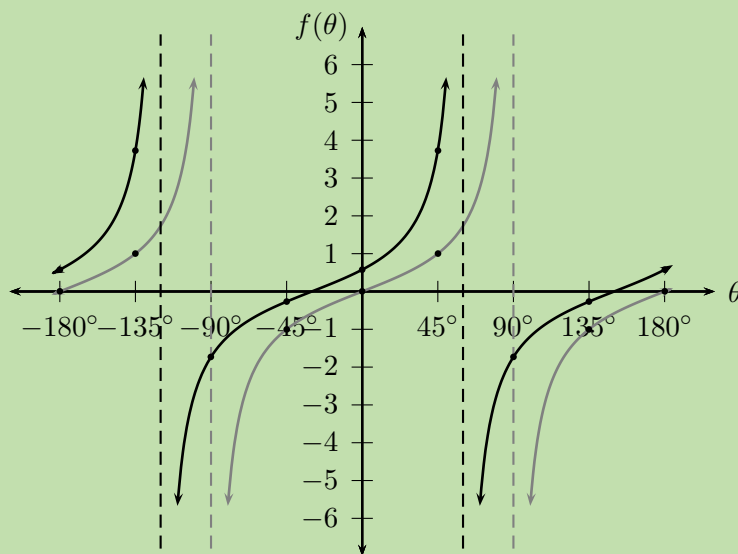
Step 1: Examine the equations of the form $y = \tan(\theta + p)$

Notice that for $y_1 = \tan \theta$ we have $p = 0^\circ$ (no phase shift) and for $y_2 = \tan(\theta + 30^\circ)$, $p > 0$ therefore the graph shifts to the left by 30° .

Step 2: Complete a table of values

θ	-180°	-135°	-90°	-45°	0°	45°	90°	135°	180°
$\tan \theta$	0	1	UNDEF	-1	0	1	UNDEF	-1	0
$\tan(\theta + 30^\circ)$	0,58	3,73	-1,73	-0,27	0,58	3,73	-1,73	-0,27	0,58

Step 3: Sketch the tangent graphs



Step 4: Complete the table

	$y_1 = \tan \theta$	$y_2 = \tan(\theta + 30^\circ)$
period	180°	180°
domain	$\{\theta : -180^\circ \leq \theta \leq 180^\circ, \theta \neq -90^\circ; 90^\circ\}$	$\{\theta : -180^\circ \leq \theta \leq 180^\circ, \theta \neq -120^\circ; 60^\circ\}$
range	$(-\infty; \infty)$	$(-\infty; \infty)$
y-intercept(s)	$(0^\circ; 0)$	$(0^\circ; 0,58)$
x-intercept(s)	$(-180^\circ; 0), (0^\circ; 0)$ and $(180^\circ; 0)$	$(-30^\circ; 0)$ and $(150^\circ; 0)$
asymptotes	$\theta = -90^\circ$ and $\theta = 90^\circ$	$\theta = -120^\circ$ and $\theta = 60^\circ$

Discovering the characteristics

For functions of the general form: $f(\theta) = y = \tan(\theta + p)$:

Domain and range

The domain of one branch is $\{\theta : \theta \in (-90^\circ - p; 90^\circ - p)\}$ because the function is undefined for $\theta = -90^\circ - p$ and $\theta = 90^\circ - p$.

The range is $\{f(\theta) : f(\theta) \in \mathbb{R}\}$.

Intercepts

The x -intercepts are determined by letting $f(\theta) = 0$ and solving for θ .

The y -intercept is calculated by letting $\theta = 0^\circ$ and solving for $f(\theta)$.

$$\begin{aligned}y &= \tan(\theta + p) \\&= \tan(0^\circ + p) \\&= \tan p\end{aligned}$$

This gives the point $(0^\circ; \tan p)$.

Exercise 5 – 30: Tangent functions of the form $y = \tan(\theta + p)$

Sketch the following functions for $-360^\circ \leq \theta \leq 360^\circ$.

For each function, determine the following:

- Period
- Domain and range
- x - and y -intercepts
- Asymptotes

1. $f(\theta) = \tan(\theta + 45^\circ)$

2. $g(\theta) = \tan(\theta - 30^\circ)$

3. $h(\theta) = \tan(\theta + 60^\circ)$

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1. 22WB 2. 22WC 3. 22WD



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Worked example 28: Sketching a tangent graph**QUESTION**

Sketch the graph of $f(\theta) = \tan \frac{1}{2}(\theta - 30^\circ)$ for $-180^\circ \leq \theta \leq 180^\circ$.

SOLUTION**Step 1: Examine the form of the equation**

From the equation we see that $0 < k < 1$, therefore the branches of the graph will be less steep than the standard tangent graph $y = \tan \theta$. We also notice that $p < 0$ so the graph will be shifted to the right on the x -axis.

Step 2: Determine the period

The period for $f(\theta) = \tan \frac{1}{2}(\theta - 30^\circ)$ is:

$$\begin{aligned}\text{Period} &= \frac{180^\circ}{|k|} \\ &= \frac{180^\circ}{\frac{1}{2}} \\ &= 360^\circ\end{aligned}$$

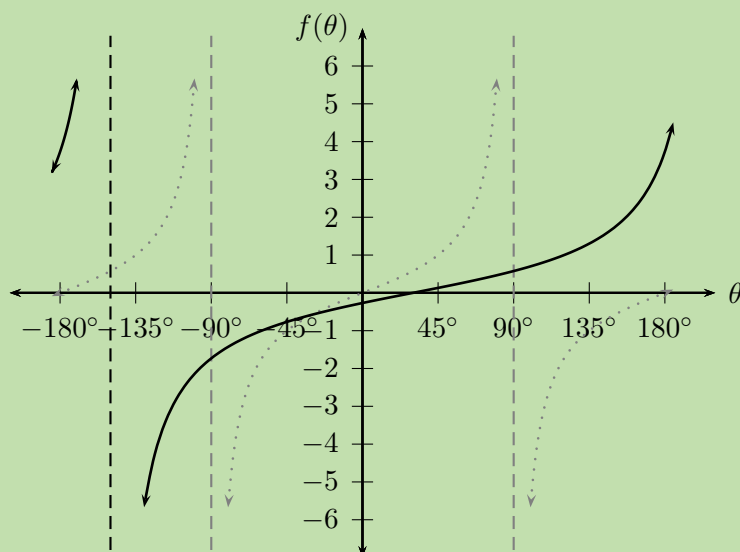
Step 3: Determine the asymptotes

The standard tangent graph, $y = \tan \theta$, for $-180^\circ \leq \theta \leq 180^\circ$ is undefined at $\theta = -90^\circ$ and $\theta = 90^\circ$. Therefore we can determine the asymptotes of $f(\theta) = \tan \frac{1}{2}(\theta - 30^\circ)$:

- $\frac{-90^\circ}{0,5} + 30^\circ = -150^\circ$
- $\frac{90^\circ}{0,5} + 30^\circ = 210^\circ$

The asymptote at $\theta = 210^\circ$ lies outside the required interval.

Step 4: Plot the points and join with a smooth curve



Period: 360°

Domain: $\{\theta : -180^\circ \leq \theta \leq 180^\circ, \theta \neq -150^\circ\}$

Range: $(-\infty; \infty)$

y -intercepts: $(0^\circ; -0,27)$

x -intercept: $(30^\circ; 0)$

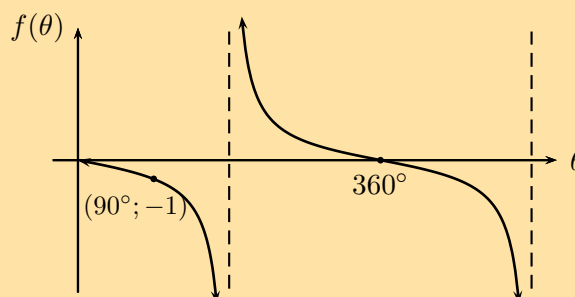
Asymptotes: $\theta = -150^\circ$

Exercise 5 – 31: The tangent function

1. Sketch the following graphs on separate axes:

- $y = \tan \theta - 1$ for $-90^\circ \leq \theta \leq 90^\circ$
- $f(\theta) = -\tan 2\theta$ for $0^\circ \leq \theta \leq 90^\circ$
- $y = \frac{1}{2} \tan(\theta + 45^\circ)$ for $0^\circ \leq \theta \leq 360^\circ$
- $y = \tan(30^\circ - \theta)$ for $-180^\circ \leq \theta \leq 180^\circ$

2. Given the graph of $y = a \tan k\theta$, determine the values of a and k .



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1a. [22WF](#) 1b. [22WG](#) 1c. [22WH](#) 1d. [22WJ](#) 2. [22WK](#)



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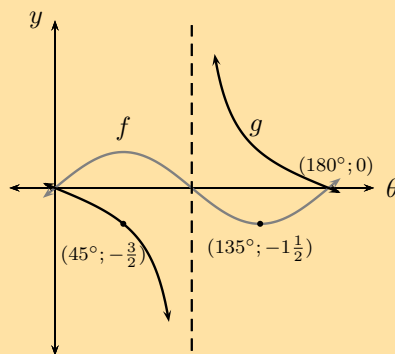


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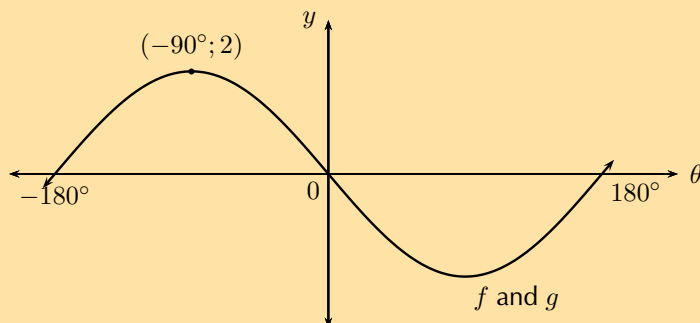
Exercise 5 – 32: Mixed exercises

1. Determine the equation for each of the following:

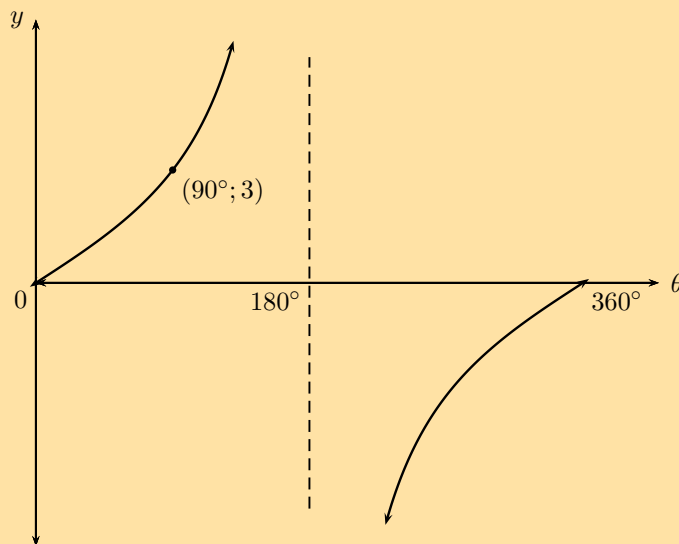
a) $f(\theta) = a \sin k\theta$ and $g(\theta) = a \tan \theta$



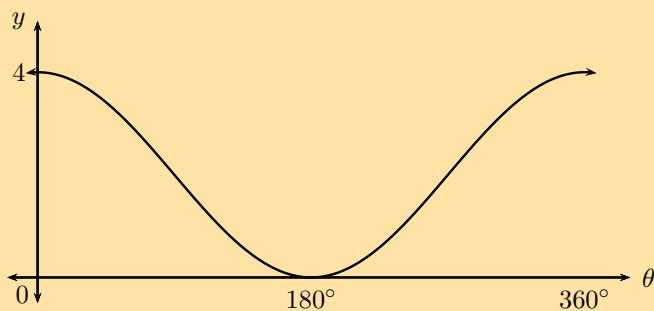
b) $f(\theta) = a \sin k\theta$ and $g(\theta) = a \cos(\theta + p)$



c) $y = a \tan k\theta$



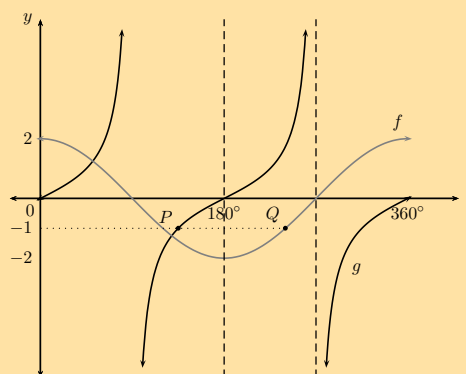
d) $y = a \cos \theta + q$



2. Given the functions $f(\theta) = 2 \sin \theta$ and $g(\theta) = \cos \theta + 1$:

- Sketch the graphs of both functions on the same system of axes, for $0^\circ \leq \theta \leq 360^\circ$. Indicate the turning points and intercepts on the diagram.
- What is the period of f ?
- What is the amplitude of g ?
- Use your sketch to determine how many solutions there are for the equation $2 \sin \theta - \cos \theta = 1$. Give one of the solutions.
- Indicate on your sketch where on the graph the solution to $2 \sin \theta = -1$ is found.

3. The sketch shows the two functions $f(\theta) = a \cos \theta$ and $g(\theta) = \tan \theta$ for $0^\circ \leq \theta \leq 360^\circ$. Points $P(135^\circ; b)$ and $Q(c; -1)$ lie on $g(\theta)$ and $f(\theta)$ respectively.



- Determine the values of a , b and c .
 - What is the period of g ?
 - Solve the equation $\cos \theta = \frac{1}{2}$ graphically and show your answer(s) on the diagram.
 - Determine the equation of the new graph if g is reflected about the x -axis and shifted to the right by 45° .
4. Sketch the graphs of $y_1 = -\frac{1}{2} \sin(\theta + 30^\circ)$ and $y_2 = \cos(\theta - 60^\circ)$, on the same system of axes for $0^\circ \leq \theta \leq 360^\circ$.

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1a. 22WM 1b. 22WN 1c. 22WP 1d. 22WQ 2. 22WR 3. 22WS
4. 22WT



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▶ See presentation: 22WV at www.everythingmaths.co.za

1. Parabolic functions:

Standard form: $y = ax^2 + bx + c$

- y -intercept: $(0; c)$
- x -intercept: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- Turning point: $\left(-\frac{b}{2a}; -\frac{b^2}{4a} + c\right)$
- Axis of symmetry: $x = -\frac{b}{2a}$

Completed square form: $y = a(x + p)^2 + q$

- Turning point: $(-p; q)$
- $p > 0$: horizontal shift left
- $p < 0$: horizontal shift right
- $q > 0$: vertical shift up
- $q < 0$: vertical shift down

2. Average gradient:

- Average gradient = $\frac{y_2 - y_1}{x_2 - x_1}$

3. Hyperbolic functions:

Standard form: $y = \frac{k}{x}$

- $k > 0$: first and third quadrant
- $k < 0$: second and fourth quadrant

Shifted form: $y = \frac{k}{x+p} + q$

- $p > 0$: horizontal shift left
- $p < 0$: horizontal shift right
- $q > 0$: vertical shift up
- $q < 0$: vertical shift down
- Asymptotes: $x = -p$ and $y = q$

4. Exponential functions:

Standard form: $y = ab^x$

- $a > 0$: above x -axis
- $a < 0$: below x -axis
- $b > 1$: increasing function if $a > 0$; decreasing function if $a < 0$
- $0 < b < 1$: decreasing function if $a > 0$; increasing function if $a < 0$

Shifted form: $y = ab^{(x+p)} + q$

- $p > 0$: horizontal shift left
- $p < 0$: horizontal shift right

- $q > 0$: vertical shift up
- $q < 0$: vertical shift down
- Asymptotes: $y = q$

5. Sine functions:

Shifted form: $y = a \sin(k\theta + p) + q$

- Period = $\frac{360^\circ}{|k|}$
- $k > 1$ or $k < -1$: period decreases
- $0 < k < 1$ or $-1 < k < 0$: period increases
- $p > 0$: horizontal shift left
- $p < 0$: horizontal shift right
- $q > 0$: vertical shift up
- $q < 0$: vertical shift down
- $\sin(-\theta) = -\sin \theta$

6. Cosine functions:

Shifted form: $y = a \cos(k\theta + p) + q$

- Period = $\frac{360^\circ}{|k|}$
- $k > 1$ or $k < -1$: period decreases
- $0 < k < 1$ or $-1 < k < 0$: period increases
- $p > 0$: horizontal shift left
- $p < 0$: horizontal shift right
- $q > 0$: vertical shift up
- $q < 0$: vertical shift down
- $\cos(-\theta) = \cos \theta$

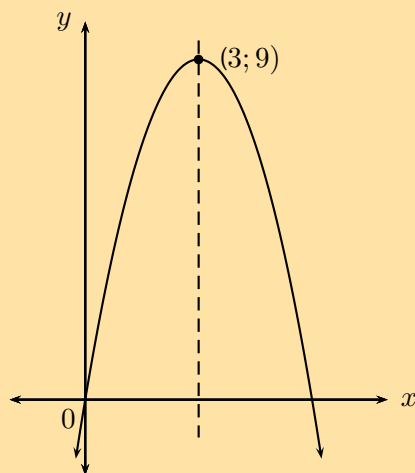
7. Tangent functions:

Shifted form: $y = a \tan(k\theta + p) + q$

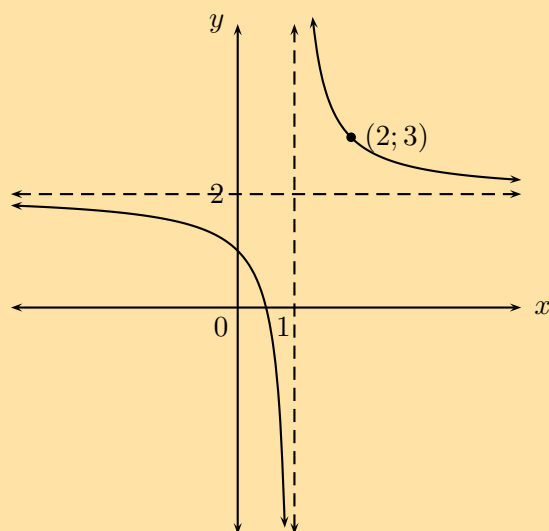
- Period = $\frac{180^\circ}{|k|}$
- $k > 1$ or $k < -1$: period decreases
- $0 < k < 1$ or $-1 < k < 0$: period increases
- $p > 0$: horizontal shift left
- $p < 0$: horizontal shift right
- $q > 0$: vertical shift up
- $q < 0$: vertical shift down
- $\tan(-\theta) = -\tan \theta$
- Asymptotes: $\frac{90^\circ - p}{k} \pm \frac{180^\circ n}{k}, n \in \mathbb{Z}$

Exercise 5 – 33: End of chapter exercises

1. Show that if $a < 0$, then the range of $f(x) = a(x+p)^2 + q$ is $\{f(x) : f(x) \in (-\infty, q]\}$.
2. If $(2; 7)$ is the turning point of $f(x) = -2x^2 - 4ax + k$, find the values of the constants a and k .
3. The following graph is represented by the equation $f(x) = ax^2 + bx$. The coordinates of the turning point are $(3; 9)$. Show that $a = -1$ and $b = 6$.



4. Given: $f(x) = x^2 - 2x + 3$. Give the equation of the new graph originating if:
 - a) the graph of f is moved three units to the left.
 - b) the x -axis is moved down three units.
5. A parabola with turning point $(-1; -4)$ is shifted vertically by 4 units upwards. What are the coordinates of the turning point of the shifted parabola?
6. Plot the graph of the hyperbola defined by $y = \frac{2}{x}$ for $-4 \leq x \leq 4$. Suppose the hyperbola is shifted 3 units to the right and 1 unit down. What is the new equation then?
7. Based on the graph of $y = \frac{k}{(x+p)} + q$, determine the equation of the graph with asymptotes $y = 2$ and $x = 1$ and passing through the point $(2; 3)$.



8. The columns in the table below give the y -values for the following functions: $y = a^x$, $y = a^{x+1}$ and $y = a^x + 1$. Match each function to the correct column.

x	A	B	C
-2	7,25	6,25	2,5
-1	3,5	2,5	1
0	2	1	0,4
1	1,4	0,4	0,16
2	1,16	0,16	0,064

9. The graph of $f(x) = 1 + a \cdot 2^x$ (a is a constant) passes through the origin.
- Determine the value of a .
 - Determine the value of $f(-15)$ correct to five decimal places.
 - Determine the value of x , if $P(x; 0,5)$ lies on the graph of f .
 - If the graph of f is shifted 2 units to the right to give the function h , write down the equation of h .
10. The graph of $f(x) = a \cdot b^x$ ($a \neq 0$) has the point $P(2; 144)$ on f .
- If $b = 0,75$, calculate the value of a .
 - Hence write down the equation of f .
 - Determine, correct to two decimal places, the value of $f(13)$.
 - Describe the transformation of the curve of f to h if $h(x) = f(-x)$.
11. Using your knowledge of the effects of p and k draw a rough sketch of the following graphs without a table of values.
- $y = \sin 3\theta$ for $-180^\circ \leq \theta \leq 180^\circ$
 - $y = -\cos 2\theta$ for $0^\circ \leq \theta \leq 180^\circ$
 - $y = \tan \frac{1}{2}\theta$ for $0^\circ \leq \theta \leq 360^\circ$
 - $y = \sin(\theta - 45^\circ)$ for $-360^\circ \leq \theta \leq 360^\circ$
 - $y = \cos(\theta + 45^\circ)$ for $0^\circ \leq \theta \leq 360^\circ$
 - $y = \tan(\theta - 45^\circ)$ for $0^\circ \leq \theta \leq 360^\circ$
 - $y = 2 \sin 2\theta$ for $-180^\circ \leq \theta \leq 180^\circ$
 - $y = \sin(\theta + 30^\circ) + 1$ for $-360^\circ \leq \theta \leq 0^\circ$

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. 22WW 2. 22WX 3. 22WY 4. 22WZ 5. 22X2 6. 22X3
 7. 22X4 8. 22X5 9. 22X6 10. 22X7 11a. 22X8 11b. 22X9
 11c. 22XB 11d. 22XC 11e. 22XD 11f. 22XF 11g. 22XC 11h. 22XH



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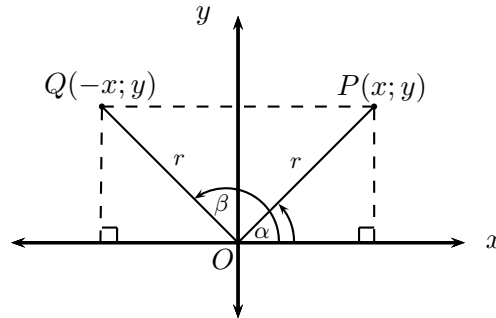
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Trigonometry

6.1	<i>Revision</i>	240
6.2	<i>Trigonometric identities</i>	247
6.3	<i>Reduction formula</i>	253
6.4	<i>Trigonometric equations</i>	266
6.5	<i>Area, sine, and cosine rules</i>	280
6.6	<i>Summary</i>	301

6.1 Revision

EMBHG

Trigonometric ratios

We plot the points $P(x; y)$ and $Q(-x; y)$ in the Cartesian plane and measure the angles from the positive x -axis to the terminal arms (OP and OQ).

$P(x; y)$ lies in the first quadrant with $\angle POX = \alpha$ and $Q(-x; y)$ lies in the second quadrant with $\angle QOX = \beta$.

Using the theorem of Pythagoras we have that

$$\begin{aligned} OP^2 &= x^2 + y^2 \\ \text{And } OQ^2 &= (-x)^2 + y^2 \\ &= x^2 + y^2 \\ \therefore OP &= OQ \end{aligned}$$

Let $OP = OQ = r$.

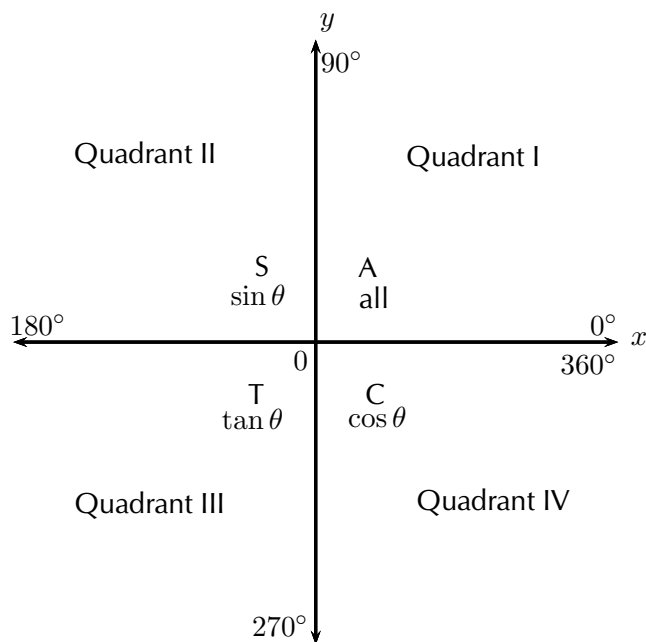
Trigonometric ratios

$$\begin{aligned} \sin \alpha &= \frac{y}{r} \\ \cos \alpha &= \frac{x}{r} \\ \tan \alpha &= \frac{y}{x} \end{aligned}$$

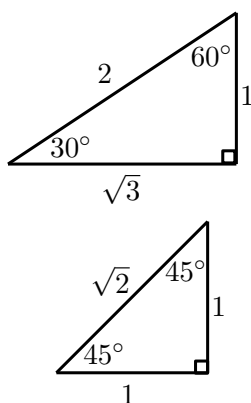
In the second quadrant we notice that $-x < 0$

$$\begin{aligned} \sin \beta &= \frac{y}{r} \\ \cos \beta &= -\frac{x}{r} \\ \tan \beta &= -\frac{y}{x} \end{aligned}$$

Similarly, in the third and fourth quadrants the sign of the trigonometric ratios depends on the signs of x and y :



Special angles

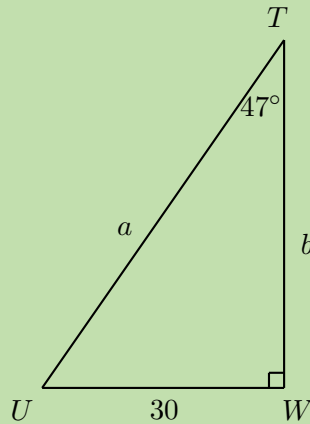


θ	0°	30°	45°	60°	90°
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undef

► See video: [22XJ](#) at www.everythingmaths.co.za

Worked example 1: Solving equations**QUESTION**

Determine the values of a and b in the right-angled triangle TUW (correct to one decimal place):

**SOLUTION**

Step 1: Identify the opposite and adjacent sides and the hypotenuse

Step 2: Determine the value of a

$$\begin{aligned}\sin \theta &= \frac{\text{opposite side}}{\text{hypotenuse}} \\ \sin 47^\circ &= \frac{30}{a} \\ a &= \frac{30}{\sin 47^\circ} \\ \therefore a &= 41,0\end{aligned}$$

Step 3: Determine the value of b

Always try to use the information that is given for calculations and not answers that you have worked out in case you have made an error. For example, avoid using $a = 41,0$ to determine the value of b .

$$\begin{aligned}\tan \theta &= \frac{\text{opposite side}}{\text{adjacent side}} \\ \tan 47^\circ &= \frac{30}{b} \\ b &= \frac{30}{\tan 47^\circ} \\ \therefore b &= 28,0\end{aligned}$$

Step 4: Write the final answer

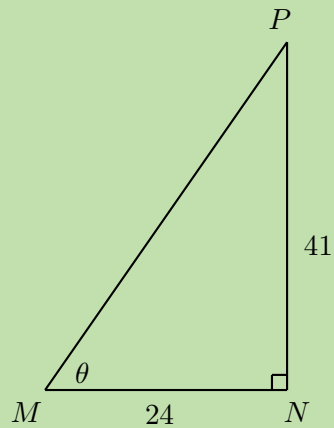
$a = 41,0$ units and $b = 28,0$ units.

Finding an angle

Worked example 2: Finding an angle

QUESTION

Calculate the value of θ in the right-angled triangle MNP (correct to one decimal place):



SOLUTION

Step 1: Identify the opposite and adjacent sides and the hypotenuse

Step 2: Determine the value of θ

$$\tan \theta = \frac{\text{opposite side}}{\text{adjacent side}}$$

$$\tan \theta = \frac{41}{24}$$

$$\therefore \theta = \tan^{-1} \left(\frac{41}{24} \right)$$

$$\theta = 59,7^\circ$$

Worked example 3: Finding an angle

QUESTION

Given $2 \sin \frac{\theta}{2} = \cos 43^\circ$, for $\theta \in [0^\circ; 90^\circ]$, determine the value of θ (correct to one decimal place).

SOLUTION

Step 1: Simplify the equation

Avoiding rounding off in calculations until you have determined the final answer. In the calculation below, the dots indicate that the number has not been rounded so that the answer is as accurate as possible.

$$\begin{aligned} 2 \sin \frac{\theta}{2} &= \cos 43^\circ \\ \sin \frac{\theta}{2} &= \frac{\cos 43^\circ}{2} \\ \frac{\theta}{2} &= \sin^{-1}(0,365 \dots) \\ \theta &= 2(21,449 \dots) \\ \therefore \theta &= 42,9^\circ \end{aligned}$$

Two-dimensional problems

Worked example 4: Flying a kite

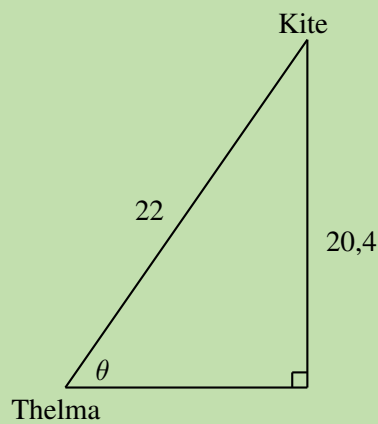
QUESTION

Thelma flies a kite on a 22 m piece of string and the height of the kite above the ground is 20,4 m. Determine the angle of inclination of the string (correct to one decimal place).

SOLUTION

Step 1: Draw a sketch and identify the opposite and adjacent sides and the hypotenuse

Let the angle of inclination of the string be θ .



Step 2: Use an appropriate trigonometric ratio to find θ

$$\begin{aligned}\sin \theta &= \frac{\text{opposite side}}{\text{hypotenuse}} \\ &= \frac{20,4}{22} \\ \theta &= \sin^{-1}(0,927 \dots) \\ \therefore \theta &= 68,0^\circ\end{aligned}$$

Exercise 6 – 1: Revision

1. If $p = 49^\circ$ and $q = 32^\circ$, use a calculator to determine whether the following statements are true or false:

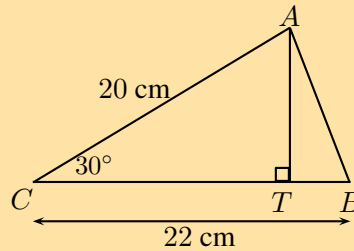
- a) $\sin p + 3 \sin p = 4 \sin p$
- b) $\frac{\sin q}{\cos q} = \tan q$
- c) $\cos(p - q) = \cos p - \cos q$
- d) $\sin(2p) = 2 \sin p \cos p$

2. Determine the following angles (correct to one decimal place):

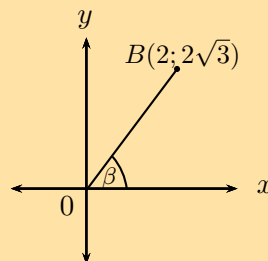
- a) $\cos \alpha = 0,64$
- b) $\sin \theta + 2 = 2,65$
- c) $\frac{1}{2} \cos 2\beta = 0,3$
- d) $\tan \frac{\theta}{3} = \sin 48^\circ$
- e) $\cos 3p = 1,03$
- f) $2 \sin 3\beta + 1 = 2,6$
- g) $\frac{\sin \theta}{\cos \theta} = 4\frac{2}{3}$

3. In $\triangle ABC$, $\hat{ACB} = 30^\circ$, $AC = 20$ cm and $BC = 22$ cm. The perpendicular line from A intersects BC at T .

Determine:



- a) the length TC
 - b) the length AT
 - c) the angle $\hat{BAT}$
4. A rhombus has a perimeter of 40 cm and one of the internal angles is 30° .
 - a) Determine the length of the sides.
 - b) Determine the lengths of the diagonals.
 - c) Calculate the area of the rhombus.
 5. Simplify the following without using a calculator:
 - a) $2 \sin 45^\circ \times 2 \cos 45^\circ$
 - b) $\cos^2 30^\circ - \sin^2 60^\circ$
 - c) $\sin 60^\circ \cos 30^\circ - \cos 60^\circ \sin 30^\circ - \tan 45^\circ$
 - d) $4 \sin 60^\circ \cos 30^\circ - 2 \tan 45^\circ + \tan 60^\circ - 2 \sin 60^\circ$
 - e) $\sin 60^\circ \times \sqrt{2 \tan 45^\circ + 1} - \sin 30^\circ$
 6. Given the diagram below.



Determine the following without using a calculator:

- a) β
 - b) $\cos \beta$
 - c) $\cos^2 \beta + \sin^2 \beta$
7. The 10 m ladder of a fire truck leans against the wall of a burning building at an angle of 60° . The height of an open window is 9 m from the ground. Will the ladder reach the window?

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- | | | | | | |
|----------|----------|----------|----------|----------|----------|
| 1. 22XK | 2a. 22XM | 2b. 22XN | 2c. 22XP | 2d. 22XQ | 2e. 22XR |
| 2f. 22XS | 2g. 22XT | 3. 22XV | 4. 22XW | 5a. 22XX | 5b. 22XY |
| 5c. 22XZ | 5d. 22Y2 | 5e. 22Y3 | 6. 22Y4 | 7. 22Y5 | |



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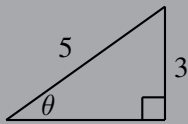
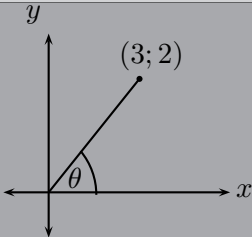
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An identity is a mathematical statement that equates one quantity with another. Trigonometric identities allow us to simplify a given expression so that it contains sine and cosine ratios only. This enables us to solve equations and also to prove other identities.

Quotient identity

Investigation: Quotient identity

1. Complete the table without using a calculator, leaving your answer in surd form where applicable:

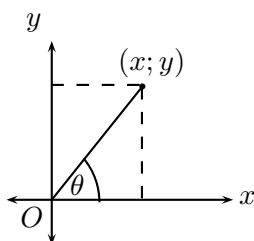
	$\theta = 45^\circ$		
$\sin \theta$			
$\cos \theta$			
$\frac{\sin \theta}{\cos \theta}$			
$\tan \theta$			

2. Examine the last two rows of the table and make a conjecture.
3. Are there any values of θ for which your conjecture would not be true? Explain your answer.

We know that $\tan \theta$ is defined as:

$$\tan \theta = \frac{\text{opposite side}}{\text{adjacent side}}$$

Using the diagram below and the theorem of Pythagoras, we can write the tangent function in terms of x , y and r :

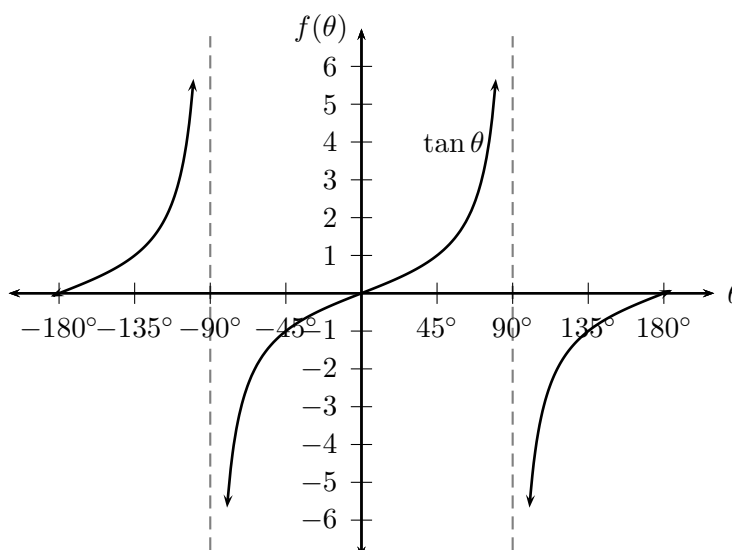


$$\begin{aligned}
 \tan \theta &= \frac{y}{x} \\
 &= \frac{y}{x} \times \frac{r}{r} \\
 &= \frac{y}{r} \times \frac{r}{x} \\
 &= \frac{y}{r} \div \frac{x}{r} \\
 &= \sin \theta \div \cos \theta \\
 &= \frac{\sin \theta}{\cos \theta}
 \end{aligned}$$

This is the quotient identity:

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Notice that $\tan \theta$ is undefined if $\cos \theta = 0$, therefore $\theta \neq k \times 90^\circ$, where k is an odd integer.



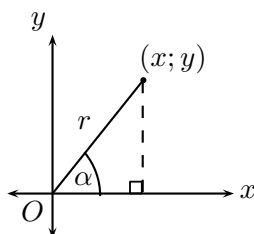
Square identity

Investigation: Square identity

1. Use a calculator to complete the following table:

$\sin^2 80^\circ + \cos^2 80^\circ =$	
$\cos^2 23^\circ + \sin^2 23^\circ =$	
$\sin 50^\circ + \cos 50^\circ =$	
$\sin^2 67^\circ - \cos^2 67^\circ =$	
$\sin^2 67^\circ + \cos^2 67^\circ =$	

2. What do you notice? Make a conjecture.
3. Draw a sketch and prove your conjecture in general terms, using x , y and r .



Using the theorem of Pythagoras, we can write the sine and cosine functions in terms of x , y and r :

$$\begin{aligned}\sin^2 \theta + \cos^2 \theta &= \left(\frac{y}{r}\right)^2 + \left(\frac{x}{r}\right)^2 \\ &= \frac{y^2}{r^2} + \frac{x^2}{r^2} \\ &= \frac{y^2 + x^2}{r^2} \\ &= \frac{r^2}{r^2} \\ &= 1\end{aligned}$$

This is the square identity:

$$\sin^2 \theta + \cos^2 \theta = 1$$

Other forms of the square identity

Complete the following:

1. $\sin^2 \theta = 1 - \dots\dots$
2. $\cos \theta = \pm \sqrt{\dots\dots}$
3. $\sin^2 \theta = (1 + \dots\dots)(1 - \dots\dots)$
4. $\cos^2 \theta - 1 = \dots\dots$

Here are some useful tips for proving identities:

- Change all trigonometric ratios to sine and cosine.
- Choose one side of the equation to simplify and show that it is equal to the other side.
- Usually it is better to choose the more complicated side to simplify.
- Sometimes we need to simplify both sides of the equation to show that they are equal.
- A square root sign often indicates that we need to use the square identity.
- We can also add to the expression to make simplifying easier:
 - replace 1 with $\sin^2 \theta + \cos^2 \theta$.
 - multiply by 1 in the form of a suitable fraction, for example $\frac{1 + \sin \theta}{1 + \sin \theta}$.

Worked example 5: Trigonometric identities

QUESTION

Simplify the following:

1. $\tan^2 \theta \times \cos^2 \theta$

2. $\frac{1}{\cos^2 \theta} - \tan^2 \theta$

SOLUTION

Step 1: Write the expression in terms of sine and cosine only

We use the square and quotient identities to write the given expression in terms of sine and cosine and then simplify as far as possible.

1.

$$\begin{aligned}\tan^2 \theta \times \cos^2 \theta &= \left(\frac{\sin \theta}{\cos \theta} \right)^2 \times \cos^2 \theta \\ &= \frac{\sin^2 \theta}{\cos^2 \theta} \times \cos^2 \theta \\ &= \sin^2 \theta\end{aligned}$$

2.

$$\begin{aligned}\frac{1}{\cos^2 \theta} - \tan^2 \theta &= \frac{1}{\cos^2 \theta} - \left(\frac{\sin \theta}{\cos \theta} \right)^2 \\ &= \frac{1}{\cos^2 \theta} - \frac{\sin^2 \theta}{\cos^2 \theta} \\ &= \frac{1 - \sin^2 \theta}{\cos^2 \theta} \\ &= \frac{\cos^2 \theta}{\cos^2 \theta} \\ &= 1\end{aligned}$$

QUESTION

Prove: $\frac{1 - \sin \alpha}{\cos \alpha} = \frac{\cos \alpha}{1 + \sin \alpha}$

SOLUTION

Step 1: Note restrictions

When working with fractions, we must be careful that the denominator does not equal 0. Therefore $\cos \theta \neq 0$ for the fraction on the left-hand side and $\sin \theta + 1 \neq 0$ for the fraction on the right-hand side.

Step 2: Simplify the left-hand side

This is not an equation that needs to be solved. We are required to show that one side of the equation is equal to the other. We can choose either of the two sides to simplify.

$$\begin{aligned}\text{LHS} &= \frac{1 - \sin \alpha}{\cos \alpha} \\ &= \frac{1 - \sin \alpha}{\cos \alpha} \times \frac{1 + \sin \alpha}{1 + \sin \alpha}\end{aligned}$$

Notice that we have not changed the equation — this is the same as multiplying by 1 since the numerator and the denominator are the same.

Step 3: Determine the lowest common denominator and simplify

$$\begin{aligned}\text{LHS} &= \frac{1 - \sin^2 \alpha}{\cos \alpha(1 + \sin \alpha)} \\ &= \frac{\cos^2 \alpha}{\cos \alpha(1 + \sin \alpha)} \\ &= \frac{\cos \alpha}{1 + \sin \alpha} \\ &= \text{RHS}\end{aligned}$$

Exercise 6 – 2: Trigonometric identities

1. Reduce the following to one trigonometric ratio:

a) $\frac{\sin \alpha}{\tan \alpha}$

b) $\cos^2 \theta \tan^2 \theta + \tan^2 \theta \sin^2 \theta$

c) $1 - \sin \theta \cos \theta \tan \theta$

d) $\left(\frac{1 - \cos^2 \beta}{\cos^2 \beta} \right) - \tan^2 \beta$

2. Prove the following identities and state restrictions where appropriate:

a) $\frac{1 + \sin \theta}{\cos \theta} = \frac{\cos \theta}{1 - \sin \theta}$

b) $\sin^2 \alpha + (\cos \alpha - \tan \alpha)(\cos \alpha + \tan \alpha) = 1 - \tan^2 \alpha$

c) $\frac{1}{\cos \theta} - \frac{\cos \theta \tan^2 \theta}{1} = \cos \theta$

d) $\frac{2 \sin \theta \cos \theta}{\sin \theta + \cos \theta} = \sin \theta + \cos \theta - \frac{1}{\sin \theta + \cos \theta}$

e) $\left(\frac{\cos \beta}{\sin \beta} + \tan \beta \right) \cos \beta = \frac{1}{\sin \beta}$

f) $\frac{1}{1 + \sin \theta} + \frac{1}{1 - \sin \theta} = d \frac{2 \tan \theta}{\sin \theta \cos \theta}$

g) $\frac{(1 + \tan^2 \alpha) \cos \alpha}{(1 - \tan \alpha)} = \frac{1}{\cos \alpha - \sin \alpha}$

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1a. 22Y7 1b. 22Y8 1c. 22Y9 1d. 22YB 2a. 22YC 2b. 22YD
2c. 22YF 2d. 22YG 2e. 22YH 2f. 22YJ 2g. 22YK



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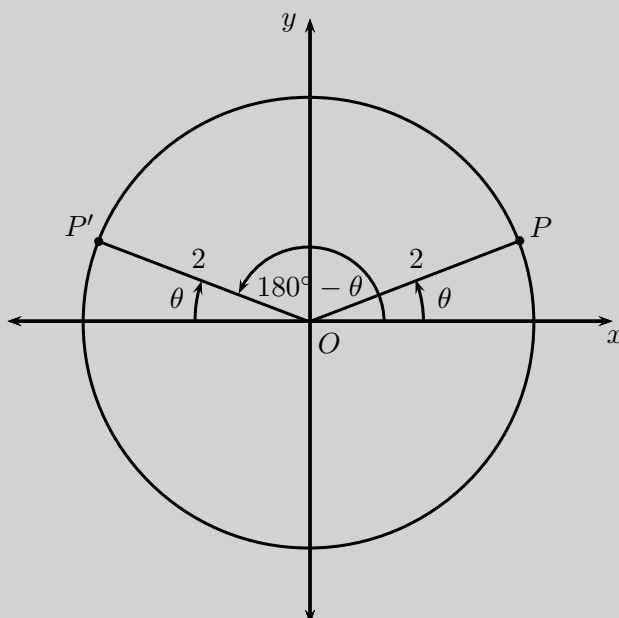
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Any trigonometric function whose argument is $90^\circ \pm \theta$; $180^\circ \pm \theta$ and $360^\circ \pm \theta$ can be written simply in terms of θ .

Investigation: Reduction formulae for function values of $180^\circ \pm \theta$

1. Function values of $180^\circ - \theta$

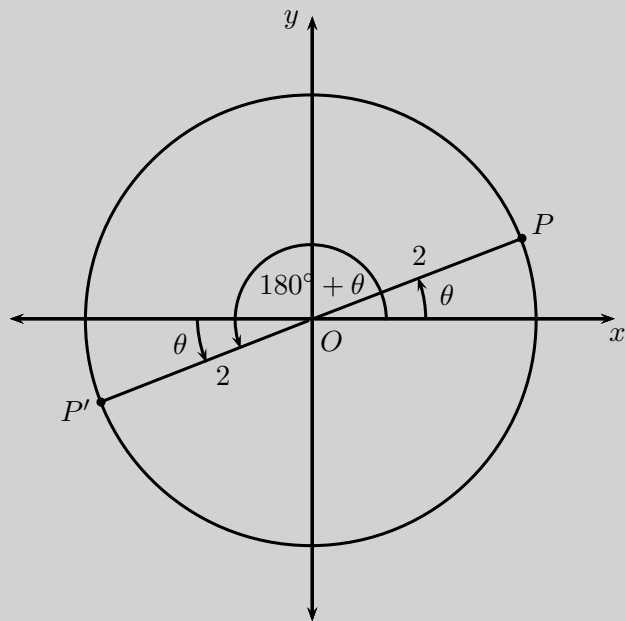
In the figure $P(\sqrt{3}; 1)$ and P' lie on the circle with radius 2. OP makes an angle $\theta = 30^\circ$ with the x -axis.



- If points P and P' are symmetrical about the y -axis, determine the coordinates of P' .
- Write down values for $\sin \theta$, $\cos \theta$ and $\tan \theta$.
- Use the coordinates for P' to determine $\sin(180^\circ - \theta)$, $\cos(180^\circ - \theta)$, $\tan(180^\circ - \theta)$.
- From your results determine a relationship between the trigonometric function values of $(180^\circ - \theta)$ and θ .

2. Function values of $180^\circ + \theta$

In the figure $P(\sqrt{3}; 1)$ and P' lie on the circle with radius 2. OP makes an angle $\theta = 30^\circ$ with the x -axis.



- If points P and P' are symmetrical about the origin (the two points are symmetrical about both the x -axis and the y -axis), determine the coordinates of P' .
- Use the coordinates for P' to determine $\sin(180^\circ + \theta)$, $\cos(180^\circ + \theta)$ and $\tan(180^\circ + \theta)$.
- From your results determine a relationship between the trigonometric function values of $(180^\circ + \theta)$ and θ .

3. Complete the following reduction formulae:

- $\sin(180^\circ - \theta) = \dots\dots$
- $\cos(180^\circ - \theta) = \dots\dots$
- $\tan(180^\circ - \theta) = \dots\dots$
- $\sin(180^\circ + \theta) = \dots\dots$
- $\cos(180^\circ + \theta) = \dots\dots$
- $\tan(180^\circ + \theta) = \dots\dots$

Worked example 7: Reduction formulae for function values of $180^\circ \pm \theta$

QUESTION

Write the following as a single trigonometric ratio:

$$\frac{\sin 163^\circ}{\cos 197^\circ} + \tan 17^\circ + \cos(180^\circ - \theta) \times \tan(180^\circ + \theta)$$

SOLUTION

Step 1: Use reduction formulae to write the trigonometric function values in terms of acute angles and θ

$$= \frac{\sin(180^\circ - 17^\circ)}{\cos(180^\circ + 17^\circ)} + \tan 17^\circ + (-\cos \theta) \times \tan \theta$$

Step 2: Simplify

$$\begin{aligned} &= \frac{\sin 17^\circ}{-\cos 17^\circ} + \tan 17^\circ - \cos \theta \times \frac{\sin \theta}{\cos \theta} \\ &= -\tan 17^\circ + \tan 17^\circ - \sin \theta \\ &= -\sin \theta \end{aligned}$$

Exercise 6 – 3: Reduction formulae for function values of $180^\circ \pm \theta$

1. Determine the value of the following expressions without using a calculator:

- a) $\tan 150^\circ \sin 30^\circ - \cos 210^\circ$
- b) $(1 + \cos 120^\circ)(1 - \sin^2 240^\circ)$
- c) $\cos^2 140^\circ + \sin^2 220^\circ$

2. Write the following in terms of a single trigonometric ratio:

- a) $\tan(180^\circ - \theta) \times \sin(180^\circ + \theta)$
- b) $\frac{\tan(180^\circ + \theta) \cos(180^\circ - \theta)}{\sin(180^\circ - \theta)}$

3. If $t = \tan 40^\circ$, express the following in terms of t :

a) $\tan 140^\circ + 3 \tan 220^\circ$

b) $\frac{\cos 220^\circ}{\sin 140^\circ}$

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1a. 22YM 1b. 22YN 1c. 22YP 2a. 22YQ 2b. 22YR 3. 22YS



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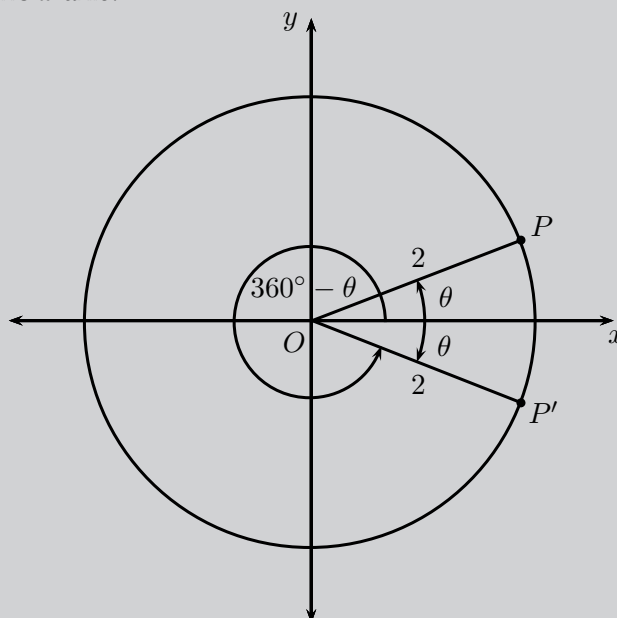
Investigation: Reduction formulae for function values of $(360^\circ \pm \theta)$ and $(-\theta)$

1. Function values of $(360^\circ - \theta)$ and $(-\theta)$

In the Cartesian plane we measure angles from the positive x -axis to the terminal arm, which means that an anti-clockwise rotation gives a positive angle. We can therefore measure negative angles by rotating in a clockwise direction.

For an acute angle θ , we know that $-\theta$ will lie in the fourth quadrant.

In the figure $P(\sqrt{3}; 1)$ and P' lie on the circle with radius 2. OP makes an angle $\theta = 30^\circ$ with the x -axis.



- If points P and P' are symmetrical about the x -axis ($y = 0$), determine the coordinates of P' .
- Use the coordinates of P' to determine $\sin(360^\circ - \theta)$, $\cos(360^\circ - \theta)$ and $\tan(360^\circ - \theta)$.
- Use the coordinates of P' to determine $\sin(-\theta)$, $\cos(-\theta)$ and $\tan(-\theta)$.
- From your results determine a relationship between the function values of $(360^\circ - \theta)$ and $-\theta$.

e) Complete the following reduction formulae:

- i. $\sin(360^\circ - \theta) = \dots\dots$
- ii. $\cos(360^\circ - \theta) = \dots\dots$
- iii. $\tan(360^\circ - \theta) = \dots\dots$
- iv. $\sin(-\theta) = \dots\dots$
- v. $\cos(-\theta) = \dots\dots$
- vi. $\tan(-\theta) = \dots\dots$

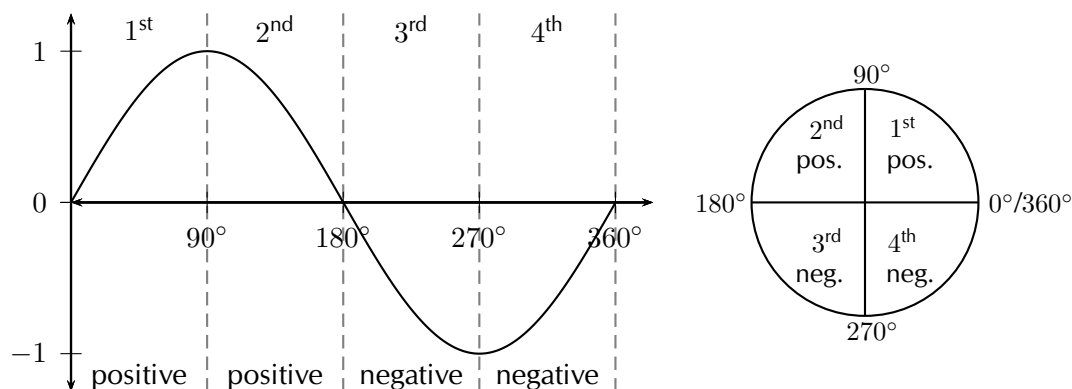
2. Function values of $360^\circ + \theta$

We can also have an angle that is larger than 360° . The angle completes a revolution of 360° and then continues to give an angle of θ .

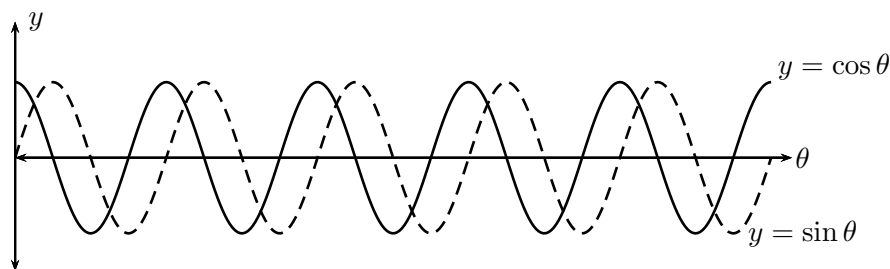
Complete the following reduction formulae:

- a) $\sin(360^\circ + \theta) = \dots\dots$
- b) $\cos(360^\circ + \theta) = \dots\dots$
- c) $\tan(360^\circ + \theta) = \dots\dots$

From working with functions, we know that the graph of $y = \sin \theta$ has a period of 360° . Therefore, one complete wave of a sine graph is the same as one complete revolution for $\sin \theta$ in the Cartesian plane.



We can also have multiple revolutions. The periodicity of the trigonometric graphs shows this clearly. A complete sine or cosine curve is completed in 360° .



If k is any integer, then

$$\sin(k \cdot 360^\circ + \theta) = \sin \theta$$

$$\cos(k \cdot 360^\circ + \theta) = \cos \theta$$

$$\tan(k \cdot 360^\circ + \theta) = \tan \theta$$

Worked example 8: Reduction formulae for function values of $360^\circ \pm \theta$

QUESTION

If $f = \tan 67^\circ$, express the following in terms of f

$$\frac{\sin 293^\circ}{\cos 427^\circ} + \tan(-67^\circ) + \tan 1147^\circ$$

SOLUTION

Step 1: Using reduction formula

$$\begin{aligned} &= \frac{\sin(360^\circ - 67^\circ)}{\cos(360^\circ + 67^\circ)} - \tan(67^\circ) + \tan(3(360^\circ) + 67^\circ) \\ &= \frac{-\sin 67^\circ}{\cos 67^\circ} - \tan 67^\circ + \tan 67^\circ \\ &= -\tan 67^\circ \\ &= -f \end{aligned}$$

Worked example 9: Using reduction formula

QUESTION

Evaluate without using a calculator:

$$\tan^2 210^\circ - (1 + \cos 120^\circ) \sin^2 405^\circ$$

SOLUTION

Step 1: Simplify the expression using reduction formulae and special angles

$$\begin{aligned}
&= \tan^2(180^\circ + 30^\circ) - (1 + \cos(180^\circ - 60^\circ)) \sin^2(360^\circ + 45^\circ) \\
&= \tan^2 30^\circ - (1 + (-\cos 60^\circ)) \sin^2 45^\circ \\
&= \left(\frac{1}{\sqrt{3}}\right)^2 - \left(1 - \frac{1}{2}\right) \left(\frac{1}{\sqrt{2}}\right)^2 \\
&= \frac{1}{3} - \left(\frac{1}{2}\right) \left(\frac{1}{2}\right) \\
&= \frac{1}{3} - \frac{1}{4} \\
&= \frac{1}{12}
\end{aligned}$$

Exercise 6 – 4: Using reduction formula

1. Simplify the following:

a) $\frac{\tan(180^\circ - \theta) \sin(360^\circ + \theta)}{\cos(180^\circ + \theta) \tan(360^\circ - \theta)}$

b) $\cos^2(360^\circ + \theta) + \cos(180^\circ + \theta) \tan(360^\circ - \theta) \sin(360^\circ + \theta)$

c) $\frac{\sin(360^\circ + \alpha) \tan(180^\circ + \alpha)}{\cos(360^\circ - \alpha) \tan^2(360^\circ + \alpha)}$

2. Write the following in terms of $\cos \beta$:

$$\frac{\cos(360^\circ - \beta) \cos(-\beta) - 1}{\sin(360^\circ + \beta) \tan(360^\circ - \beta)}$$

3. Simplify the following without using a calculator:

a) $\frac{\cos 300^\circ \tan 150^\circ}{\sin 225^\circ \cos(-45^\circ)}$

b) $3 \tan 405^\circ + 2 \tan 330^\circ \cos 750^\circ$

c) $\frac{\cos 315^\circ \cos 405^\circ + \sin 45^\circ \sin 135^\circ}{\sin 750^\circ}$

d) $\tan 150^\circ \cos 390^\circ - 2 \sin 510^\circ$

e) $\frac{2 \sin 120^\circ + 3 \cos 765^\circ - 2 \sin 240^\circ - 3 \cos 45^\circ}{5 \sin 300^\circ + 3 \tan 225^\circ - 6 \cos 60^\circ}$

4. Given $90^\circ < \alpha < 180^\circ$, use a sketch to help explain why:

a) $\sin(-\alpha) = -\sin \alpha$

b) $\cos(-\alpha) = -\cos \alpha$

5. If $t = \sin 43^\circ$, express the following in terms of t :

- a) $\sin 317^\circ$
- b) $\cos^2 403^\circ$
- c) $\tan(-43^\circ)$

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1a. [22YT](#) 1b. [22YV](#) 1c. [22YW](#) 2. [22YX](#) 3a. [22YY](#) 3b. [22YZ](#)
3c. [22Z2](#) 3d. [22Z3](#) 3e. [22Z4](#) 4. [22Z5](#) 5. [22Z6](#)



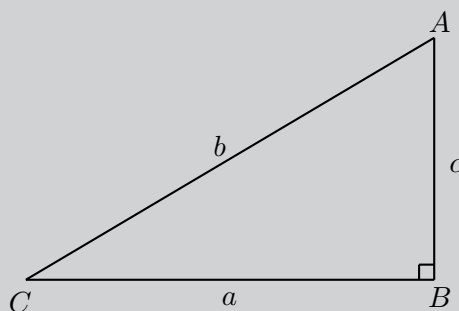
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Investigation: Reduction formulae for function values of $90^\circ \pm \theta$

In any right-angled triangle, the two acute angles are complements of each other, $\hat{A} + \hat{C} = 90^\circ$



Complete the following:

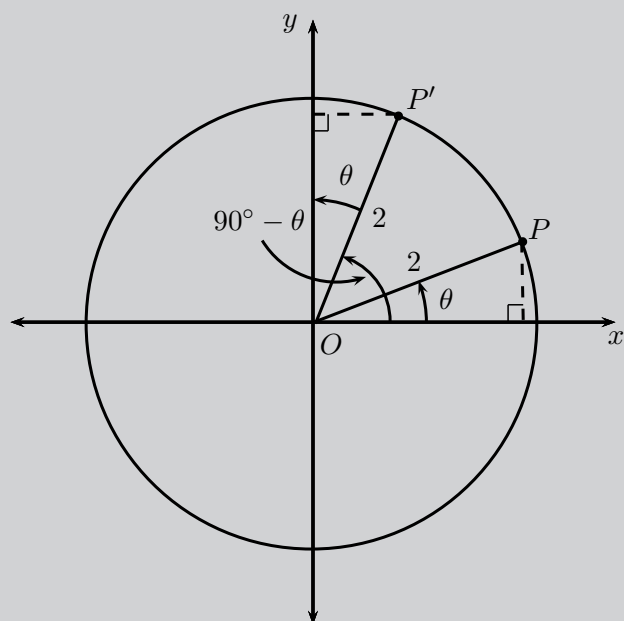
In $\triangle ABC$

$$\sin \hat{C} = \frac{c}{b} = \cos \dots$$

$$\cos \hat{C} = \frac{a}{b} = \sin \dots$$

Complementary angles are positive acute angles that add up to 90° . For example 20° and 70° are complementary angles.

In the figure $P(\sqrt{3}; 1)$ and P' lie on a circle with radius 2. OP makes an angle of $\theta = 30^\circ$ with the x -axis.

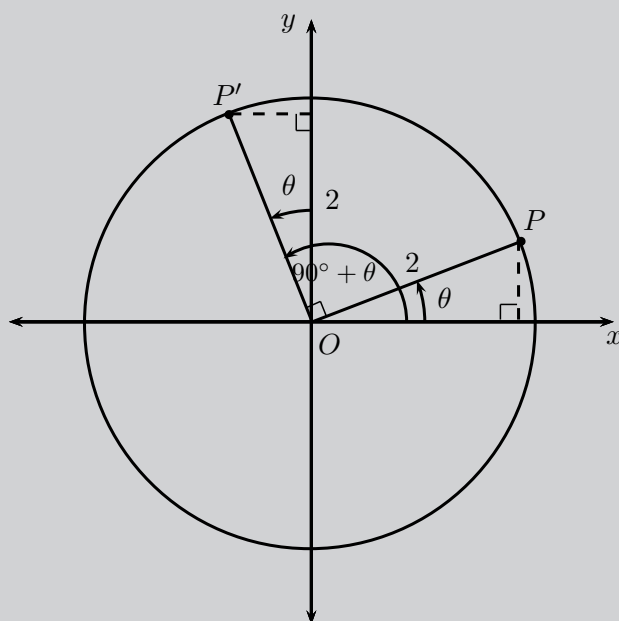


1. Function values of $90^\circ - \theta$

- If points P and P' are symmetrical about the line $y = x$, determine the coordinates of P' .
- Use the coordinates for P' to determine $\sin(90^\circ - \theta)$ and $\cos(90^\circ - \theta)$.
- From your results determine a relationship between the function values of $(90^\circ - \theta)$ and θ .

2. Function values of $90^\circ + \theta$

In the figure $P(\sqrt{3}; 1)$ and P' lie on the circle with radius 2. OP makes an angle $\theta = 30^\circ$ with the x -axis.



- a) If point P is rotated through 90° to get point P' , determine the coordinates of P' .
- b) Use the coordinates for P' to determine $\sin(90^\circ + \theta)$ and $\cos(90^\circ + \theta)$.
- c) From your results determine a relationship between the function values of $(90^\circ + \theta)$ and θ .

3. Complete the following reduction formulae:

a) $\sin(90^\circ - \theta) = \dots\dots$

b) $\cos(90^\circ - \theta) = \dots\dots$

c) $\sin(90^\circ + \theta) = \dots\dots$

d) $\cos(90^\circ + \theta) = \dots\dots$

Sine and cosine are known as **co-functions**. Two functions are called co-functions if $f(A) = g(B)$ whenever $A + B = 90^\circ$ (that is, A and B are complementary angles).

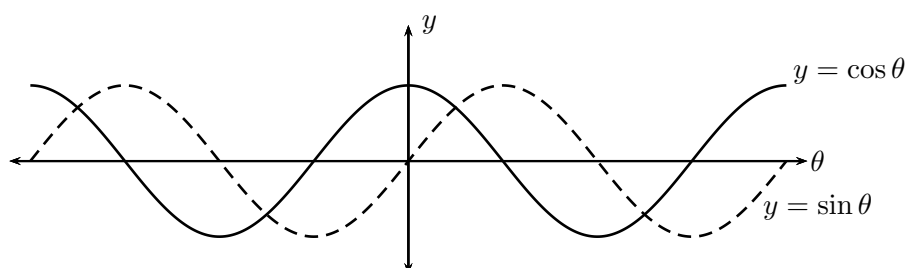
The function value of an angle is equal to the co-function of its complement.

Thus for sine and cosine we have

$$\sin(90^\circ - \theta) = \cos \theta$$

$$\cos(90^\circ - \theta) = \sin \theta$$

The sine and cosine graphs illustrate this clearly: the two graphs are identical except that they have a 90° phase difference.



Worked example 10: Using the co-function rule

QUESTION

Write each of the following in terms of $\sin 40^\circ$:

1. $\cos 50^\circ$

3. $\cos 230^\circ$

2. $\sin 320^\circ$

4. $\cos 130^\circ$

SOLUTION

1. $\cos 50^\circ = \sin(90^\circ - 50^\circ) = \sin 40^\circ$

2. $\sin 320^\circ = \sin(360^\circ - 40^\circ) = -\sin 40^\circ$

3. $\cos 230^\circ = \cos(180^\circ + 50^\circ) = -\cos 50^\circ = -\cos(90^\circ - 40^\circ) = -\sin 40^\circ$

4. $\cos 130^\circ = \cos(90^\circ + 40^\circ) = -\sin 40^\circ$

Function values of $\theta - 90^\circ$

We can write $\sin(\theta - 90^\circ)$ as

$$\begin{aligned}\sin(\theta - 90^\circ) &= \sin [-(90^\circ - \theta)] \\ &= -\sin(90^\circ - \theta) \\ &= -\cos \theta\end{aligned}$$

similarly, we can show that $\cos(\theta - 90^\circ) = \sin \theta$

Therefore, $\sin(\theta - 90^\circ) = -\cos \theta$ and $\cos(\theta - 90^\circ) = \sin \theta$.

Worked example 11: Co-functions

QUESTION

Express the following in terms of t if $t = \sin \theta$:

$$\frac{\cos(\theta - 90^\circ) \cos(720^\circ + \theta) \tan(\theta - 360^\circ)}{\sin^2(\theta + 360^\circ) \cos(\theta + 90^\circ)}$$

SOLUTION

Step 1: Simplify the expression using reduction formulae and co-functions

Use the CAST diagram to check in which quadrants the trigonometric ratios are positive and negative.

$$\begin{aligned}
 & \frac{\cos(\theta - 90^\circ) \cos(720^\circ + \theta) \tan(\theta - 360^\circ)}{\sin^2(\theta + 360^\circ) \cos(\theta + 90^\circ)} \\
 &= \frac{\cos[-(90^\circ - \theta)] \cos[2(360^\circ) + \theta] \tan[-(360^\circ - \theta)]}{\sin^2(360^\circ + \theta) \cos(90^\circ + \theta)} \\
 &= \frac{\sin \theta \cos \theta \tan \theta}{\sin^2 \theta (-\sin \theta)} \\
 &= -\frac{\cos \theta \left(\frac{\sin \theta}{\cos \theta}\right)}{\sin^2 \theta} \\
 &= -\frac{1}{\sin \theta} \\
 &= -\frac{1}{t}
 \end{aligned}$$

Exercise 6 – 5: Co-functions

1. Simplify the following:

a) $\frac{\cos(90^\circ + \theta) \sin(\theta + 90^\circ)}{\sin(-\theta)}$

b) $\frac{2 \sin(90^\circ - x) + \sin(90^\circ + x)}{\sin(90^\circ - x) + \cos(180^\circ + x)}$

2. Given $\cos 36^\circ = p$, express the following in terms on p :

a) $\sin 54^\circ$

c) $\tan 126^\circ$

b) $\sin 36^\circ$

d) $\cos 324^\circ$

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1a. 22Z7 1b. 22Z8 2. 22Z9



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Reduction formulae and co-functions:

1. The reduction formulae hold for any angle θ . For convenience, we assume θ is an acute angle ($0^\circ < \theta < 90^\circ$).
2. When determining function values of $(180^\circ \pm \theta)$, $(360^\circ \pm \theta)$ and $(-\theta)$ the function does not change.
3. When determining function values of $(90^\circ \pm \theta)$ and $(\theta \pm 90^\circ)$ the function changes to its co-function.

second quadrant ($180^\circ - \theta$) or ($90^\circ + \theta$)	first quadrant (θ) or ($90^\circ - \theta$)
$\sin(180^\circ - \theta) = +\sin \theta$	all trig functions are positive
$\cos(180^\circ - \theta) = -\cos \theta$	$\sin(360^\circ + \theta) = \sin \theta$
$\tan(180^\circ - \theta) = -\tan \theta$	$\cos(360^\circ + \theta) = \cos \theta$
$\sin(90^\circ + \theta) = +\cos \theta$	$\tan(360^\circ + \theta) = \tan \theta$
$\cos(90^\circ + \theta) = -\sin \theta$	$\sin(90^\circ - \theta) = \cos \theta$
	$\cos(90^\circ - \theta) = \sin \theta$
third quadrant ($180^\circ + \theta$)	fourth quadrant ($360^\circ - \theta$)
$\sin(180^\circ + \theta) = -\sin \theta$	$\sin(360^\circ - \theta) = -\sin \theta$
$\cos(180^\circ + \theta) = -\cos \theta$	$\cos(360^\circ - \theta) = +\cos \theta$
$\tan(180^\circ + \theta) = +\tan \theta$	$\tan(360^\circ - \theta) = -\tan \theta$

Exercise 6 – 6: Reduction formulae

1. Write A and B as a single trigonometric ratio:
 - a) $A = \sin(360^\circ - \theta) \cos(180^\circ - \theta) \tan(360^\circ + \theta)$
 - b) $B = \frac{\cos(360^\circ + \theta) \cos(-\theta) \sin(-\theta)}{\cos(90^\circ + \theta)}$
 - c) Hence, determine:
 - i. $A + B = \dots$
 - ii. $\frac{A}{B} = \dots$
2. Write the following as a function of an acute angle:
 - a) $\sin 163^\circ$
 - b) $\cos 327^\circ$
 - c) $\tan 248^\circ$
 - d) $\cos(-213^\circ)$
3. Determine the value of the following, without using a calculator:
 - a) $\frac{\sin(-30^\circ)}{\tan(150^\circ)} + \cos 330^\circ$
 - b) $\tan 300^\circ \cos 120^\circ$
 - c) $(1 - \cos 30^\circ)(1 - \cos 210^\circ)$
 - d) $\cos 780^\circ - (\sin 315^\circ)(\cos 405^\circ)$
4. Prove that the following identity is true and state any restrictions:

$$\frac{\sin(180^\circ + \alpha) \tan(360^\circ + \alpha) \cos \alpha}{\cos(90^\circ - \alpha)} = \sin \alpha$$

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1. [22ZB](#) 2a. [22ZC](#) 2b. [22ZD](#) 2c. [22ZF](#) 2d. [22ZG](#) 3a. [22ZH](#)
 3b. [22ZJ](#) 3c. [22ZK](#) 3d. [22ZM](#) 4. [22ZN](#)

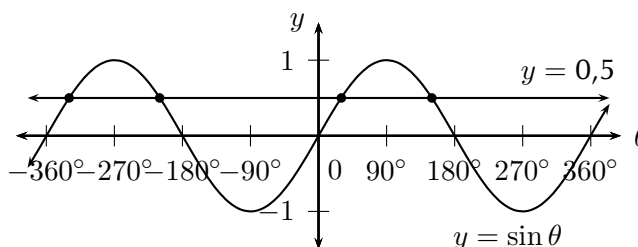


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Solving trigonometric equations requires that we find the value of the angles that satisfy the equation. If a specific interval for the solution is given, then we need only find the value of the angles within the given interval that satisfy the equation. If no interval is given, then we need to find the general solution. The periodic nature of trigonometric functions means that there are many values that satisfy a given equation, as shown in the diagram below.



Worked example 12: Solving trigonometric equations

QUESTION

Solve for θ (correct to one decimal place), given $\tan \theta = 5$ and $\theta \in [0^\circ; 360^\circ]$.

SOLUTION

Step 1: Use a calculator to solve for θ

$$\begin{aligned}\tan \theta &= 5 \\ \therefore \theta &= \tan^{-1} 5 \\ &= 78,7^\circ\end{aligned}$$

This value of θ is an acute angle which lies in the first quadrant and is called the reference angle.

Step 2: Use the CAST diagram to determine in which quadrants $\tan \theta$ is positive

The CAST diagram indicates that $\tan \theta$ is positive in the first and third quadrants, therefore we must determine the value of θ such that $180^\circ < \theta < 270^\circ$.

Using reduction formulae, we know that $\tan(180^\circ + \theta) = \tan \theta$

$$\begin{aligned}\theta &= 180^\circ + 78,7^\circ \\ \therefore \theta &= 258,7^\circ\end{aligned}$$

Step 3: Use a calculator to check that the solution satisfies the original equation

Step 4: Write the final answer

$$\theta = 78,7^\circ \text{ or } \theta = 258,7^\circ.$$

Worked example 13: Solving trigonometric equations

QUESTION

Solve for α (correct to one decimal place), given $\cos \alpha = -0,7$ and $\theta \in [0^\circ; 360^\circ]$.

SOLUTION

Step 1: Use a calculator to find the reference angle

To determine the reference angle, we do not include the negative sign. The reference angle must be an acute angle in the first quadrant, where all the trigonometric functions are positive.

$$\begin{aligned}\text{ref } \angle &= \cos^{-1} 0,7 \\ &= 45,6^\circ\end{aligned}$$

Step 2: Use the CAST diagram to determine in which quadrants $\cos \alpha$ is negative

The CAST diagram indicates that $\cos \alpha$ is negative in the second and third quadrants, therefore we must determine the value of α such that $90^\circ < \alpha < 270^\circ$.

Using reduction formulae, we know that $\cos(180^\circ - \alpha) = -\cos \alpha$ and $\cos(180^\circ + \alpha) = -\cos \alpha$

In the second quadrant:

$$\begin{aligned}\alpha &= 180^\circ - 45,6^\circ \\ &= 134,4^\circ\end{aligned}$$

In the third quadrant:

$$\begin{aligned}\alpha &= 180^\circ + 45,6^\circ \\ &= 225,6^\circ\end{aligned}$$

Note: the reference angle ($45,6^\circ$) does not form part of the solution.

Step 3: Use a calculator to check that the solution satisfies the original equation

Step 4: Write the final answer

$$\alpha = 134,4^\circ \text{ or } \alpha = 225,6^\circ.$$

Worked example 14: Solving trigonometric equations

QUESTION

Solve for β (correct to one decimal place), given $\sin \beta = -0,5$ and $\beta \in [-360^\circ; 360^\circ]$.

SOLUTION

Step 1: Use a calculator to find the reference angle

To determine the reference angle, we use a positive value.

$$\begin{aligned}\text{ref } \angle &= \sin^{-1} 0,5 \\ &= 30^\circ\end{aligned}$$

Step 2: Use the CAST diagram to determine in which quadrants $\sin \beta$ is negative

The CAST diagram indicates that $\sin \beta$ is negative in the third and fourth quadrants. We also need to find the values of β such that $-360^\circ \leq \beta \leq 360^\circ$.

Using reduction formulae, we know that $\sin(180^\circ + \beta) = -\sin \beta$ and $\sin(360^\circ - \beta) = -\sin \beta$

In the third quadrant:

$$\begin{aligned}\beta &= 180^\circ + 30^\circ \\ &= 210^\circ \\ \text{or } \beta &= -180^\circ + 30^\circ \\ &= -150^\circ\end{aligned}$$

In the fourth quadrant:

$$\begin{aligned}\beta &= 360^\circ - 30^\circ \\ &= 330^\circ \\ \text{or } \beta &= 0^\circ - 30^\circ \\ &= -30^\circ\end{aligned}$$

Notice: the reference angle (30°) does not form part of the solution.

Step 3: Use a calculator to check that the solution satisfies the original equation

Step 4: Write the final answer

$$\beta = -150^\circ, -30^\circ, 210^\circ \text{ or } 330^\circ.$$

Exercise 6 – 7: Solving trigonometric equations

1. Determine the values of α for $\alpha \in [0^\circ; 360^\circ]$ if:

a) $4 \cos \alpha = 2$

d) $\cos \alpha + 0,939 = 0$

b) $\sin \alpha + 3,65 = 3$

e) $5 \sin \alpha = 3$

c) $\tan \alpha = 5\frac{1}{4}$

f) $\frac{1}{2} \tan \alpha = -1,4$

2. Determine the values of θ for $\theta \in [-360^\circ; 360^\circ]$ if:

a) $\sin \theta = 0,6$

d) $\sin \theta = \cos 180^\circ$

b) $\cos \theta + \frac{3}{4} = 0$

e) $2 \cos \theta = \frac{4}{5}$

c) $3 \tan \theta = 20$

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1a. [22ZP](#) 1b. [22ZQ](#) 1c. [22ZR](#) 1d. [22ZS](#) 1e. [22ZT](#) 1f. [22ZV](#)
2a. [22ZW](#) 2b. [22ZX](#) 2c. [22ZY](#) 2d. [22ZZ](#) 2e. [2322](#)



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The general solution

EMBHN

In the previous worked example, the solution was restricted to a certain interval. However, the periodicity of the trigonometric functions means that there are an infinite number of positive and negative angles that satisfy an equation. If we do not restrict the solution, then we need to determine the general solution to the equation. We know that the sine and cosine functions have a period of 360° and the tangent function has a period of 180° .

Method for finding the general solution:

1. Determine the reference angle (use a positive value).
2. Use the CAST diagram to determine where the function is positive or negative (depending on the given equation).
3. Find the angles in the interval $[0^\circ; 360^\circ]$ that satisfy the equation and add multiples of the period to each answer.
4. Check answers using a calculator.

Worked example 15: Finding the general solution

QUESTION

Determine the general solution for $\sin \theta = 0,3$ (correct to one decimal place).

SOLUTION

Step 1: Use a calculator to find the reference angle

$$\begin{aligned}\sin \theta &= 0,3 \\ \therefore \text{ref } \angle &= \sin^{-1} 0,3 \\ &= 17,5^\circ\end{aligned}$$

Step 2: Use CAST diagram to determine in which quadrants $\sin \theta$ is positive

The CAST diagram indicates that $\sin \theta$ is positive in the first and second quadrants.

Using reduction formulae, we know that $\sin(180^\circ - \theta) = \sin \theta$.

In the first quadrant:

$$\begin{aligned}\theta &= 17,5^\circ \\ \therefore \theta &= 17,5^\circ + k \cdot 360^\circ\end{aligned}$$

In the second quadrant:

$$\begin{aligned}\theta &= 180^\circ - 17,5^\circ \\ \therefore \theta &= 162,5^\circ + k \cdot 360^\circ\end{aligned}$$

where $k \in \mathbb{Z}$.

Step 3: Check that the solution satisfies the original equation

We can select random values of k to check that the answers satisfy the original equation.

Let $k = 4$:

$$\begin{aligned}\theta &= 17,5^\circ + 4(360)^\circ \\ \therefore \theta &= 1457,5^\circ \\ \text{And } \sin 1457,5^\circ &= 0,3007 \dots\end{aligned}$$

This solution is correct.

Similarly, if we let $k = -2$:

$$\begin{aligned}\theta &= 162,5^\circ - 2(360)^\circ \\ \therefore \theta &= -557,5^\circ \\ \text{And } \sin(-557,5^\circ) &= 0,3007 \dots\end{aligned}$$

This solution is also correct.

Step 4: Write the final answer

$$\theta = 17,5^\circ + k \cdot 360^\circ \text{ or } \theta = 162,5^\circ + k \cdot 360^\circ.$$

QUESTION

Determine the general solution for $\cos 2\theta = -0,6427$ (give answers correct to one decimal place).

SOLUTION

Step 1: Use a calculator to find the reference angle

$$\begin{aligned}\text{ref } \angle &= \sin^{-1} 0,6427 \\ &= 50,0^\circ\end{aligned}$$

Step 2: Use CAST diagram to determine in which quadrants $\cos \theta$ is negative

The CAST diagram shows that $\cos \theta$ is negative in the second and third quadrants.

Therefore we use the reduction formulae $\cos(180^\circ - \theta) = -\cos \theta$ and $\cos(180^\circ + \theta) = -\cos \theta$.

In the second quadrant:

$$\begin{aligned}2\theta &= 180^\circ - 50^\circ + k \cdot 360^\circ \\ &= 130^\circ + k \cdot 360^\circ \\ \therefore \theta &= 65^\circ + k \cdot 180^\circ\end{aligned}$$

In the third quadrant:

$$\begin{aligned}2\theta &= 180^\circ + 50^\circ + k \cdot 360^\circ \\ &= 230^\circ + k \cdot 360^\circ \\ \therefore \theta &= 115^\circ + k \cdot 180^\circ\end{aligned}$$

where $k \in \mathbb{Z}$.

Remember: also divide the period (360°) by the coefficient of θ .

Step 3: Check that the solution satisfies the original equation

We can select random values of k to check that the answers satisfy the original equation.

Let $k = 2$:

$$\begin{aligned}\theta &= 65^\circ + 2(180^\circ) \\ \therefore \theta &= 425^\circ \\ \text{And } \cos 2(425)^\circ &= -0,6427 \dots\end{aligned}$$

This solution is correct.

Similarly, if we let $k = -5$:

$$\theta = 115^\circ - 5(180^\circ)$$

$$\therefore \theta = -785^\circ$$

$$\text{And } \cos 2(-785^\circ) = -0,6427 \dots$$

This solution is also correct.

Step 4: Write the final answer

$$\theta = 65^\circ + k \cdot 180^\circ \text{ or } \theta = 115^\circ + k \cdot 180^\circ.$$

Worked example 17: Finding the general solution

QUESTION

Determine the general solution for $\tan(2\alpha - 10^\circ) = 2,5$ such that $-180^\circ \leq \alpha \leq 180^\circ$ (give answers correct to one decimal place).

SOLUTION

Step 1: Make a substitution

To solve this equation, it can be useful to make a substitution: let $x = 2\alpha - 10^\circ$.

$$\tan(x) = 2,5$$

Step 2: Use a calculator to find the reference angle

$$\tan x = 2,5$$

$$\begin{aligned}\therefore \text{ref } \angle &= \tan^{-1} 2,5 \\ &= 68,2^\circ\end{aligned}$$

Step 3: Use CAST diagram to determine in which quadrants the tangent function is positive

We see that $\tan x$ is positive in the first and third quadrants, so we use the reduction formula $\tan(180^\circ + x) = \tan x$. It is also important to remember that the period of the tangent function is 180° .

In the first quadrant:

$$x = 68,2^\circ + k \cdot 180^\circ$$

$$\text{Substitute } x = 2\alpha - 10^\circ$$

$$2\alpha - 10^\circ = 68,2^\circ + k \cdot 180^\circ$$

$$2\alpha = 78,2^\circ + k \cdot 180^\circ$$

$$\therefore \alpha = 39,1^\circ + k \cdot 90^\circ$$

In the third quadrant:

$$x = 180^\circ + 68,2^\circ + k \cdot 180^\circ$$

$$= 248,2^\circ + k \cdot 180^\circ$$

$$\text{Substitute } x = 2\alpha - 10^\circ$$

$$2\alpha - 10^\circ = 248,2^\circ + k \cdot 180^\circ$$

$$2\alpha = 258,2^\circ + k \cdot 180^\circ$$

$$\therefore \alpha = 129,1^\circ + k \cdot 90^\circ$$

where $k \in \mathbb{Z}$.

Remember: to divide the period (180°) by the coefficient of α .

Step 4: Find the answers within the given interval

Substitute suitable values of k to determine the values of α that lie within the interval $(-180^\circ \leq \alpha \leq 180^\circ)$.

	I: $\alpha = 39,1^\circ + k \cdot 90^\circ$	III: $\alpha = 129,1^\circ + k \cdot 90^\circ$
$k = 0$	$39,1^\circ$	$129,1^\circ$
$k = 1$	$129,1^\circ$	$219,1^\circ$ (outside)
$k = 2$	$219,1^\circ$ (outside)	
$k = -1$	$-50,9^\circ$	$39,1^\circ$
$k = -2$	$-140,9^\circ$	$-50,9^\circ$
$k = -3$	$-230,9^\circ$ (outside)	$-140,9^\circ$
$k = -4$		$-230,9^\circ$ (outside)

Notice how some of the values repeat. This is because of the periodic nature of the tangent function. Therefore we need only determine the solution:

$$\alpha = 39,1^\circ + k \cdot 90^\circ$$

for $k \in \mathbb{Z}$.

Step 5: Write the final answer

$$\alpha = -140,9^\circ; -50,9^\circ; 39,1^\circ \text{ or } 129,1^\circ.$$

QUESTION

Determine the general solution for $\sin(\theta - 20^\circ) = \cos 2\theta$.

SOLUTION

Step 1: Use co-functions to simplify the equation

$$\begin{aligned}\sin(\theta - 20^\circ) &= \cos 2\theta \\ &= \sin(90^\circ - 2\theta) \\ \therefore \theta - 20^\circ &= 90^\circ - 2\theta + k \cdot 360^\circ, \quad k \in \mathbb{Z} \\ 3\theta &= 110^\circ + k \cdot 360^\circ \\ \therefore \theta &= 36,7^\circ + k \cdot 120^\circ\end{aligned}$$

Step 2: Use the CAST diagram to determine the correct quadrants

Since the original equation equates a sine and cosine function, we need to work in the quadrant where both functions are positive or in the quadrant where both functions are negative so that the equation holds true. We therefore determine the solution using the first and third quadrants.

In the first quadrant: $\theta = 36,7^\circ + k \cdot 120^\circ$.

In the third quadrant:

$$\begin{aligned}3\theta &= 180^\circ + 110^\circ + k \cdot 360^\circ \\ &= 290^\circ + k \cdot 360^\circ \\ \therefore \theta &= 96,6^\circ + k \cdot 120^\circ\end{aligned}$$

where $k \in \mathbb{Z}$.

Step 3: Check that the solution satisfies the original equation

Step 4: Write the final answer

$$\theta = 36,7^\circ + k \cdot 120^\circ \text{ or } \theta = 96,6^\circ + k \cdot 120^\circ$$

Exercise 6 – 8: General solution

1. • Find the general solution for each equation.
• Hence, find all the solutions in the interval $[-180^\circ; 180^\circ]$.

a) $\cos(\theta + 25^\circ) = 0,231$

f) $\cos \theta = -1$

b) $\sin 2\alpha = -0,327$

g) $\tan \frac{\theta}{2} = 0,9$

c) $2 \tan \beta = -2,68$

h) $4 \cos \theta + 3 = 1$

d) $\cos \alpha = 1$

i) $\sin 2\theta = -\frac{\sqrt{3}}{2}$

e) $4 \sin \theta = 0$

2. Find the general solution for each equation.

a) $\cos(\theta + 20^\circ) = 0$

d) $\cos(\alpha - 25^\circ) = 0,707$

b) $\sin 3\alpha = -1$

e) $2 \sin \frac{3\theta}{2} = -1$

c) $\tan 4\beta = 0,866$

f) $5 \tan(\beta + 15^\circ) = \frac{5}{\sqrt{3}}$

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1a. 2323 1b. 2324 1c. 2325 1d. 2326 1e. 2327 1f. 2328
1g. 2329 1h. 232B 1i. 232C 2a. 232D 2b. 232F 2c. 232G
2d. 232H 2e. 232J 2f. 232K



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Solving quadratic trigonometric equations

We can use our knowledge of algebraic equations to solve quadratic trigonometric equations.

Worked example 19: Quadratic trigonometric equations

QUESTION

Find the general solution of $4 \sin^2 \theta = 3$.

SOLUTION

Step 1: Simplify the equation and determine the reference angle

$$4 \sin^2 \theta = 3$$

$$\sin^2 \theta = \frac{3}{4}$$

$$\begin{aligned}\therefore \sin \theta &= \pm \sqrt{\frac{3}{4}} \\ &= \pm \frac{\sqrt{3}}{2}\end{aligned}$$

$$\therefore \text{ref } \angle = 60^\circ$$

Step 2: Determine in which quadrants the sine function is positive and negative

The CAST diagram shows that $\sin \theta$ is positive in the first and second quadrants and negative in the third and fourth quadrants.

Positive in the first and second quadrants:

$$\begin{aligned}\theta &= 60^\circ + k \cdot 360^\circ \\ \text{or } \theta &= 180^\circ - 60^\circ + k \cdot 360^\circ \\ &= 120^\circ + k \cdot 360^\circ\end{aligned}$$

Negative in the third and fourth quadrants:

$$\begin{aligned}\theta &= 180^\circ + 60^\circ + k \cdot 360^\circ \\ &= 240^\circ + k \cdot 360^\circ \\ \text{or } \theta &= 360^\circ - 60^\circ + k \cdot 360^\circ \\ &= 300^\circ + k \cdot 360^\circ\end{aligned}$$

where $k \in \mathbb{Z}$.

Step 3: Check that the solution satisfies the original equation**Step 4: Write the final answer**

$$\theta = 60^\circ + k \cdot 360^\circ \text{ or } 120^\circ + k \cdot 360^\circ \text{ or } 240^\circ + k \cdot 360^\circ \text{ or } 300^\circ + k \cdot 360^\circ$$

Worked example 20: Quadratic trigonometric equations**QUESTION**

Find θ if $2 \cos^2 \theta - \cos \theta - 1 = 0$ for $\theta \in [-180^\circ; 180^\circ]$.

SOLUTION**Step 1: Factorise the equation**

$$\begin{aligned}2 \cos^2 \theta - \cos \theta - 1 &= 0 \\ (2 \cos \theta + 1)(\cos \theta - 1) &= 0 \\ \therefore 2 \cos \theta + 1 = 0 \text{ or } \cos \theta - 1 &= 0\end{aligned}$$

Step 2: Simplify the equations and solve for θ

$$2 \cos \theta + 1 = 0$$

$$2 \cos \theta = -1$$

$$\cos \theta = -\frac{1}{2}$$

$$\therefore \text{ref } \angle = 60^\circ$$

$$\begin{aligned}\text{II quadrant: } \theta &= 180^\circ - 60^\circ + k \cdot 360^\circ \\ &= 120^\circ + k \cdot 360^\circ\end{aligned}$$

$$\begin{aligned}\text{III quadrant: } \theta &= 180^\circ + 60^\circ + k \cdot 360^\circ \\ &= 240^\circ + k \cdot 360^\circ\end{aligned}$$

or

$$\cos \theta - 1 = 0$$

$$\cos \theta = 1$$

$$\therefore \text{ref } \angle = 0^\circ$$

$$\text{II and IV quadrants: } \theta = k \cdot 360^\circ$$

where $k \in \mathbb{Z}$.

Step 3: Substitute suitable values of k

Determine the values of θ that lie within the given interval $\theta \in [-180^\circ; 180^\circ]$ by substituting suitable values of k .

If $k = -1$,

$$\begin{aligned}\theta &= 240^\circ + k \cdot 360^\circ \\ &= 240^\circ - (360^\circ) \\ &= -120^\circ\end{aligned}$$

If $k = 0$,

$$\begin{aligned}\theta &= 120^\circ + k \cdot 360^\circ \\ &= 120^\circ + 0(360^\circ) \\ &= 120^\circ\end{aligned}$$

If $k = 1$,

$$\begin{aligned}\theta &= k \cdot 360^\circ \\ &= 0(360^\circ) \\ &= 0^\circ\end{aligned}$$

Step 4: Alternative method: substitution

We can simplify the given equation by letting $y = \cos \theta$ and then factorising as:

$$\begin{aligned}2y^2 - y - 1 &= 0 \\(2y + 1)(y - 1) &= 0 \\ \therefore y &= -\frac{1}{2} \text{ or } y = 1\end{aligned}$$

We substitute $y = \cos \theta$ back into these two equations and solve for θ .

Step 5: Write the final answer

$$\theta = -120^\circ; 0^\circ; 120^\circ$$

Worked example 21: Quadratic trigonometric equations**QUESTION**

Find α if $2 \sin^2 \alpha - \sin \alpha \cos \alpha = 0$ for $\alpha \in [0^\circ; 360^\circ]$.

SOLUTION**Step 1: Factorise the equation by taking out a common factor**

$$\begin{aligned}2 \sin^2 \alpha - \sin \alpha \cos \alpha &= 0 \\ \sin \alpha (2 \sin \alpha - \cos \alpha) &= 0 \\ \therefore \sin \alpha &= 0 \text{ or } 2 \sin \alpha - \cos \alpha = 0\end{aligned}$$

Step 2: Simplify the equations and solve for α

$$\begin{aligned}\sin \alpha &= 0 \\ \therefore \text{ref } \angle &= 0^\circ \\ \therefore \alpha &= 0^\circ + k \cdot 360^\circ \\ \text{or } \alpha &= 180^\circ + k \cdot 360^\circ \\ \text{and since } 360^\circ &= 2 \times 180^\circ \\ \text{we therefore have } \alpha &= k \cdot 180^\circ\end{aligned}$$

or

$$2 \sin \alpha - \cos \alpha = 0$$

$$2 \sin \alpha = \cos \alpha$$

To simplify further, we divide both sides of the equation by $\cos \alpha$.

$$\frac{2 \sin \alpha}{\cos \alpha} = \frac{\cos \alpha}{\cos \alpha} \quad (\cos \alpha \neq 0)$$

$$2 \tan \alpha = 1$$

$$\tan \alpha = \frac{1}{2}$$

$$\therefore \text{ref } \angle = 26,6^\circ$$

$$\therefore \alpha = 26,6^\circ + k \cdot 180^\circ$$

where $k \in \mathbb{Z}$.

Step 3: Substitute suitable values of k

Determine the values of α that lie within the the given interval $\alpha \in [0^\circ; 360^\circ]$ by substituting suitable values of k .

If $k = 0$:

$$\alpha = 0^\circ$$

$$\text{or } \alpha = 26,6^\circ$$

If $k = 1$:

$$\alpha = 180^\circ$$

$$\text{or } \alpha = 26,6^\circ + 180^\circ$$

$$= 206,6^\circ$$

If $k = 2$:

$$\alpha = 360^\circ$$

Step 4: Write the final answer

$$\alpha = 0^\circ; 26,6^\circ; 180^\circ; 206,6^\circ; 360^\circ$$

Exercise 6 – 9: Solving trigonometric equations

1. Find the general solution for each of the following equations:

a) $\cos 2\theta = 0$

f) $3 \sin \alpha = -1,5$

b) $\sin(\alpha + 10^\circ) = \frac{\sqrt{3}}{2}$

g) $\sin 2\beta = \cos(\beta + 20^\circ)$

c) $2 \cos \frac{\theta}{2} - \sqrt{3} = 0$

h) $0,5 \tan \theta + 2,5 = 1,7$

d) $\frac{1}{2} \tan(\beta - 30^\circ) = -1$

i) $\sin(3\alpha - 10^\circ) = \sin(\alpha + 32^\circ)$

e) $5 \cos \theta = \tan 300^\circ$

j) $\sin 2\beta = \cos 2\beta$

2. Find θ if $\sin^2 \theta + \frac{1}{2} \sin \theta = 0$ for $\theta \in [0^\circ; 360^\circ]$.

3. Determine the general solution for each of the following:

a) $2 \cos^2 \theta - 3 \cos \theta = 2$

e) $\sin(\alpha + 15^\circ) = 2 \cos(\alpha + 15^\circ)$

b) $3 \tan^2 \theta + 2 \tan \theta = 0$

f) $\sin^2 \theta - 4 \cos^2 \theta = 0$

c) $\cos^2 \alpha = 0,64$

d) $\sin(4\beta + 35^\circ) = \cos(10^\circ - \beta)$

g) $\frac{\cos(2\theta + 30^\circ)}{2} + 0,38 = 0$

4. Find β if $\frac{1}{3} \tan \beta = \cos 200^\circ$ for $\beta \in [-180^\circ; 180^\circ]$.

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6.5 Area, sine, and cosine rules

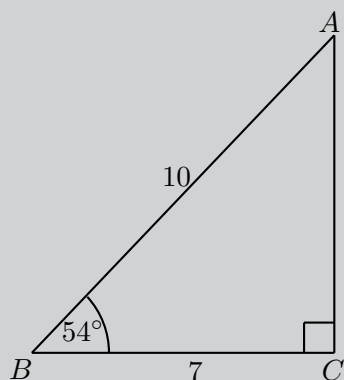
EMBHP

There are three identities relating to the trigonometric functions that make working with triangles easier:

1. the area rule
2. the sine rule
3. the cosine rule

Investigation: The area rule

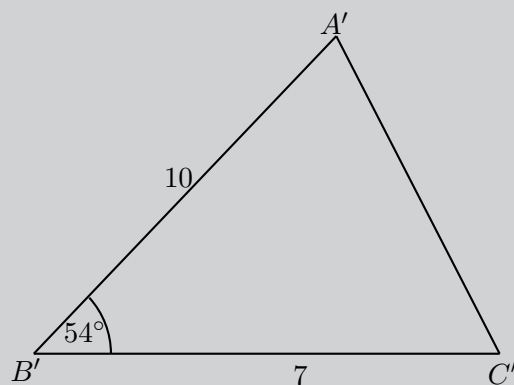
1. Consider $\triangle ABC$:



Complete the following:

- Area $\triangle ABC = \frac{1}{2} \times \dots \times AC$
- $\sin \hat{B} = \dots$ and $AC = \dots \times \dots$
- Therefore area $\triangle ABC = \dots \times \dots \times \dots \times \dots$

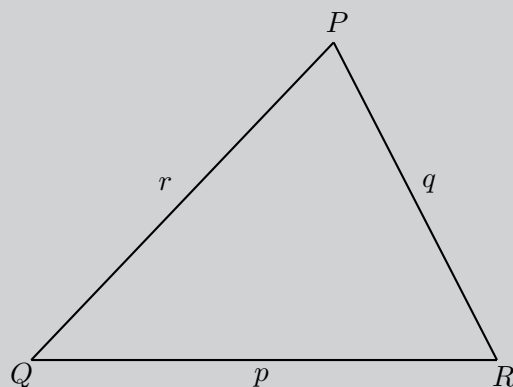
2. Consider $\triangle A'B'C'$:



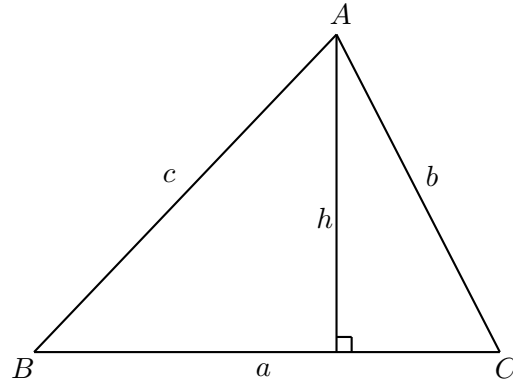
Complete the following:

- How is $\triangle A'B'C'$ different from $\triangle ABC$?
- Calculate area $\triangle A'B'C'$.

3. Use your results to write a general formula for determining the area of $\triangle PQR$:



For any $\triangle ABC$ with $AB = c$, $BC = a$ and $AC = b$, we can construct a perpendicular height (h) from vertex A to the line BC :



In $\triangle ABC$:

$$\begin{aligned}\sin \hat{B} &= \frac{h}{c} \\ \therefore h &= c \sin \hat{B}\end{aligned}$$

And we know that

$$\begin{aligned}\text{Area } \triangle ABC &= \frac{1}{2} \times a \times h \\ &= \frac{1}{2} \times a \times c \sin \hat{B} \\ \therefore \text{Area } \triangle ABC &= \frac{1}{2} ac \sin \hat{B}\end{aligned}$$

Alternatively, we could write that

$$\begin{aligned}\sin \hat{C} &= \frac{h}{b} \\ \therefore h &= b \sin \hat{C}\end{aligned}$$

And then we would have that

$$\begin{aligned}\text{Area } \triangle ABC &= \frac{1}{2} \times a \times h \\ &= \frac{1}{2} ab \sin \hat{C}\end{aligned}$$

Similarly, by constructing a perpendicular height from vertex B to the line AC , we can also show that $\text{area } \triangle ABC = \frac{1}{2} bc \sin \hat{A}$.

The area rule

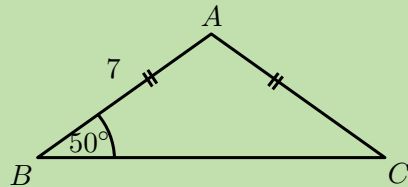
In any $\triangle ABC$:

$$\begin{aligned}\text{Area } \triangle ABC &= \frac{1}{2} bc \sin \hat{A} \\ &= \frac{1}{2} ac \sin \hat{B} \\ &= \frac{1}{2} ab \sin \hat{C}\end{aligned}$$

Worked example 22: The area rule

QUESTION

Find the area of $\triangle ABC$ (correct to two decimal places):



SOLUTION

Step 1: Use the given information to determine unknown angles and sides

$$\begin{aligned}AB &= AC = 7 && \text{(given)} \\ \therefore \hat{B} &= \hat{C} = 50^\circ && (\angle\text{s opp. equal sides}) \\ \text{And } \hat{A} &= 180^\circ - 50^\circ - 50^\circ && (\angle\text{s sum of } \triangle ABC) \\ \therefore \hat{A} &= 80^\circ\end{aligned}$$

Step 2: Use the area rule to calculate the area of $\triangle ABC$

Notice that we do not know the length of side a and must therefore choose the form of the area rule that does not include this side of the triangle.

In $\triangle ABC$:

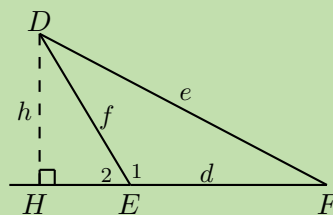
$$\begin{aligned}\text{Area} &= \frac{1}{2}bc \sin \hat{A} \\ &= \frac{1}{2}(7)(7) \sin 80^\circ \\ &= 24,13\end{aligned}$$

Step 3: Write the final answer

Area of $\triangle ABC = 24,13$ square units.

QUESTION

Show that the area of $\triangle DEF = \frac{1}{2}df \sin \hat{E}$.



SOLUTION

Step 1: Construct a perpendicular height h

Draw DH such that $DH \perp EF$ and let $DH = h$, $\hat{DEF} = \hat{E}_1$ and $\hat{DEH} = \hat{E}_2$.

In $\triangle DHE$:

$$\sin \hat{E}_2 = \frac{h}{f}$$

$$h = f \sin(180^\circ - \hat{E}_1) \quad (\angle \text{s on str. line})$$

$$= f \sin \hat{E}_1$$

Step 2: Use the area rule to calculate the area of $\triangle DEF$

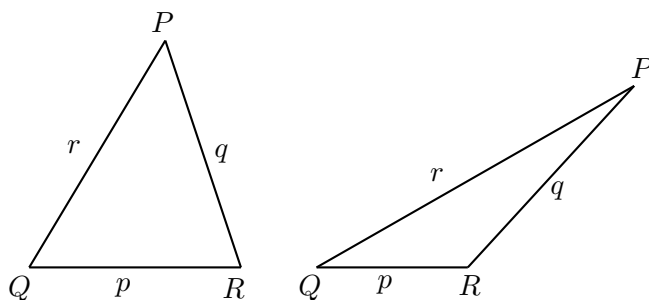
In $\triangle DEF$:

$$\text{Area} = \frac{1}{2}d \times h$$

$$= \frac{1}{2}df \sin \hat{E}_1$$

The area rule

In any $\triangle PQR$:



$$\begin{aligned}\text{Area } \triangle PQR &= \frac{1}{2}qr \sin \hat{P} \\ &= \frac{1}{2}pr \sin \hat{Q} \\ &= \frac{1}{2}pq \sin \hat{R}\end{aligned}$$

The area rule states that the area of any triangle is equal to half the product of the lengths of the two sides of the triangle multiplied by the sine of the angle included by the two sides.

Exercise 6 – 10: The area rule

1. Draw a sketch and calculate the area of $\triangle PQR$ given:

a) $\hat{Q} = 30^\circ$; $r = 10$ and $p = 7$

b) $\hat{R} = 110^\circ$; $p = 8$ and $q = 9$

2. Find the area of $\triangle XYZ$ given $XZ = 52$ cm, $XY = 29$ cm and $\hat{X} = 58,9^\circ$.

3. Determine the area of a parallelogram in which two adjacent sides are 10 cm and 13 cm and the angle between them is 55° .

4. If the area of $\triangle ABC$ is 5000 m^2 with $a = 150$ m and $b = 70$ m, what are the two possible sizes of $\hat{C}$?

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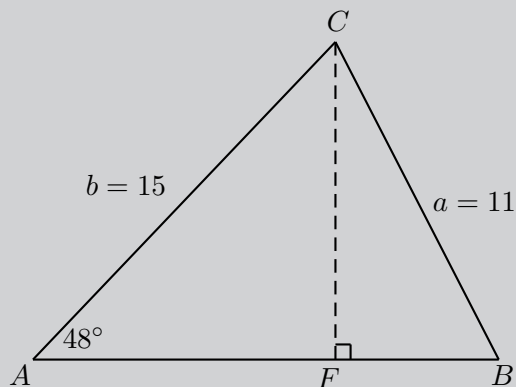


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So far we have only applied the trigonometric ratios to right-angled triangles. We now expand the application of the trigonometric ratios to triangles that do not have a right angle:

Investigation: The sine rule

In $\triangle ABC$, $AC = 15$, $BC = 11$ and $\hat{A} = 48^\circ$. Find $\hat{B}$.



1. Method 1: using the sine ratio

- Draw a sketch of $\triangle ABC$.
- Construct $CF \perp AB$.
- In $\triangle CBF$:

$$\begin{aligned}\frac{CF}{\dots} &= \sin \hat{B} \\ \therefore CF &= \dots \times \sin \hat{B}\end{aligned}$$

- In $\triangle CAF$:

$$\begin{aligned}\frac{CF}{15} &= \dots \\ \therefore CF &= 15 \times \dots\end{aligned}$$

- Therefore we have that:

$$\begin{aligned}CF &= 15 \times \dots \\ \text{and } CF &= \dots \times \sin \hat{B} \\ \therefore 15 \times \dots &= \dots \times \sin \hat{B} \\ \therefore \sin \hat{B} &= \dots \\ \therefore \hat{B} &= \dots\end{aligned}$$

2. Method 2: using the area rule

a) In $\triangle ABC$:

$$\begin{aligned}\text{Area } \triangle ABC &= \frac{1}{2} AB \times AC \times \dots \\ &= \frac{1}{2} AB \times \dots \times \dots\end{aligned}$$

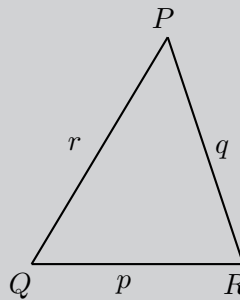
b) And we also know that

$$\text{Area } \triangle ABC = \frac{1}{2} AB \times \dots \times \sin \hat{B}$$

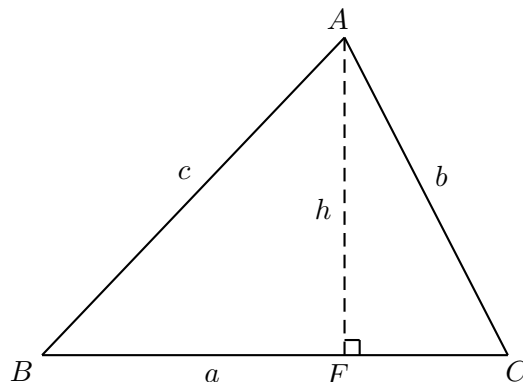
c) We can equate these two equations and solve for $\hat{B}$:

$$\begin{aligned}\frac{1}{2} AB \times \dots \times \sin \hat{B} &= \frac{1}{2} AB \times \dots \times \dots \\ \therefore \dots \times \sin \hat{B} &= \dots \times \dots \\ \therefore \sin \hat{B} &= \dots \times \dots \\ \therefore \hat{B} &= \dots\end{aligned}$$

3. Use your results to write a general formula for the sine rule given $\triangle PQR$:



For any triangle ABC with $AB = c$, $BC = a$ and $AC = b$, we can construct a perpendicular height (h) at F :



Method 1: using the sine ratio

In $\triangle ABF$:

$$\begin{aligned}\sin \hat{B} &= \frac{h}{c} \\ \therefore h &= c \sin \hat{B}\end{aligned}$$

In $\triangle ACF$:

$$\sin \hat{C} = \frac{h}{b}$$
$$\therefore h = b \sin \hat{C}$$

We can equate the two equations

$$c \sin \hat{B} = b \sin \hat{C}$$
$$\therefore \frac{\sin \hat{B}}{b} = \frac{\sin \hat{C}}{c}$$
$$\text{or } \frac{b}{\sin \hat{B}} = \frac{c}{\sin \hat{C}}$$

Similarly, by constructing a perpendicular height from vertex B to the line AC , we can also show that:

$$\frac{\sin \hat{A}}{a} = \frac{\sin \hat{C}}{c}$$
$$\text{or } \frac{a}{\sin \hat{A}} = \frac{c}{\sin \hat{C}}$$

Similarly, by constructing a perpendicular height from vertex B to the line AC , we can also show that $\text{area } \triangle ABC = \frac{1}{2}bc \sin \hat{A}$.

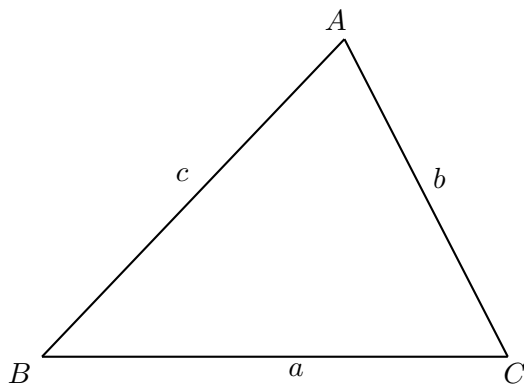
Method 2: using the area rule

In $\triangle ABC$:

$$\text{Area } \triangle ABC = \frac{1}{2}ac \sin \hat{B}$$
$$= \frac{1}{2}ab \sin \hat{C}$$
$$\therefore \frac{1}{2}ac \sin \hat{B} = \frac{1}{2}ab \sin \hat{C}$$
$$c \sin \hat{B} = b \sin \hat{C}$$
$$\frac{\sin \hat{B}}{b} = \frac{\sin \hat{C}}{c}$$
$$\text{or } \frac{b}{\sin \hat{B}} = \frac{c}{\sin \hat{C}}$$

The sine rule

In any $\triangle ABC$:



$$\frac{\sin \hat{A}}{a} = \frac{\sin \hat{B}}{b} = \frac{\sin \hat{C}}{c}$$

$$\frac{a}{\sin \hat{A}} = \frac{b}{\sin \hat{B}} = \frac{c}{\sin \hat{C}}$$

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Worked example 24: The sine rule

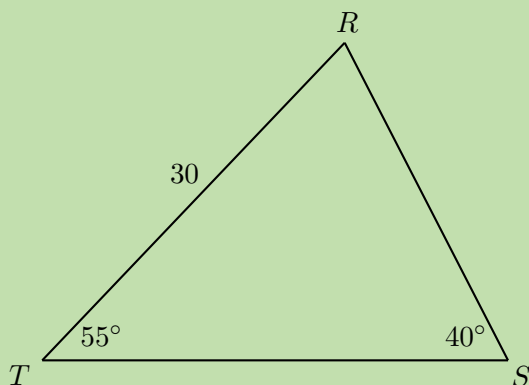
QUESTION

Given $\triangle TRS$ with $\hat{S}TR = 55^\circ$, $TR = 30$ and $\hat{R}ST = 40^\circ$, determine RS , ST and $\hat{T}RS$.

SOLUTION

Step 1: Draw a sketch

Let $RS = t$, $ST = r$ and $TR = s$.



Step 2: Find $\hat{T}RS$ using angles in a triangle

$$\hat{T}RS + \hat{R}ST + \hat{S}TR = 180^\circ \quad (\angle \text{sum of } \triangle TRS)$$

$$\therefore \hat{T}RS = 180^\circ - 40^\circ - 55^\circ$$

$$= 85^\circ$$

Step 3: Determine t and r using the sine rule

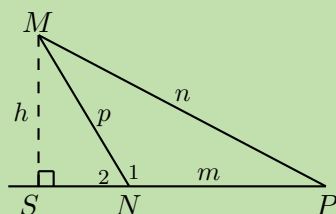
$$\begin{aligned}\frac{t}{\sin \hat{T}} &= \frac{s}{\sin \hat{S}} \\ \frac{t}{\sin 55^\circ} &= \frac{30}{\sin 40^\circ} \\ \therefore t &= \frac{30}{\sin 40^\circ} \times \sin 55^\circ \\ &= 38,2\end{aligned}$$

$$\begin{aligned}\frac{r}{\sin \hat{R}} &= \frac{s}{\sin \hat{S}} \\ \frac{r}{\sin 85^\circ} &= \frac{30}{\sin 40^\circ} \\ \therefore r &= \frac{30}{\sin 40^\circ} \times \sin 85^\circ \\ &= 46,5\end{aligned}$$

Worked example 25: The sine rule

QUESTION

Prove the sine rule for $\triangle MNP$ with $MS \perp NP$.



SOLUTION

Step 1: Use the sine ratio to express the angles in the triangle in terms of the length of the sides

In $\triangle MSN$:

$$\sin \hat{N}_2 = \frac{h}{p}$$

$$\therefore h = p \sin \hat{N}_2$$

$$\text{and } \hat{N}_2 = 180^\circ - \hat{N}_1 \quad \angle\text{s on str. line}$$

$$\therefore h = p \sin(180^\circ - \hat{N}_1)$$

$$= p \sin \hat{N}_1$$

In $\triangle MSP$:

$$\sin \hat{P} = \frac{h}{n}$$

$$\therefore h = n \sin \hat{P}$$

Step 2: Equate the two equations to derive the sine rule

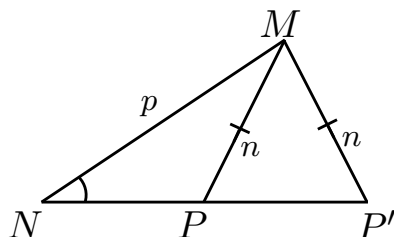
$$p \sin \hat{N}_1 = n \sin \hat{P}$$

$$\therefore \frac{\sin \hat{N}_1}{n} = \frac{\sin \hat{P}}{p}$$

$$\text{or } \frac{n}{\sin \hat{N}_1} = \frac{p}{\sin \hat{P}}$$

The ambiguous case

If two sides and an interior angle of a triangle are given, and the side opposite the given angle is the shorter of the two sides, then we can draw two different triangles ($\triangle NMP$ and $\triangle NMP'$), both having the given dimensions. We call this the ambiguous case because there are two ways of interpreting the given information and it is not certain which is the required solution.



Worked example 26: The ambiguous case

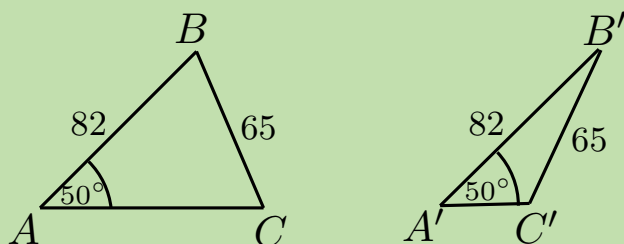
QUESTION

In $\triangle ABC$, $AB = 82$, $BC = 65$ and $\hat{A} = 50^\circ$. Draw $\triangle ABC$ and find $\hat{C}$ (correct to one decimal place).

SOLUTION

Step 1: Draw a sketch and identify the ambiguous case

We notice that for the given dimensions of $\triangle ABC$, the side BC opposite $\hat{A}$ is shorter than AB . This means that we can draw two different triangles with the given dimensions.



Step 2: Solve for unknown angle using the sine rule

In $\triangle ABC$:

$$\begin{aligned}\frac{\sin \hat{A}}{BC} &= \frac{\sin \hat{C}}{AB} \\ \frac{\sin 50^\circ}{65} &= \frac{\sin \hat{C}}{82} \\ \therefore \frac{\sin 50^\circ}{65} \times 82 &= \sin \hat{C} \\ \therefore \hat{C} &= 75,1^\circ\end{aligned}$$

In $\triangle A'B'C'$:

We know that $\sin(180 - \hat{C}) = \sin \hat{C}$, which means we can also have the solution

$$\begin{aligned}\hat{C}' &= 180^\circ - 75,1^\circ \\ &= 104,9^\circ\end{aligned}$$

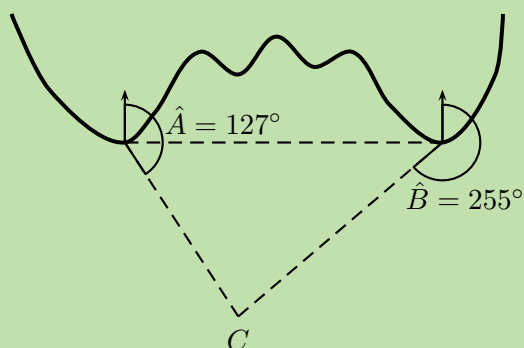
Both solutions are correct.

Worked example 27: Lighthouses

QUESTION

There is a coastline with two lighthouses, one on either side of a beach. The two lighthouses are 0,67 km apart and one is exactly due east of the other. The lighthouses tell how close a boat is by taking bearings to the boat (a bearing is an angle measured clockwise from north). These bearings are shown on the diagram below.

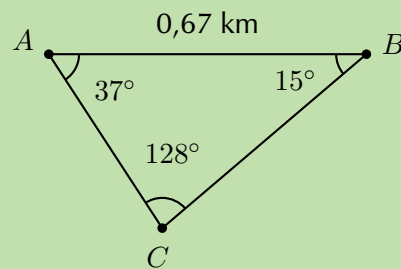
Calculate how far the boat is from each lighthouse.



SOLUTION

We see that the two lighthouses and the boat form a triangle. Since we know the distance between the lighthouses and we have two angles we can use trigonometry

to find the remaining two sides of the triangle, the distance of the boat from the two lighthouses.



We need to determine the lengths of the two sides AC and BC . We can use the sine rule to find the missing lengths.

$$\begin{aligned}\frac{BC}{\sin \hat{A}} &= \frac{AB}{\sin \hat{C}} \\ BC &= \frac{AB \cdot \sin \hat{A}}{\sin \hat{C}} \\ &= \frac{(0,67 \text{ km}) \sin 37^\circ}{\sin 128^\circ} \\ &= 0,51 \text{ km}\end{aligned}$$

$$\begin{aligned}\frac{AC}{\sin \hat{B}} &= \frac{AB}{\sin \hat{C}} \\ AC &= \frac{AB \cdot \sin \hat{B}}{\sin \hat{C}} \\ &= \frac{(0,67 \text{ km}) \sin 15^\circ}{\sin 128^\circ} \\ &= 0,22 \text{ km}\end{aligned}$$

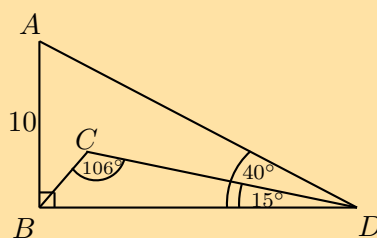
Exercise 6 – 11: Sine rule

1. Find all the unknown sides and angles of the following triangles:

- a) $\triangle PQR$ in which $\hat{Q} = 64^\circ$; $\hat{R} = 24^\circ$ and $r = 3$
- b) $\triangle KLM$ in which $\hat{K} = 43^\circ$; $\hat{M} = 50^\circ$ and $m = 1$
- c) $\triangle ABC$ in which $\hat{A} = 32,7^\circ$; $\hat{C} = 70,5^\circ$ and $a = 52,3$
- d) $\triangle XYZ$ in which $\hat{X} = 56^\circ$; $\hat{Z} = 40^\circ$ and $x = 50$

- 2. In $\triangle ABC$, $\hat{A} = 116^\circ$; $\hat{C} = 32^\circ$ and $AC = 23$ m. Find the lengths of the sides AB and BC .
- 3. In $\triangle RST$, $\hat{R} = 19^\circ$; $\hat{S} = 30^\circ$ and $RT = 120$ km. Find the length of the side ST .
- 4. In $\triangle KMS$, $\hat{K} = 20^\circ$; $\hat{M} = 100^\circ$ and $s = 23$ cm. Find the length of the side m .

5. In $\triangle ABD$, $\hat{B} = 90^\circ$, $AB = 10$ cm and $\hat{ADB} = 40^\circ$. In $\triangle BCD$, $\hat{C} = 106^\circ$ and $\hat{CDB} = 15^\circ$. Determine BC .



6. In $\triangle ABC$, $\hat{A} = 33^\circ$, $AC = 21$ mm and $AB = 17$ mm. Can you determine BC ?

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4. [233Q](#) 5. [233R](#) 6. [233S](#)



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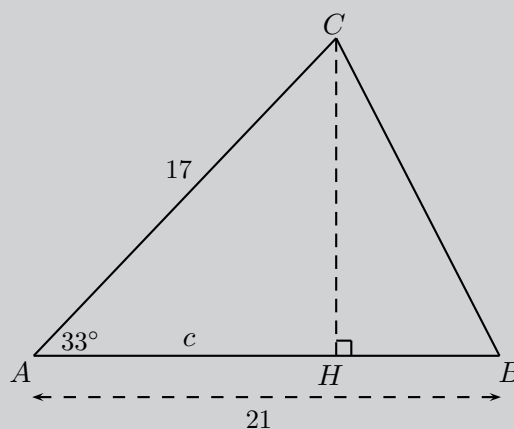
The cosine rule

EMBHS

Investigation: The cosine rule

If a triangle is given with two sides and the included angle known, then we can not solve for the remaining unknown sides and angles using the sine rule. We therefore investigate the cosine rule:

In $\triangle ABC$, $AB = 21$, $AC = 17$ and $\hat{A} = 33^\circ$. Find $\hat{B}$.



1. Determine CB :

- Construct $CH \perp AB$.
- Let $AH = c$ and therefore $HB = \dots$

c) Applying the theorem of Pythagoras in the right-angled triangles:

In $\triangle CHB$:

$$\begin{aligned} CB^2 &= BH^2 + CH^2 \\ &= (\dots)^2 + CH^2 \\ &= 21^2 - (2)(21)c + c^2 + CH^2 \dots\dots (1) \end{aligned}$$

In $\triangle CHA$:

$$\begin{aligned} CA^2 &= c^2 + CH^2 \\ 17^2 &= c^2 + CH^2 \dots\dots (2) \end{aligned}$$

Substitute equation (2) into equation (1):

$$CB^2 = 21^2 - (2)(21)c + 17^2$$

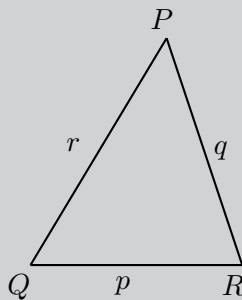
Now c is the only remaining unknown. In $\triangle CHA$:

$$\begin{aligned} \frac{c}{17} &= \cos 33^\circ \\ \therefore c &= 17 \cos 33^\circ \end{aligned}$$

Therefore we have that

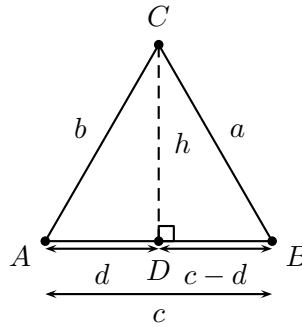
$$\begin{aligned} CB^2 &= 21^2 - (2)(21)c + 17^2 \\ &= 21^2 - (2)(21)(17 \cos 33^\circ) + 17^2 \\ &= 21^2 + 17^2 - (2)(21)(17) \cos 33^\circ \\ &= 131,189\dots \\ \therefore CB &= 11,5 \end{aligned}$$

2. Use your results to write a general formula for the cosine rule given $\triangle PQR$:



The cosine rule relates the length of a side of a triangle to the angle opposite it and the lengths of the other two sides.

Consider $\triangle ABC$ with $CD \perp AB$:



In $\triangle DCB$: $a^2 = (c - d)^2 + h^2$ from the theorem of Pythagoras.

In $\triangle ACD$: $b^2 = d^2 + h^2$ from the theorem of Pythagoras.

Since h^2 is common to both equations we can write:

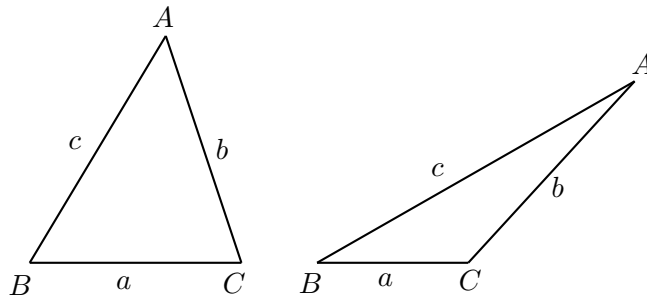
$$\begin{aligned} a^2 &= (c - d)^2 + h^2 \\ \therefore h^2 &= a^2 - (c - d)^2 \\ \text{And } b^2 &= d^2 + h^2 \\ \therefore h^2 &= b^2 - d^2 \\ \therefore b^2 - d^2 &= a^2 - (c - d)^2 \\ a^2 &= b^2 + (c^2 - 2cd + d^2) - d^2 \\ &= b^2 + c^2 - 2cd \end{aligned}$$

In order to eliminate d we look at $\triangle ACD$, where we have: $\cos \hat{A} = \frac{d}{b}$. So, $d = b \cos \hat{A}$.

Substituting back we get: $a^2 = b^2 + c^2 - 2bc \cos \hat{A}$.

The cosine rule

In any $\triangle ABC$:



$$a^2 = b^2 + c^2 - 2bc \cos \hat{A}$$

$$b^2 = a^2 + c^2 - 2ac \cos \hat{B}$$

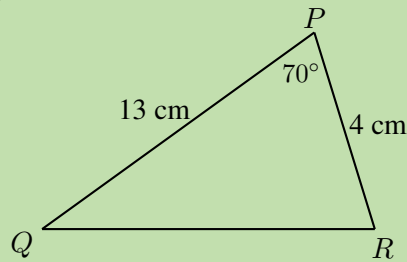
$$c^2 = a^2 + b^2 - 2ab \cos \hat{C}$$

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Worked example 28: The cosine rule

QUESTION

Determine the length of QR .



SOLUTION

Step 1: Use the cosine rule to solve for the unknown side

$$\begin{aligned} QR^2 &= PR^2 + QP^2 - 2(PR)(QP) \cos \hat{P} \\ &= 4^2 + 13^2 - 2(4)(13) \cos 70^\circ \\ &= 149,42 \dots \\ \therefore QR &= 12,2 \end{aligned}$$

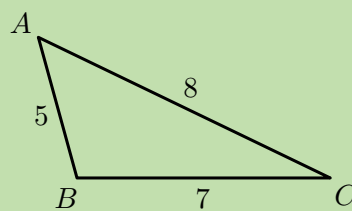
Step 2: Write the final answer

$$QR = 12,2 \text{ cm}$$

Worked example 29: The cosine rule

QUESTION

Determine $\hat{A}$.



SOLUTION

Applying the cosine rule:

$$\begin{aligned} a^2 &= b^2 + c^2 - 2bc \cos \hat{A} \\ \therefore \cos \hat{A} &= \frac{b^2 + c^2 - a^2}{2bc} \\ &= \frac{8^2 + 5^2 - 7^2}{2 \cdot 8 \cdot 5} \\ &= 0,5 \\ \therefore \hat{A} &= 60^\circ \end{aligned}$$

It is very important:

- not to round off before the final answer as this will affect accuracy;
- to take the square root;
- to remember to give units where applicable.

How to determine which rule to use:

1. Area rule:

- if no perpendicular height is given

2. Sine rule:

- if no right angle is given
- if two sides and an angle are given (not the included angle)
- if two angles and a side are given

3. Cosine rule:

- if no right angle is given
- if two sides and the included angle are given
- if three sides are given

Exercise 6 – 12: The cosine rule

1. Solve the following triangles (that is, find all unknown sides and angles):

- $\triangle ABC$ in which $\hat{A} = 70^\circ$; $b = 4$ and $c = 9$
- $\triangle RST$ in which $RS = 14$; $ST = 26$ and $RT = 16$
- $\triangle KLM$ in which $KL = 5$; $LM = 10$ and $KM = 7$
- $\triangle JHK$ in which $\hat{H} = 130^\circ$; $JH = 13$ and $HK = 8$
- $\triangle DEF$ in which $d = 4$; $e = 5$ and $f = 7$

2. Find the length of the third side of the $\triangle XYZ$ where:

- $\hat{X} = 71,4^\circ$; $y = 3,42$ km and $z = 4,03$ km
- $x = 103,2$ cm; $\hat{Y} = 20,8^\circ$ and $z = 44,59$ cm

3. Determine the largest angle in:

- $\triangle JHK$ in which $JH = 6$; $HK = 4$ and $JK = 3$
- $\triangle PQR$ where $p = 50$; $q = 70$ and $r = 60$

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1a. 233V 1b. 233W 1c. 233X 1d. 233Y 1e. 233Z 2a. 2342
2b. 2343 3a. 2344 3b. 2345



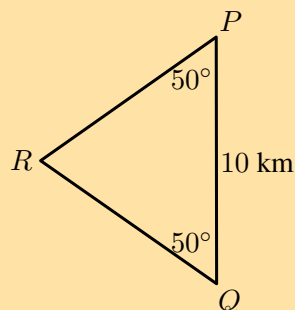
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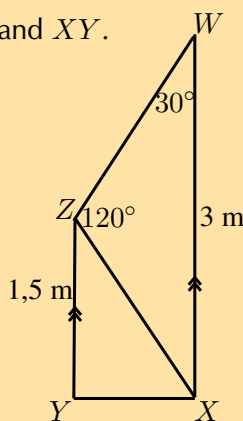
Exercise 6 – 13: Area, sine and cosine rule

1. Q is a ship at a point 10 km due south of another ship P . R is a lighthouse on the coast such that $\hat{P} = \hat{Q} = 50^\circ$.

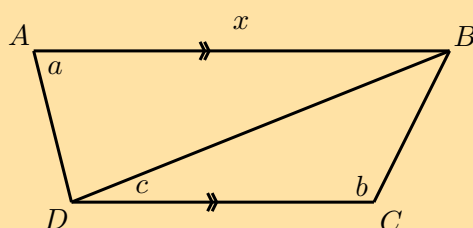


Determine:

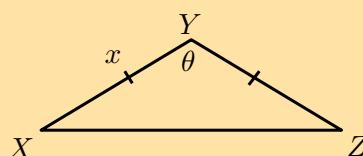
- the distance QR
 - the shortest distance from the lighthouse to the line joining the two ships (PQ).
2. $WXYZ$ is a trapezium, $WX \parallel YZ$ with $WX = 3$ m; $YZ = 1,5$ m; $\hat{Z} = 120^\circ$ and $\hat{W} = 30^\circ$.



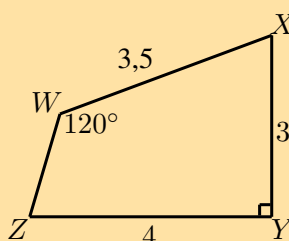
3. On a flight from Johannesburg to Cape Town, the pilot discovers that he has been flying 3° off course. At this point the plane is 500 km from Johannesburg. The direct distance between Cape Town and Johannesburg airports is 1552 km. Determine, to the nearest km:
- The distance the plane has to travel to get to Cape Town and hence the extra distance that the plane has had to travel due to the pilot's error.
 - The correction, to one hundredth of a degree, to the plane's heading (or direction).
4. $ABCD$ is a trapezium (meaning that $AB \parallel CD$). $AB = x$; $\hat{BAD} = a$; $\hat{BCD} = b$ and $\hat{BDC} = c$.
Find an expression for the length of CD in terms of x , a , b and c .



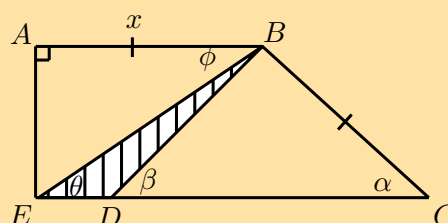
5. A surveyor is trying to determine the distance between points X and Z . However the distance cannot be determined directly as a ridge lies between the two points. From a point Y which is equidistant from X and Z , he measures the angle $X\hat{Y}Z$.



- a) If $XY = x$ and $X\hat{Y}Z = \theta$, show that $XZ = x\sqrt{2(1 - \cos \theta)}$.
 b) Calculate XZ (to the nearest kilometre) if $x = 240$ km and $\theta = 132^\circ$.
6. Find the area of $WXYZ$ (to two decimal places):



7. Find the area of the shaded triangle in terms of x , α , β , θ and ϕ :



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1. 2347 2. 2348 3. 2349 4. 234B 5. 234C 6. 234D
 7. 234F



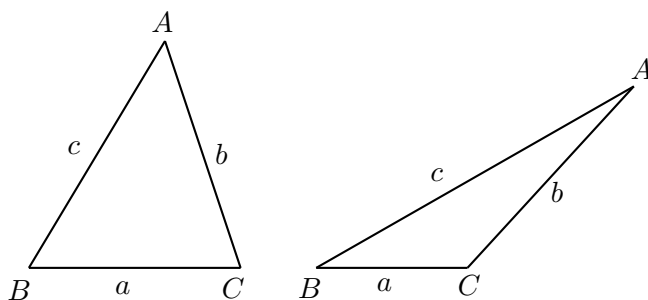
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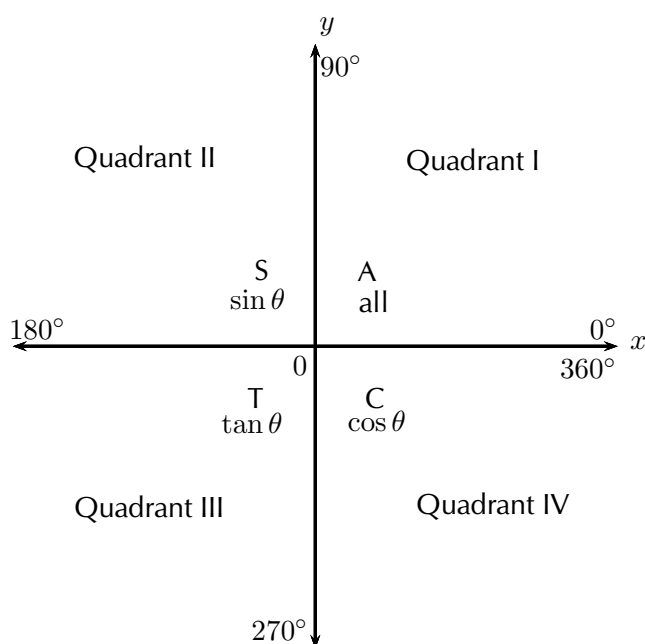
🔗 See presentation: 234G at www.everythingmaths.co.za

square identity	quotient identity
$\cos^2 \theta + \sin^2 \theta = 1$	$\tan \theta = \frac{\sin \theta}{\cos \theta}$



negative angles	periodicity identities	co-function identities
$\sin(-\theta) = -\sin \theta$	$\sin(\theta \pm 360^\circ) = \sin \theta$	$\sin(90^\circ - \theta) = \cos \theta$
$\cos(-\theta) = \cos \theta$	$\cos(\theta \pm 360^\circ) = \cos \theta$	$\cos(90^\circ - \theta) = \sin \theta$

sine rule	area rule	cosine rule
$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$	area $\triangle ABC = \frac{1}{2}bc \sin A$	$a^2 = b^2 + c^2 - 2bc \cos A$
$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$	area $\triangle ABC = \frac{1}{2}ac \sin B$	$b^2 = a^2 + c^2 - 2ac \cos B$
	area $\triangle ABC = \frac{1}{2}ab \sin C$	$c^2 = a^2 + b^2 - 2ab \cos C$



General solution:

1.

$$\text{If } \sin \theta = x$$

$$\theta = \sin^{-1} x + k \cdot 360^\circ$$

$$\text{or } \theta = (180^\circ - \sin^{-1} x) + k \cdot 360^\circ$$

2.

$$\text{If } \cos \theta = x$$

$$\theta = \cos^{-1} x + k \cdot 360^\circ$$

$$\text{or } \theta = (360^\circ - \cos^{-1} x) + k \cdot 360^\circ$$

3.

$$\text{If } \tan \theta = x$$

$$\theta = \tan^{-1} x + k \cdot 180^\circ$$

for $k \in \mathbb{Z}$.

How to determine which rule to use:

1. Area rule:

- no perpendicular height is given

2. Sine rule:

- no right angle is given
- two sides and an angle are given (not the included angle)
- two angles and a side are given

3. Cosine rule:

- no right angle is given
- two sides and the included angle are given
- three sides are given

Exercise 6 – 14: End of chapter exercises

1. Write the following as a single trigonometric ratio:

$$\frac{\cos(90^\circ - A) \sin 20^\circ}{\sin(180^\circ - A) \cos 70^\circ} + \cos(180^\circ + A) \sin(90^\circ + A)$$

2. Determine the value of the following expression without using a calculator:

$$\sin 240^\circ \cos 210^\circ - \tan^2 225^\circ \cos 300^\circ \cos 180^\circ$$

3. Simplify:

$$\frac{\sin(180^\circ + \theta) \sin(\theta + 360^\circ)}{\sin(-\theta) \tan(\theta - 360^\circ)}$$

4. Without the use of a calculator, evaluate:

$$\frac{3 \sin 55^\circ \sin^2 325^\circ}{\cos(-145^\circ)} - 3 \cos 395^\circ \sin 125^\circ$$

5. Prove the following identities:

a) $\frac{1}{(\cos x - 1)(\cos x + 1)} = \frac{-1}{\tan^2 x \cos^2 x}$

b) $(1 - \tan \alpha) \cos \alpha = \sin(90^\circ + \alpha) + \cos(90^\circ + \alpha)$

6. a) Prove: $\tan y + \frac{1}{\tan y} = \frac{1}{\cos^2 y \tan y}$

b) For which values of $y \in [0^\circ; 360^\circ]$ is the identity above undefined?

7. a) Simplify: $\frac{\sin(180^\circ + \theta) \tan(360^\circ - \theta)}{\sin(-\theta) \tan(180^\circ + \theta)}$

b) Hence, solve the equation

$$\frac{\sin(180^\circ + \theta) \tan(360^\circ - \theta)}{\sin(-\theta) \tan(180^\circ + \theta)} = \tan \theta$$

for $\theta \in [0^\circ; 360^\circ]$.

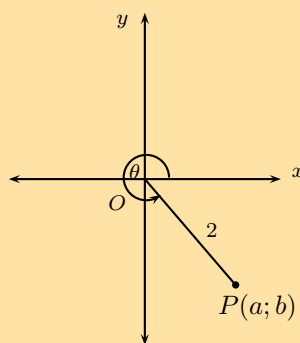
8. Given $12 \tan \theta = 5$ and $\theta > 90^\circ$.

a) Draw a sketch.

b) Determine without using a calculator $\sin \theta$ and $\cos(180^\circ + \theta)$.

c) Use a calculator to find θ (correct to two decimal places).

9.



In the figure, P is a point on the Cartesian plane such that $OP = 2$ units and $\theta = 300^\circ$. Without the use of a calculator, determine:

a) the values of a and b

b) the value of $\sin(180^\circ - \theta)$

10. Solve for x with $x \in [-180^\circ; 180^\circ]$ (correct to one decimal place):

a) $2 \sin \frac{x}{2} = 0,86$

b) $\tan(x + 10^\circ) = \cos 202,6^\circ$

c) $\cos^2 x - 4 \sin^2 x = 0$

11. Find the general solution for the following equations:

a) $\frac{1}{2} \sin(x - 25^\circ) = 0,25$

b) $\sin^2 x + 2 \cos x = -2$

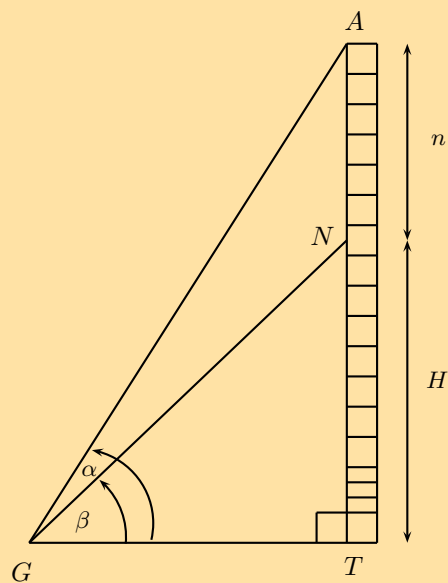
12. Given the equation: $\sin 2\alpha = 0,84$

a) Find the general solution of the equation.

b) Illustrate how this equation could be solved graphically for $\alpha \in [0^\circ; 360^\circ]$.

c) Write down the solutions for $\sin 2\alpha = 0,84$ for $\alpha \in [0^\circ; 360^\circ]$.

13.



A is the highest point of a vertical tower AT. At point N on the tower, n metres from the top of the tower, a bird has made its nest. The angle of inclination from G to point A is α and the angle of inclination from G to point N is β .

a) Express $\hat{AGN}$ in terms of α and β .

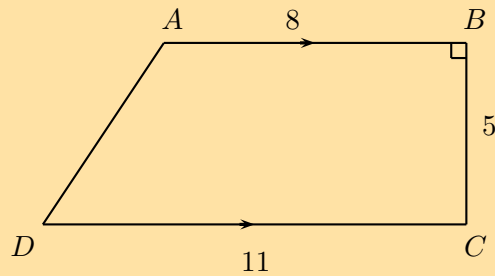
b) Express $\hat{A}$ in terms of α and/or β .

c) Show that the height of the nest from the ground (H) can be determined by the formula

$$H = \frac{n \cos \alpha \sin \beta}{\sin(\alpha - \beta)}$$

d) Calculate the height of the nest H if $n = 10$ m, $\alpha = 68^\circ$ and $\beta = 40^\circ$ (give your answer correct to the nearest metre).

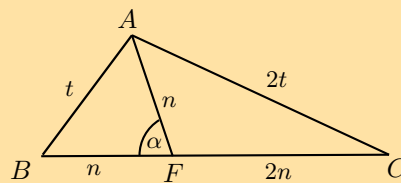
14.



Mr. Collins wants to pave his trapezium-shaped backyard, $ABCD$. $AB \parallel DC$ and $\hat{B} = 90^\circ$. $DC = 11$ m, $AB = 8$ m and $BC = 5$ m.

- Calculate the length of the diagonal AC .
- Calculate the length of the side AD .
- Calculate the area of the patio using geometry.
- Calculate the area of the patio using trigonometry.

15.



In $\triangle ABC$, $AC = 2AB$, $AF = BF$, $\hat{AFB} = \alpha$ and $FC = 2AF$. Prove that $\cos \alpha = \frac{1}{4}$.

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- | | | | | | |
|---------------------------|---------------------------|---------------------------|--------------------------|---------------------------|---------------------------|
| 1. 234H | 2. 234J | 3. 234K | 4. 234M | 5a. 234N | 5b. 234P |
| 6. 234Q | 7. 234R | 8. 234S | 9. 234T | 10a. 234V | 10b. 234W |
| 10c. 234X | 11a. 234Y | 11b. 234Z | 12. 2352 | 13. 2353 | 14. 2354 |
| 15. 2355 | | | | | |



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